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MANUAL FOR
OVERHAUL, REPAIR AND HANDLING OF
HAMILTON SHIP CHRONOMETER
WITH PARTS CATALOG

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HAMILTON SHIP CHRONOMETER

This publication will be known as: "Manual for Overhaul, Repair and Handling of Hamilton Ship Chronometer, with Parts Catalog." It is published by the Chief of the Bureau of Ships as directed by the Secretary of the Navy.

The directives contained in this manual are issued for guidance, and are binding upon all Naval and Civilian Personnel of Naval Establishments engaged in the servicing, repairing and testing of chronometers. They have the same force and effect as orders issued by the Secretary of the Navy and the Chief of the Bureau of Ships.

Reproductions of, and extracts from, this manual may be made for Naval use as directed by local competent authority.

This manual may be made available to private activities when it is in the interests of the United States Navy, provided that authority is obtained from the Chief of the Bureau of Ships, and a local record of distribution is maintained.

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This manual may be carried in aircraft.

Since this is the first comprehensive manual to be published on the subject, recipients are especially encouraged to recommend changes hereto which are based upon service experience.

This manual will become effective 1 October 1948.



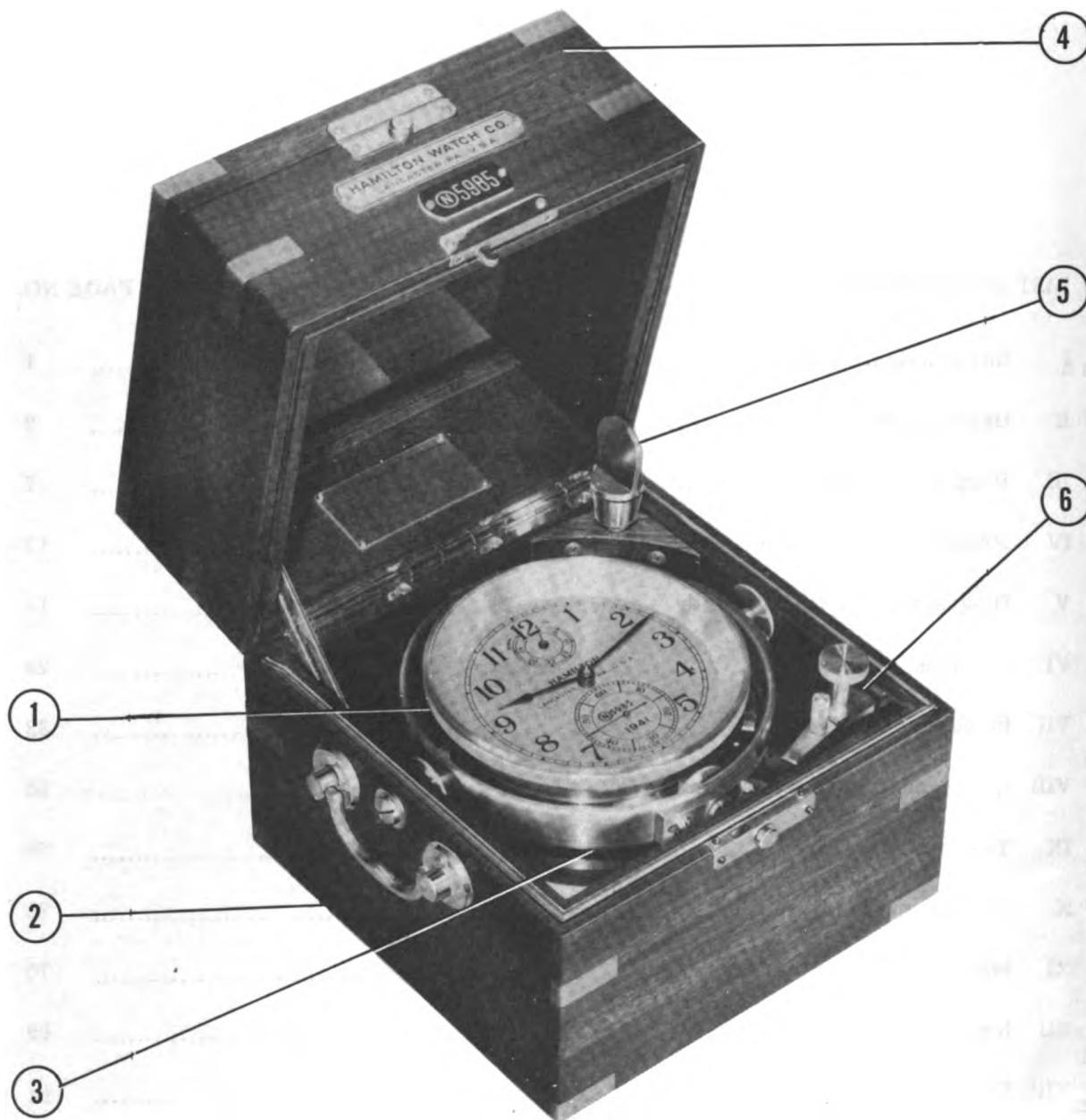
C. D. Wheelock
Chief of the Bureau of Ships (Acting)

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HAMILTON SHIP CHRONOMETER



1 CHRONOMETER

2 MOUNTING BOX

3 GIMBAL MOUNT

4 UPPER LID

5 WINDING KEY

6 GIMBAL LATCH

Figure 1 — Chronometer in Mounting Box

SECTION I

INTRODUCTION

This handbook contains descriptive data and instructions for the operation, maintenance, repair and overhaul of the Ship Chronometer - Hamilton make, manufactured to U. S. Navy Specification 18-c-7. The Standard Stock Catalog number is 18-C-815.

This chronometer was designed and is manufactured by the Hamilton Watch Company, Lancaster, Pa., U. S. A. Certain instruments have been modified by the Navy to include a balance stop, which serves instead of wedges to immobilize the balance. The plan is to provide all of these chronometers with balance stops, but instructions herein cover the overhaul of a chronometer not so equipped. The device itself is described in Section III of this book. See Fig. 9.

The manufacturer's part number for the complete assembly is 42213, which covers the instrument in mounting box and carrying case. Manufacturer's part number 42301 includes the instrument and mounting box only. Since the carrying case is discontinued as an item of general issue, the combination which will be found in general use is covered by the last mentioned part number. Serial numbers as assigned by the U. S. Navy are engraved on each chronometer dial and the barrel bridge.

Capital ships of the U. S. Navy (battleships, carriers, cruisers) are allotted two or more of these chronometers. Thus, the navigator can compare readings for more accurate determination of time and longitude. For safety, the chronometers are located in different stations aboard ship, according to the discretion of the ship's Commanding Officer.

Fighting ships other than capital ships (destroyers through submarines) and seagoing ships (such as supply ships, tenders and hospital ships) are allotted one or more chronometers and one or more gimbal chronometer watches. Usage is similar to that on capital ships. For the exact navigating timepiece allowance on a particular type of vessel, reference should be made to the latest timepiece allowance list.

Purpose

The primary objective of this publication is to provide a means for uniform control of repair procedures for the Hamilton make of Ship Chronometer by giving instruction in standardized servicing practices. It is intended that every instrument on which work is done will be completely reconditioned, and that worn, damaged or broken parts will be replaced rather than repaired.

The methods and techniques to be used are those developed in connection with Hamilton special service tools, gages and fixtures which were the result of concentrated analysis and study. For further information on service tools, see the introduction to Section XIII entitled "Service Tools."

Portions of certain sections in the book are prepared mainly for purposes of familiarization and are useful even to operating personnel or others who have little or no watchmaking skill. These sections include Description, Preparation for Use, Operation, and Storage and Shipment.

Other sections, like Disassembly, Cleaning, Inspection and Repair, Reassembly, and Test and Adjustment, are of principal utility to skilled workmen. These sections are built one upon another, the supposition being that increased knowledge and skill are required for the work of each succeeding section as it appears in the manual. The Parts Catalog will prove helpful not only to the novice but also to the expert.

CAUTION

Adjustment, repair or assembly of chronometers, as treated in this book, should be undertaken only by qualified, experienced personnel skilled in the watchmaker's trade. Experience with instruments of Hamilton manufacture is not essential if strict attention is paid to the methods herein set forth.

SECTION II

DESCRIPTION

The Hamilton make of Ship Chronometer has an 85-size, 14-jewel movement, known as manufacturer's model 21, part number 42000. It is characterized by essential interchangeability of parts and its design is such as to obviate the need for arduous and time-consuming application of personal skills in adjustment of the individual movement.

With dial and hands, the movement is fitted into its brass case and set in gimbals to maintain horizontal position in the mounting box. Fig. 1, the frontispiece, shows the instrument in its mounting box.

While the appearance of this chronometer is conventional, a detailed examination of the instrument itself reveals many innovations not previously used. A description of these appears in subsequent paragraphs.

NOTE

Carrying Case No. 42202, as formerly used, consisted of a brass-fitted mahogany box designed to hold the mounting box with a snug fit. See indication of this arrangement in Section XI, Fig. 106. It was padded inside, provided with a hinged mahogany cover, and equipped with leather carrying strap. A safety latch fastened the cover when closed. Case No. 42130 was identical to No. 42202 except that it had a hook and eye fastener. Few of these carrying cases are still in use, though some will continue to be issued for special purposes. A part of their utility is now served by the newly-developed shipping container, illustrated in Section X, Fig. 103.

MOUNTING BOX

Mounting Box No. 42201, shown in Fig. 1, is of mahogany with brass fittings and with two hinged covers. This box contains the gimbal mountings for the chronometer, with latch to immobilize the gimbals when required. Appliqued felt protects the outside bottom of the mounting box. The upper lid is hinged to permit the instrument to be read through the glass top of the second cover which is usually kept closed to protect the instrument, except

when opened for winding or setting. Latches fasten the upper and lower covers.

Mounting Box No. 42129 is identical to the box described except that it is equipped with lock and key for the lower lid. Both styles of box have provision for mounting a winding key.

CAUTION

If the instrument is not in its mounting box, the gimbal system must be locked in place with its latch before cover is closed, or the glass top will be broken.

GIMBAL MOUNT

The gimbal mount, well illustrated in Fig. 1, consists of a brass ring pivoted at two points 180 degrees apart. These pivots are fastened to opposite sides of the mounting box. The brass case containing the chronometer movement is pivoted from the brass ring at two points 180 degrees apart and 90 degrees away from the two pivots holding the ring to the mounting box.

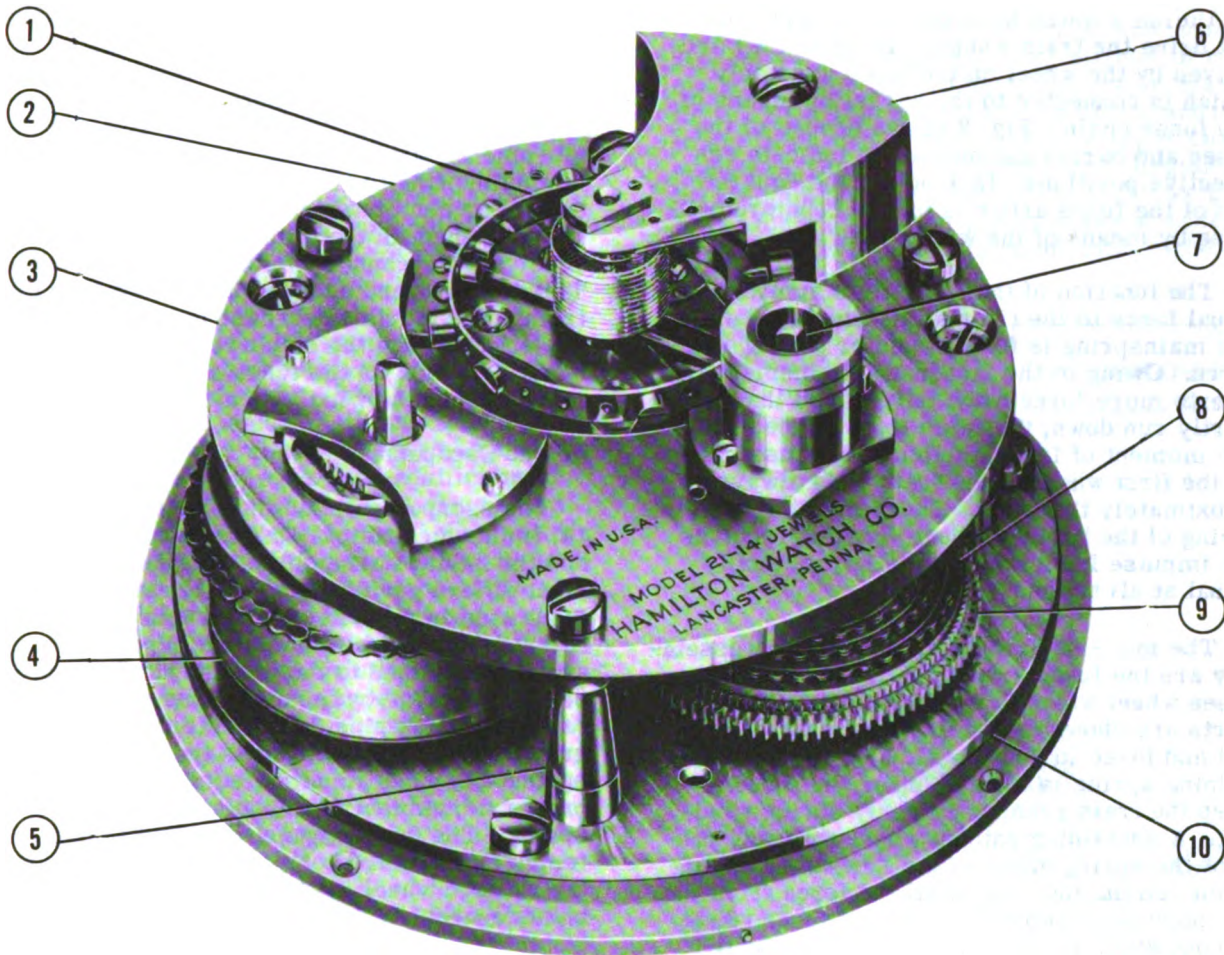
The chronometer is leveled when originally inserted in the gimbals by adjusting the position of the case brackets and gimbal support straps. Because of the gimbal construction, the mounting box may be tilted in any direction and the chronometer movement will tend to remain level. See exploded view in Section XI, Fig. 106.

CHRONOMETER CASE

The brass chronometer case consists of the bezel and brass bowl to which are mounted the case support brackets, latch keeper and shield plate. See exploded view in Section XI, Fig. 107. The winding-hole shield plate, located on the bottom of the brass case, is turned clockwise to permit insertion of the winding key. When the key is removed, a spring rotates the plate back into position.

DIAL

The dial has numerals, graduations, and letters all in black on a silver white background. The hour and minute scale is divided into 60 graduations and is marked from one to



- 1 BALANCE AND HAIRSPRING ASSEMBLY
- 2 TRAIN BRIDGE
- 3 BARREL BRIDGE
- 4 BARREL
- 5 PILLAR

- 6 BALANCE COCK
- 7 FUSEE ARBOR
- 8 FUSEE
- 9 FUSEE WHEEL
- 10 PILLAR PLATE

Figure 2 — Chronometer Movement

twelve hours. A wind indicator scale marked "UP-DOWN" is located below the 12 hour mark with graduations in multiples of 8, from 8 to 48. At the place of numeral "6" is a seconds scale divided into 60 parts with principal graduations in multiples of five, from 5 to 60. See Fig. 107 in Section XI.

FUNCTIONAL DESCRIPTION OF MOVEMENT

Plates and Bridges

As with standard watches, the pillar plate is the foundation of the movement. The barrel bridge, train bridge, and balance cock perform functions similar to those in a watch

except that the barrel bridge also holds the fusee in addition to the barrel. An additional bridge, the balance lower bridge (see exploded view in Section XI, Fig. 110) supports the lower cap jewel for the balance staff and the fourth upper setting. A lower train bridge contains the third and fourth lower settings.

As shown in Fig. 2, the barrel and train bridges are mounted off the pillar plate by means of four pillars. These serve the same purpose as the posts or pillars used to hold the plates of an old-fashioned watch movement. The balance cock and balance lower bridge are mounted to the train bridge. The barrel bridge is shaped to accommodate the balance wheel, digitized by Google

Power Source

Unlike a watch movement, the barrel does not drive the train wheels. Instead, they are driven by the wheel on the fusee assembly which is connected to the barrel by means of the fusee chain. Fig. 2 shows enough of the fusee and barrel assemblies to indicate respective positions. In winding, the squared top of the fusee arbor is turned counterclockwise by means of the winding key.

The function of the fusee is to apply an equal force to the train at all times, whether the mainspring is fully wound or almost run down. Owing to the fact that a mainspring exerts more force when fully wound than when partly run down, the fusee is so shaped that the moment of force applied to the fusee wheel — the first wheel of the train — is always approximately the same. Therefore, the arc of swing of the balance wheel — which receives its impulse from the train — is practically equal at all times.

The more obvious parts of the fusee assembly are the fusee arbor, the fusee, and the fusee wheel which drives the train. Other parts are shown on the exploded view of barrel and fusee in Section XI, Fig. 109. A sustaining spring is located under the fusee to keep the train running properly during winding. A sustaining ratchet wheel and pawl prevent the spring effect of the sustaining spring from coming into play until the force from the mainspring is removed from the fusee wheel during winding. A winding stop-bar located in the top of the fusee prevents overwinding. By the action of the fusee chain at full wind, this bar moves out to engage the winding stop on the bottom of the barrel bridge. For further detail, see Section IV entitled "Principles of Operation."

The mainspring barrel is supported by its arbor between the pillar plate and barrel bridge. See Fig. 2. As in watch movements, the barrel contains the mainspring and the mainspring brace. In place of the ratchet wheel and click on a watch movement, there is a setup ratchet wheel mounted to the barrel arbor where it protrudes through the barrel bridge and, on the barrel bridge, there is a setup pawl and setup spring with setup cover plate. The function of these parts is to prevent the barrel arbor from turning during winding and running. See exploded view in Section XI, Fig. 108. The top of the arbor is squared to allow the use of a let-down key to disarm the mainspring before servicing. The only time that the mainspring arbor turns is when manipulated during assembly, disassembly or adjustment.

Train

The train, shown in Section IV, Fig. 13, is a set of wheels and pinions which transfer and step up the rotation from the fusee wheel and deliver it to the escapement. The chronometer train is similar to that in the usual watch movement except that the first wheel is not on the mainspring barrel but on the fusee. The second or center wheel has a long arbor which projects through the pillar plate and dial, and receives the cannon pinion and hour wheel. Third, fourth, and escape wheels complete the train. The fourth wheel has a long arbor which projects through the dial to receive the second hand. A pinion on the fusee arbor controls movement of the wind indicator wheel.

Detent Escapement

The escapement, shown in Section IV, Fig. 14, is the mechanical link between the train and the balance assembly. This detent escapement differs greatly from the lever escapement used in modern watch movements. The detent is mounted to the detent support block which is fastened to the underside of the upper train bridge by means of one screw and two positioning pins. Mounted to the detent by means of an angle bracket is the detent trip spring. Various adjusting screws permit the proper adjustment of the detent and trip spring. The escape wheel is unlocked and an impulse is received only during the swing of the balance wheel in one direction. When the balance wheel returns, it is independent of the

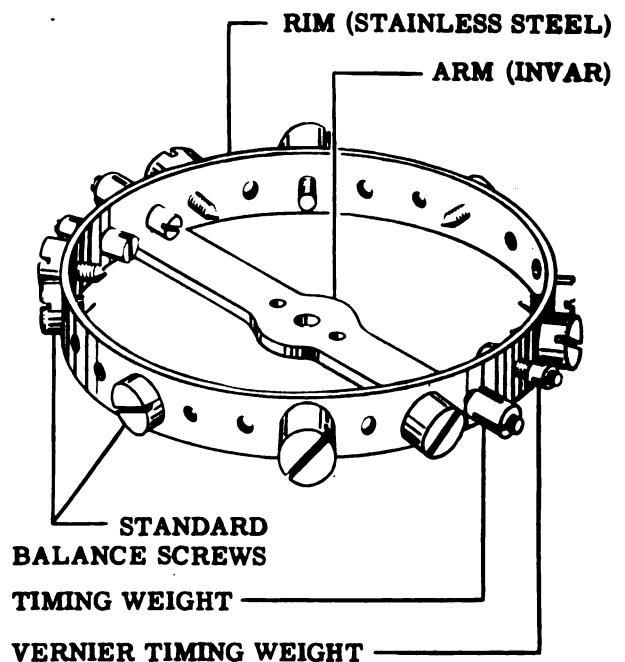


Figure 3

escapement. See Fig. 110 in Section XI for detent construction details. Operation is explained in Section IV.

Balance and Hairspring Assembly

The balance and hairspring assembly, shown in Fig. 2, is supported between the balance cock and balance lower bridge. This assembly consists of balance wheel, hub and staff, hairspring with stud and collet, impulse roller and unlocking roller. See exploded view in Section XI, Fig. 108.

The balance wheel is novel in that it is very simple in appearance compared to most of the previous designs. See Fig. 3. It consists of a solid, uncut, stainless steel rim to which is securely silver-soldered an invar arm. The thermal expansion coefficient of invar is practically zero, which means that the diameter of the wheel rim at the ends of the spoke remains almost constant with variable temperature. Other portions of the rim move outward with increases in temperature and inward with decreases in temperature. This permits temperature adjustments by moving screws either toward or away from the arm of the wheel, as required to take advantage of greater or less temperature movement. Uniformly spaced tapped holes are provided around the entire circumference of the balance rim for this purpose. With this arrangement, adjustment of compensation to errors of less than 0.02 seconds per day per degree F can be accomplished. Since this solid balance is free from centrifugal effects apparent in split wheels, motions of 1-3/8 to 1-1/2 turns are possible. Also, for the same reason, the wheel could be made larger and 35 percent lighter, without sacrificing moment of inertia. This condition reduces pivot load and friction, and improves rate.

With the balance wheel is used a variety of balance screws to allow for proper regulation. Balance screws are available in six weights ranging from 100 mgs. to 300 mgs. See Parts Catalog, page 82. Timing washers are available in a range of six weights from 4 mgs. to 20 mgs. In addition, for fine adjustment, there are two timing weights and screws, and two vernier timing weights and screws. See Section IX "Test and Adjustment."

When Hamilton first produced chronometers, the balance wheel was assembled to the hub and staff in the conventional manner. This was found to introduce lateral stresses between the staff and the balance arm. This design was then replaced with that shown in Fig. 4. By this means the lateral forces were isolated from the balance wheel.

Unlike a watch movement, the hairspring is cylindrical in shape. See Fig. 5. It is manufactured of Hamilton Elinvar, which is non-corrodible. Design of spring and terminals, as well as material and method of manufacture contribute to high and stable elasticity, good thermo-elastic qualities and minimum isochronal error. As with other chronometers, there is no regulator. Regulation is accomplished by proper use of the weights in the form of screws and washers attached to the balance wheel, as mentioned in the foregoing paragraphs.

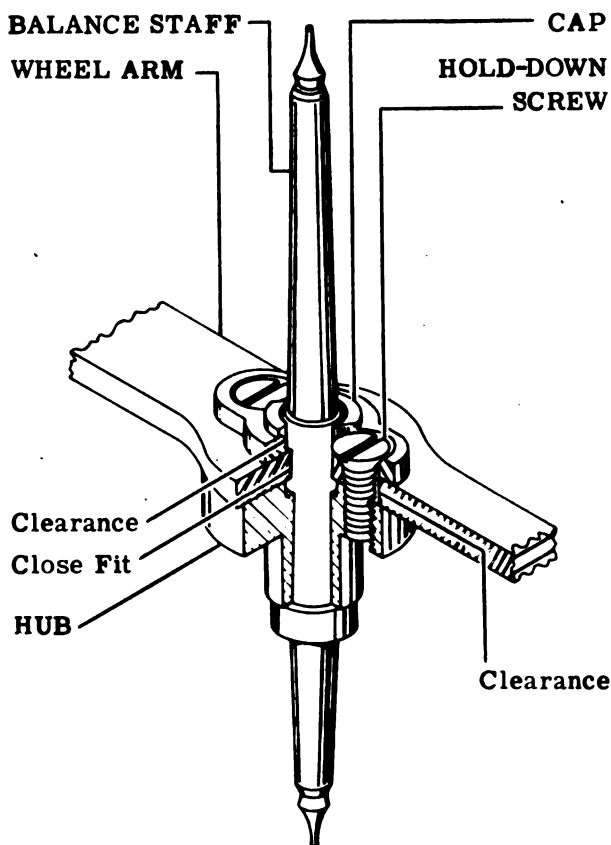


Figure 4

The hairspring stud and collet, employed to couple the hairspring to the balance wheel assembly and to the balance cock, are entirely different in design from those used in watch movements and provide an improved method for colletting and studding. The curious shape of the collet is the result of counterpoising experiments designed to eliminate slight position errors. The stud is of interest because of the unusual means of attaching it to the balance cock. Both collet and stud are shown in Fig. 5. The method of attachment of the hairspring to the collet is the same as to the stud. Fig. 6 illustrates the spring in relation to collet, collet clamp and wedge pin. It will be noted that no deformation of the spring occurs

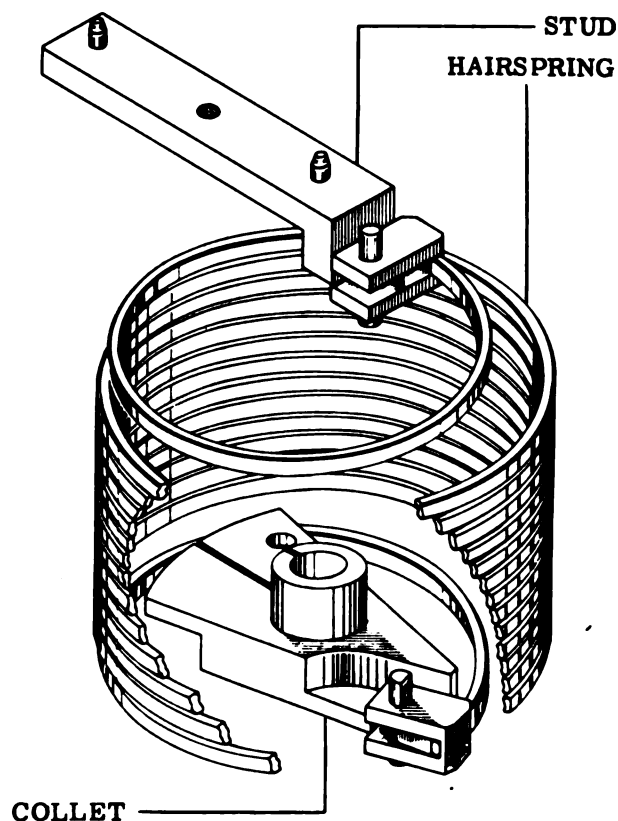


Figure 5

in the pinning operation. The active length of the spring is the same in both winding and unwinding directions, which contributes to reduction of isochronal error. The collet is slotted to facilitate assembly to the balance staff. See Figs. 5 and 6.

Instead of a watch-type roller table and safety roller mounted on the balance staff, under the balance wheel there is an impulse roller and an unlocking roller, the angle between which is adjustable. Chronometers put into the field before 1944 may still have a one-piece roller which combines the impulse and unlocking rollers and permits no change in angular relationship. At the time of overhaul, these one-piece rollers, if in need of exchange, should be replaced by two-piece rollers. See illustration of both in Section XI, Fig. 108.

Train Blocking Screw

The train blocking screw, shown in exploded view Fig. 110, Section XI, provides a simple means for blocking the gear train and retaining the motive power in the mainspring when the balance and escapement are removed for servicing or overhaul. All of the later chronometers are equipped with this device which eliminates need for a wire loop to serve the same purpose.

The train blocking screw is mounted in the balance lower bridge and is accessible through a hole in the upper train bridge. In order to block the train, the screw is turned until its head stops against a counterbored seat in the balance lower bridge, and the dog point on the screw extends between the spokes of the fourth wheel. When the screw is raised, the chamfer on the head seats against the countersunk edge of the access hole in the train bridge. The train blocking screw thus remains a permanent part of the movement and is useful in service work. See details of this construction in Section XI, Fig. 110.

Jeweling

There are fourteen jewels in the chronometer: One olive-hole jewel and one cap jewel at each end of the balance staff, one olive-hole jewel and one cap jewel at each end of the escape wheel staff, one impulse and one unlocking jewel mounted on rollers on the balance staff, one locking jewel on the detent escapement, one bar-hole jewel at each end of the fourth wheel, and one bar-hole jewel for the dial end of the third wheel staff. These jewels are all shown in the exploded views of Section XI, Figs. 108 and 110.

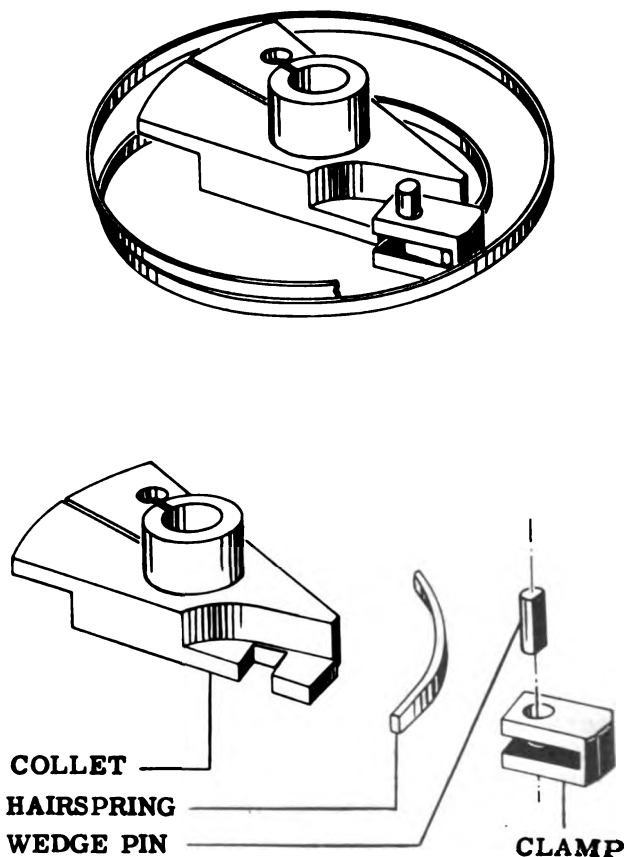


Figure 6

SECTION III

PREPARATION FOR USE

CAUTION

1. Do not move a chronometer when the mainspring is run down and the balance wheel is in a free-swinging condition without adhering to the instructions given in the following paragraphs under "Care of a Run-Down Chronometer."
2. Never touch or interfere with the second hand. Never move the hour and minute hands in a counterclockwise direction.
3. Unless absolutely necessary, do not take the chronometer movement from its case.

CARE OF A RUN-DOWN CHRONOMETER

A chronometer in service should never be permitted to run down, for in this state it is in a condition highly susceptible to damage if not properly handled. Personnel not fully experienced in the care and repair of chronometers are warned to be especially cautious in observing the following routine in the order noted, when restarting a run-down instrument.

Do not move the chronometer from its existing position which, presumably, is in its mounting box. Slowly invert the instrument and proceed with winding as described subsequently under "Winding." Note that when the chronometer is completely run down, 17-1/2 half turns will be required for a full winding. After winding, gently return the chronometer to its normal position.

To start the chronometer, grasp the mounting box with both hands, letting the box remain on its table or other support, and give the whole unit a single quick twist to the right or the left. When the instrument is out of its mounting box, the operation is the same except that the twist is given by one hand placed on the bezel. If the chronometer is started in this manner when the second hand of a master time source and that of the chronometer are in agreement, the chronometer can be set very closely to the exact time. It will, of course, be necessary to set the hour and minute hands as described in following paragraphs under "Setting."

WINDING

While the Hamilton make of chronometer is designed to run a maximum of 56 hours on each full winding, these instruments should preferably be wound at the same time every day. The following routine in the daily winding of the chronometers is recommended:

Gently turn the chronometer face down; hold it steady in this position. Open the key-hole by turning the shield plate on the bottom of the case to the right (clockwise) until the hole in the shield plate is aligned with the hole in the case bottom. Then, insert the winding key and gently push it down. See Fig. 7.

Wind to the left (counterclockwise). Seven half turns are required if the chronometer has been running 24 hours since the last winding. On the last half turn slow the winding motion

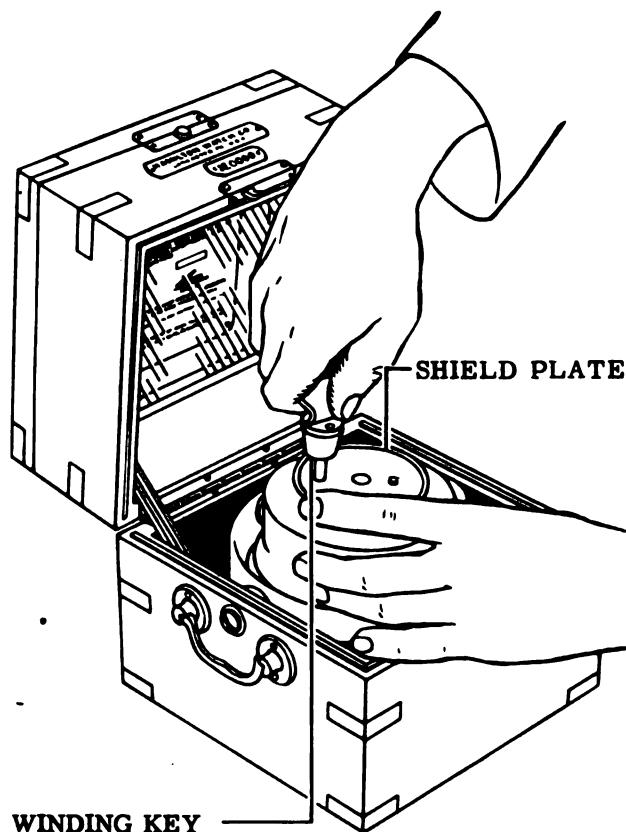


Figure 7

so that contact with the winding stop control is made gently. Care should be exercised that, whenever winding is performed, actual contact be made with the winding stop.

Remove the key from the movement and return the shield plate to its natural position. Gently place the chronometer in its operating position, face up in its gimbals.

Check the wind indicator on the dial; the winding should have moved the hand back (counterclockwise) from about "24" to "0".

SETTING

When used as a marine navigational instrument, the hands of a chronometer are never set except when the instrument is started. As more fully explained in Section IX under "Recording the Rate," a record of the chronometer rate and the accumulated error is maintained and applied to the dial reading at the time of observation. Hence, for navigational purposes, setting of the hands is unnecessary. If resetting of the chronometer becomes necessary or desirable under other circumstances, there are two possible alternate methods which may be employed.

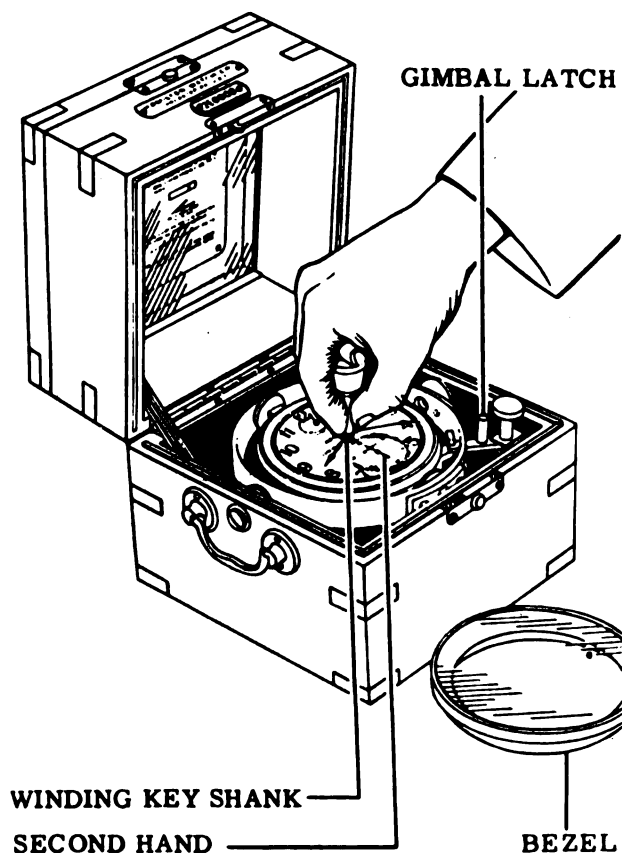


Figure 8

CAUTION

In the use of either method described in the following, these two cautions should always be observed.

- (1) Do not permit anything to touch the second hand.
- (2) Never move the hour and minute hand counterclockwise.

Setting While Running

By the first method, it is assumed that the chronometer will not be stopped to permit synchronization of the second hand with a master time source; hence, only the hour and minute hands will be set. This method permits a maximum possible error of 30 seconds. The procedure is as follows:

Latch the gimbal suspension. Unscrew the bezel containing the crystal and place it to one side. Place the winding key on the bright, square arbor at the center of the dial. As illustrated in Fig. 8, turn the key by its shank — not by the handle.

Observe the correct hour and minute from the master time source. Turn the hour and minute hands forward to a point exactly one-half minute behind the time indicated by the master time source by setting the minute hand of the chronometer on its proper marker just as the second hand of the master time source passes its "30" marker (30 seconds ahead of the chronometer minute hand).

Finally, synchronize the minute and second hands of the chronometer by advancing the minute hand to the next minute marker at the same instant the second hand passes the "60" or ("00") marker on its orbit.

Obviously, considerable inconvenience results by use of the above described method if the chronometer is fast, since the hands may be turned only in the clockwise direction. If this situation exists or if it is desirable to set the chronometer to the exact second, another method is recommended.

Setting When Stopped

Remove the chronometer movement from its case and stop the balance wheel. Familiarity with these operations can be acquired from succeeding paragraphs in this section, entitled "Operation When Received." Also see "Preparing for Shipment" in Section X.

If the chronometer is fast, let it rest until the time indicated by the chronometer hands

is "overtaken" by the correct time as determined from an accurate master time source. At this instant, start the chronometer by twisting it to the right or left. Or, it may be desirable to wait until the chronometer reading is "slow" and proceed as described in the following paragraph.

NOTE

If the escape wheel is in locked position when the chronometer is started, these methods will place the instrument in practically perfect agreement with the master time source. This is because, when the escape wheel is locked, the second hand is exactly upon the second or or half second.

If the chronometer is slow, simply restart it at that instant when the chronometer second hand and that of the master time source are in agreement. Then, using the winding key, turn the hour and minute hands forward to correct time, being certain to place the minute hand on the proper marker as the second hand passes "60" on its orbit.

REGULATING

Unlike most timekeeping instruments, the chronometer is not equipped with a conventional regulator and cannot, therefore, be regulated while it is running. Regulation of the rate can be made by adjustment of the timing weights on the balance wheel — a rather delicate operation. Any attempt at regulation must not be made by other than a fully experienced repairman as this is extremely hazardous to the chronometer. See Section IX "Test and Adjustment."

OPERATION WHEN RECEIVED

To and from a ship, a chronometer is personally transported in its mounting box, but when the instrument is received in shipment, the following conditions are generally found to hold true. (1) One shipping container comprises the shipment. This contains the movement in its brass case. See Section X, Fig. 103. (2) The instrument is 48 hours down, as shown by the wind indicator hand on the dial. (3) The balance wheel is immobilized by a locking arm.

NOTE

Chronometers received after overhaul are equipped with this locking arm. See Bureau of Ships Sketch No. 023263 dated 7 November 1947, or superseding drawings.

Prior to the installation of this attachment, chronometers were shipped with two wedges located between the rim of the balance wheel and the top surface of the train bridge. Generally, these wedges were made of folded strips of red plastic material, although sometimes of tissue paper or cork.

After the chronometer is taken from its shipping container, it is necessary to wind fully, then take the movement from its case, release the balance wheel, start and set the chronometer. The detailed procedure is given in the following paragraphs.

CAUTION

On a chronometer equipped with balance wheel locking arm, it is important to make certain that the screw

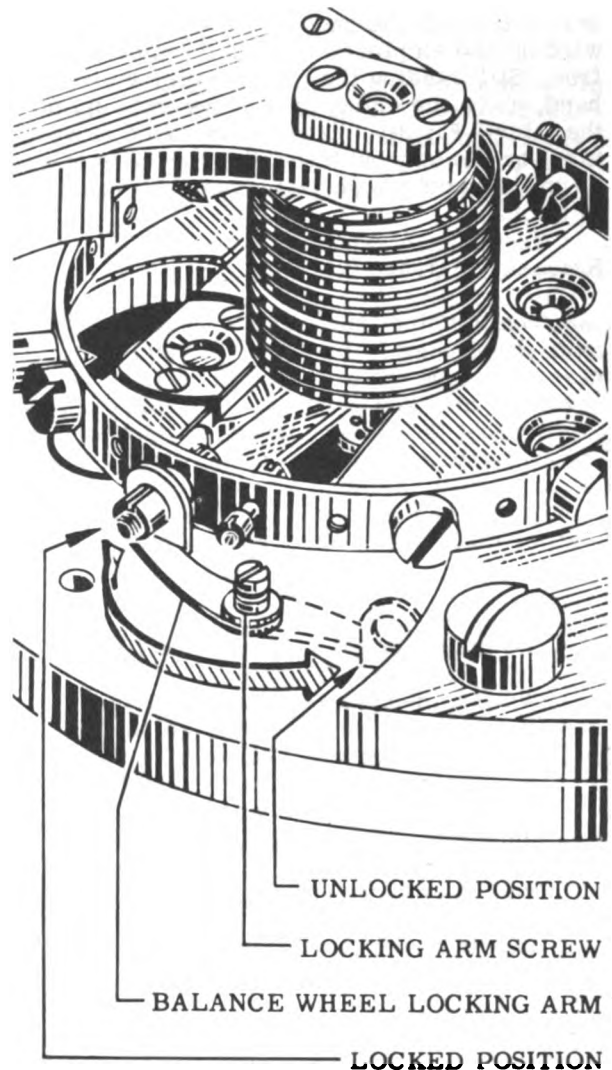


Figure 9

holding this arm is tight, with arm in either the locked or unlocked position. See Fig. 9.

Taking Movement from Case

After fully winding the instrument, remove the bezel containing the crystal from the case by turning it in the counterclockwise direction. Then, holding the chronometer above a table with one hand under the case, place fingertips of other hand around edge of the dial on the top side. Slowly invert the case, maintaining pressure upon the edge of the dial with the fingertips so that the chronometer movement is not inadvertently allowed to fall from the case and be seriously damaged.

Supporting the movement with the fingertips against the edge of the dial, raise the case with the other hand, removing it from the movement. If the movement tends to stick in the case, the winding key may be inserted through the back of the case as in winding and employed to push the movement free. Still holding the chronometer with one hand, place the empty case, open side up, on the table. Finally, place the inverted movement on the case with the dial carefully located in the shoulder recess around the top edge of the case.

Releasing Balance Wheel

Gently unlock the balance wheel locking arm, illustrated in Fig. 9, by loosening its screw, turning the arm, and tightening the screw in unlocked position. To avoid the possibility that the balance wheel might start when unlocked, it should be maintained in position by using a clean, non-metallic pointer or any dry, soft material which will injure neither the balance wheel nor its pivots.

NOTE

Do not exert pressure against the balance wheel. Former practice, when folded wedges were used under the balance wheel, called for their removal with tweezers in such a manner that the fold of the wedge would not be forced under the balance rim. See Fig. 10. The extra thickness, occurring at the fold in such a case, is sufficient to distort the wheel. The balance wheel should never be allowed to start during removal of a wedge.

Returning Movement to Case

Taking care to avoid imparting any rotary motion to the movement or the balance

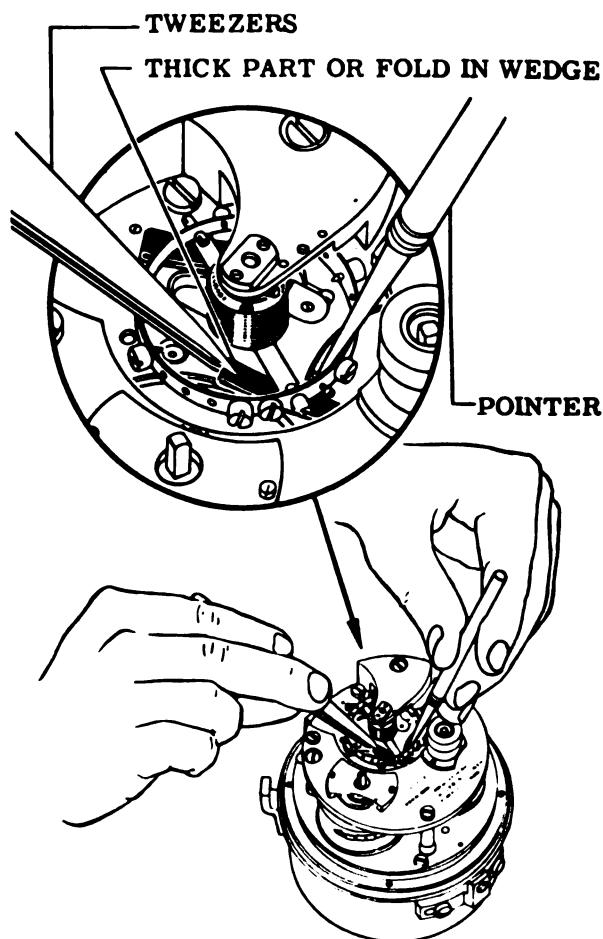


Figure 10

wheel, restore the chronometer movement to its case. Be certain that the alignment pin protruding beyond the movement enters its slot in the edge of the case.

Starting and Setting

Before the bezel is replaced, the chronometer should first be started and set. Holding the chronometer with the dial horizontal, give it a single quick twist to either the right or left. Observation of the second hand or the audible ticking of the instrument will indicate the successful performance of this operation. If the chronometer is so started when its second hand and that of a master time source are in agreement, the chronometer can be set very closely to the exact time. Then, to set the hour and minute hands, observe the instructions previously given in this section under "Setting." After this operation, assemble the bezel to the case.

Placement in Gimbal Ring

In assembling the chronometer to its gimbal ring in the mounting box, first retract the

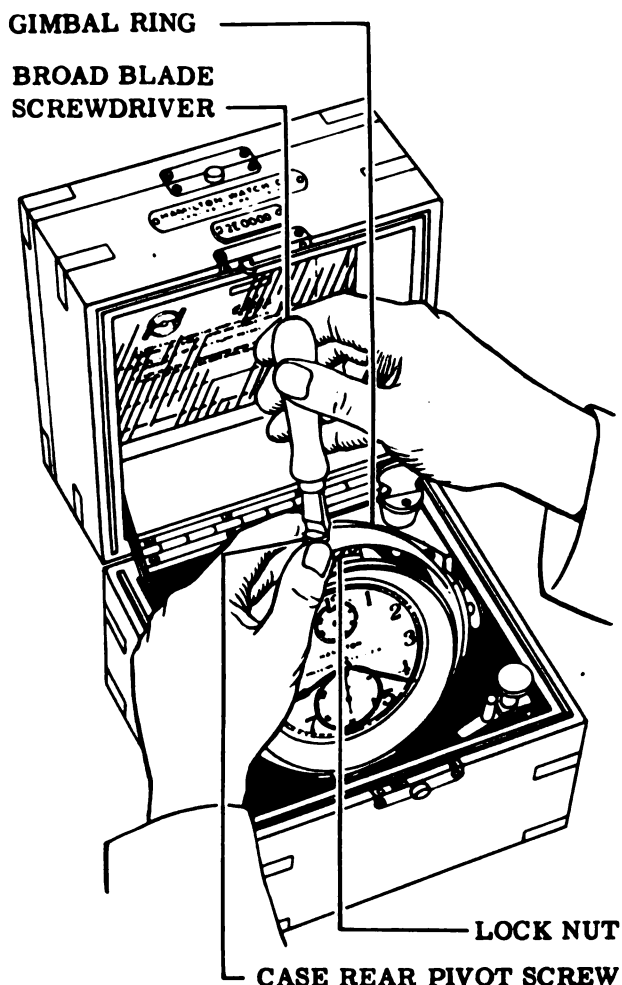


Figure 11

latch lever freeing the empty gimbal ring. Tilt the gimbal ring forward, free the case rear pivot screw in the gimbal ring by backing off the large, knurled lock nut several turns, and turn the pivot screw outward through several turns to permit entry of the chronometer case between the front and rear screws. With the gimbal ring tilted forward, carefully insert the cased chronometer into the ring, locating the end of the front pivot screw in its mating hole in the support bracket at the front of the case. Then, still holding the case with the rear pivot screw aligned with its hole in the support bracket at the rear of the case, turn the pivot screw clockwise until it enters its bushing in the rear case support

bracket. See Fig. 11. The pivot screw should be turned inward until very little clearance remains between the bushing in the bracket and the shoulder of the screw. Secure the pivot screw in position by turning the lock nut against the gimbal ring. The completed assembly should have only a small amount of shake but the cased movement should oscillate freely.

Generally, adjustments of the gimbal assembly should be confined to the case rear pivot screw, if the level of the chronometer as previously established is to be maintained.

POSITIONING

A ship chronometer should, preferably, be placed on a firm supporting base in a location where it will be subject to a minimum of jars, vibration, wide ranges of temperature, dampness and magnetic fields.

PERFORMANCE

The Ship Chronometer, Hamilton make, is a highly precise timekeeper. Its continued accuracy and reliability of performance are dependent upon the following of certain procedures which are set forth in these pages.

These chronometers have proved that they are rugged and dependable in the service of the U. S. Navy. They have survived without ill effect the greatest naval bombardments of all time. Yet, because of the inherent characteristics of this mechanism, damage may easily occur to the instrument in the hands of those unskilled in the care of chronometers. Accordingly, for the safety of the chronometer, the necessary precautions to be observed have been stressed throughout this section.

Under ordinary, as opposed to emergency, conditions of operation, these chronometers have remarkable stability of rate. No lengthy running-in period is required. No settling period between tests at different temperatures is found necessary. If stopped momentarily and restarted, one of these chronometers readily resumes its original rate. After shipment, resumption of original rate is almost exact, and is usually within tenths or hundredths of a second. Isochronal properties are excellent as are also temperature compensation qualities.

SECTION IV

PRINCIPLES OF OPERATION

The Hamilton Ship Chronometer movement, like that of any other timepiece, comprises four basic mechanisms — driving, transmitting, controlling and indicating. The principles of operation of each of these mechanisms are discussed in this section with particular reference to differences from the conventional watch movement.

The four mechanisms are held in place by various structural members — a pillar plate which supports, by means of posts, a barrel bridge and an upper train bridge. The upper train bridge in turn mounts the balance cock, the balance lower bridge, and the escape upper bridge. A lower train bridge is mounted directly to the pillar plate.

DRIVING MECHANISM

The driving mechanism consists of a long and powerful mainspring coiled inside its barrel, which is connected by means of a flexible chain to the fusee, a device designed to equalize mainspring torque. See Fig. 12. Instead of the mainspring barrel driving the center wheel, as in a conventional watch, the fusee assembly mounts the main driving wheel which transmits motion to the center wheel. In addition to equalizing mainspring torque and transferring power to the train, the fusee assembly provides sustaining or maintaining power to keep the chronometer running while the mainspring is being wound. Also, it is equipped with a winding stop-bar to prevent overwinding the mainspring.

Fig. 12 shows schematically the operation of the fusee and mainspring barrel during the running and winding cycles. For further details of the parts themselves, see the exploded view in Section XI, Fig. 109.

Running Cycle

As shown in Fig. 12, Running Cycle, the mainspring, through its barrel and the fusee chain, drives the fusee which, in turn, drives the fusee wheel (main driving wheel). During the running cycle, the barrel turns in a clockwise direction; this motion is transmitted through the fusee chain to the fusee. The fusee, rotating with its arbor, transmits the

clockwise motion to the winding ratchet wheel, the teeth of which engage the pawls of the sustaining ratchet wheel assembly. Underneath the sustaining ratchet wheel assembly (which is free on the fusee arbor) is the sustaining spring, one end of which is pinned (one pin) to the sustaining ratchet wheel assembly and the other end of which is pinned (one pin) to the top of the fusee wheel (driving wheel) which is also free on the fusee arbor. The sustaining ratchet wheel assembly transmits the clockwise rotation from the winding ratchet wheel, through the sustaining spring, under load during the entire running cycle, to the fusee wheel. The fusee wheel, in turn, transmits the torque (in a clockwise direction) to the train. A pinion on the lower end of the fusee arbor drives the wind indicator wheel mounted on the dial side of the pillar plate. The sustaining pawl serves no useful purpose in the running cycle.

Winding Cycle

The mainspring is wound by turning the fusee arbor in a counterclockwise direction. This would be expected to interrupt the transmission of power to the chronometer train, thus tending to stop the instrument. However, this effect is prevented and the chronometer is kept running, while the mainspring is being wound, by supplying sustaining or maintaining power to the train through the sustaining spring fastened to the fusee wheel.

When winding the fusee arbor with a winding key in a counterclockwise direction, the teeth of the winding ratchet wheel do not engage the pawls of the sustaining ratchet wheel assembly, but turn freely in a counterclockwise direction. See Fig. 12, Winding Cycle. This action, being directly opposite in direction to the pull of the mainspring, interrupts the flow of power to the fusee wheel. As shown in Fig. 12, Running Cycle, when power is being transmitted by the mainspring barrel, the sustaining spring serves (under load) as the connecting link between the sustaining ratchet wheel assembly and the fusee wheel. Now that power is no longer being applied by the winding ratchet wheel through the sustaining ratchet wheel assembly to the fusee wheel, the sustaining spring is free to expand.

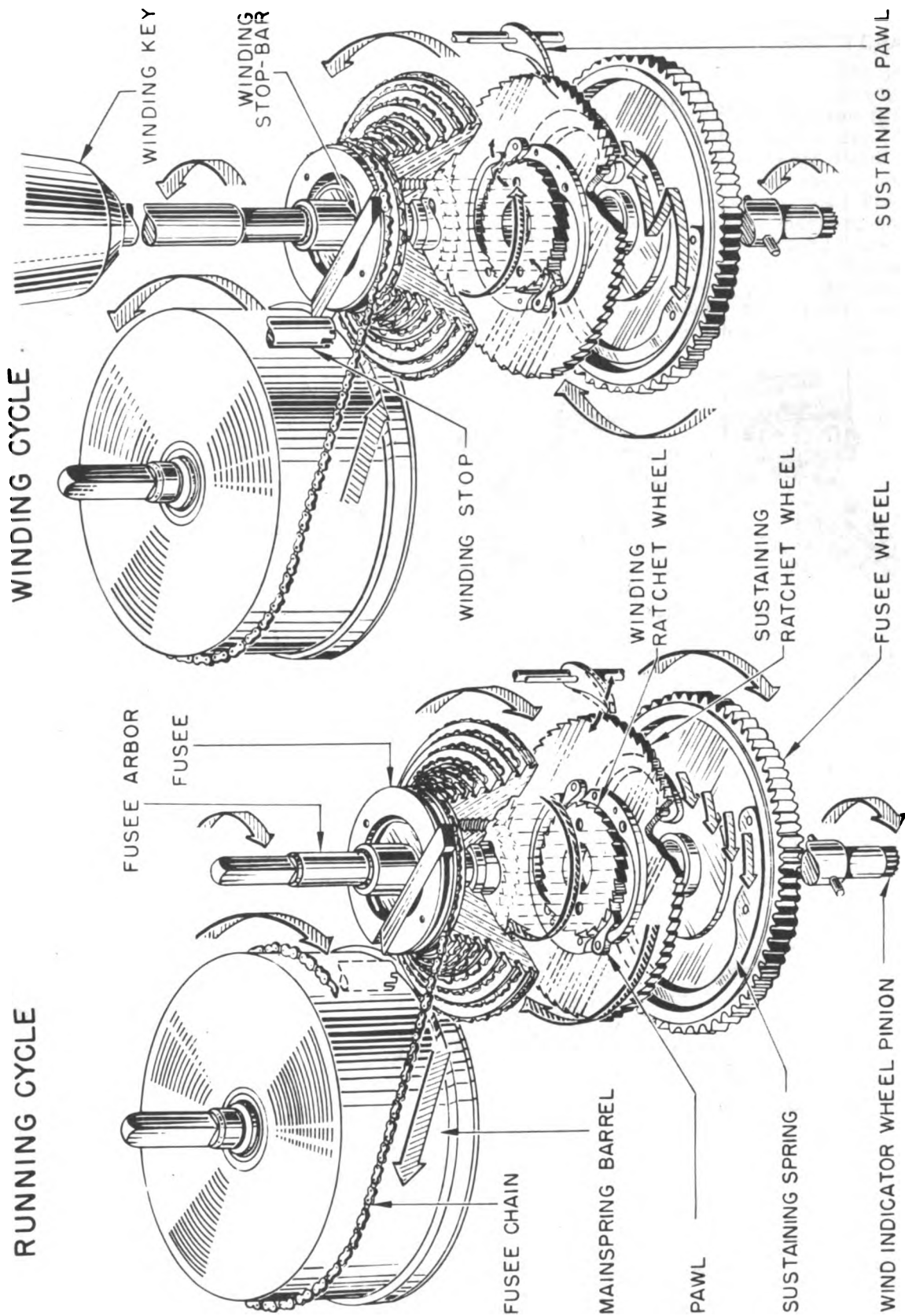


Figure 12

HAMILTON SHIP CHRONOMETER

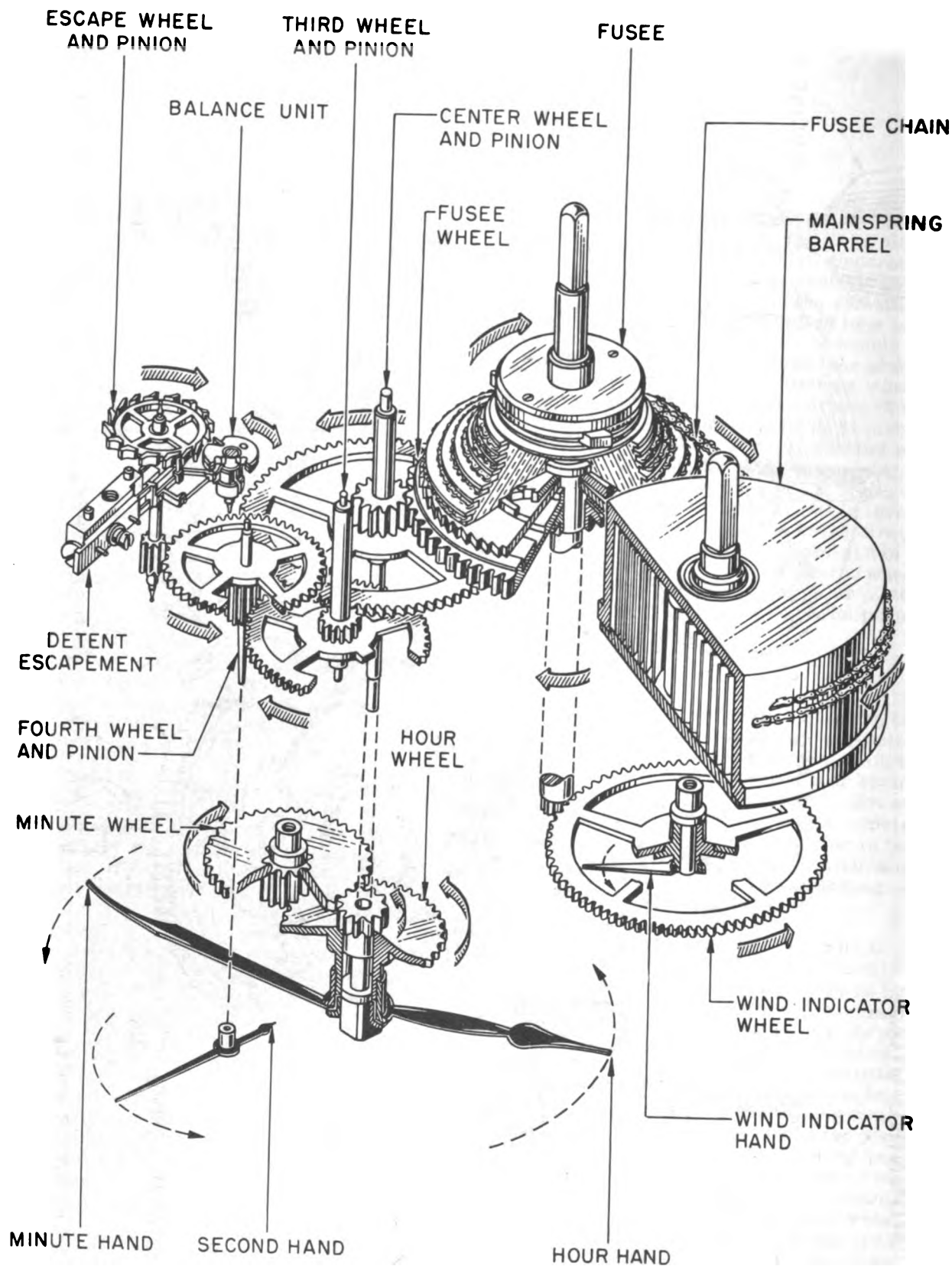


Figure 13

To follow this operation, it is necessary to consider the effect of the spring-activated sustaining pawl, which is provided to engage the teeth of the sustaining ratchet wheel. Due to the sustaining spring, which is always under load, the sustaining ratchet wheel and its pawls would follow the winding ratchet wheel in a counterclockwise direction were it not for the sustaining pawl. However, because of the action of the sustaining pawl, the power of the sustaining spring can be released in one direction only — clockwise through the fusee wheel. The sustaining spring is designed to supply enough power to drive the chronometer for five to ten minutes.

To prevent overwinding of the mainspring, a winding stop-bar is provided at the top of the fusee. When the mainspring is close to full wind, the fusee chain assembly, in being wrapped around the fusee, bears against one end of the spring-activated winding stop-bar, the opposite end of which moves out to engage the winding stop on the bottom of the barrel bridge — stopping any further turning of the fusee and, hence, any further winding of the mainspring.

TRANSMITTING MECHANISM

The chronometer train, as shown in Fig. 13, is closely analogous to a conventional watch type train. The mainspring, through the fusee wheel, drives the center wheel pinion which, through the center wheel, drives the third wheel pinion. The third wheel drives the fourth wheel pinion, and the fourth wheel meshes with the escape pinion, thus driving the escape wheel. The escape wheel, and hence, the flow of power, is regulated by the balance unit through the medium of the detent escapement as described in following paragraphs.

CONTROLLING MECHANISM

The controlling mechanism of the chronometer is the balance unit which governs the escape wheel through the detent.

As described in Section II "Description," the detent escapement (sometimes called the "chronometer escapement") differs greatly from the detached lever escapement, such as found in modern watches. No attempt will be made here to describe all the various design features of the Hamilton make of chronometer escapement. However, a series of operational schematics is provided to show the overall operation of the escapement and familiarize the reader with the functions of the main components.

General Arrangement

Fig. 14 shows the principal parts of the detent escapement. The detent (42087)* is mounted to the detent support block (42086) by means of the detent clamp screw (37024), a washer (42251) and two steady pins. The detent support block mounts the detent adjusting screw, the locking jewel button, the lock-adjusting screw and clamp screw. The detent (42087) comprises the detent spring, abutment arm (horn) and locking jewel; and mounts the detent trip spring (42088) sometimes called the "gold" spring, by means of an angle bracket (42092). The impulse roller and jewel (42263) and the unlocking roller and jewel (42252) are mounted on the balance staff (42186). The escape wheel (42076) completes the arrangement. For further details on escapement parts, see exploded view Fig. 110 in Section XI. Rollers are illustrated on exploded view Fig. 108 in Section XI.

Detent Operation — Step 1

To explain the overall operation of the detent escapement, we will assume, for illustration's sake, that the balance wheel has reached the end of its counterclockwise swing and is beginning its clockwise swing. See Step 1, Fig. 14. The unlocking jewel is approaching the detent trip spring; the impulse jewel is swinging toward teeth B and C of the escape wheel, which are stationary because escape wheel tooth A, and the escape wheel, are held in position by the locking jewel.

Detent Operation — Step 2

Continuing the clockwise swing in Step 2, Fig. 14, the unlocking jewel has reached the detent trip spring which flexes to allow the unlocking jewel to pass to the other side of the trip spring. The impulse jewel has passed from between escape wheel teeth B and C; and the escape wheel (note tooth A) is still held in its locked position by the locking jewel.

Detent Operation — Step 3

In step 3, Fig. 14, the balance wheel has passed the extremity of its clockwise swing, and has started in the counterclockwise direction. The unlocking jewel is about to engage the detent trip spring. The impulse jewel is approaching escape wheel tooth C; escape wheel (note tooth A) is still held in its locked position by the locking jewel.

*Parenthetical figures are references to Hamilton Part Numbers

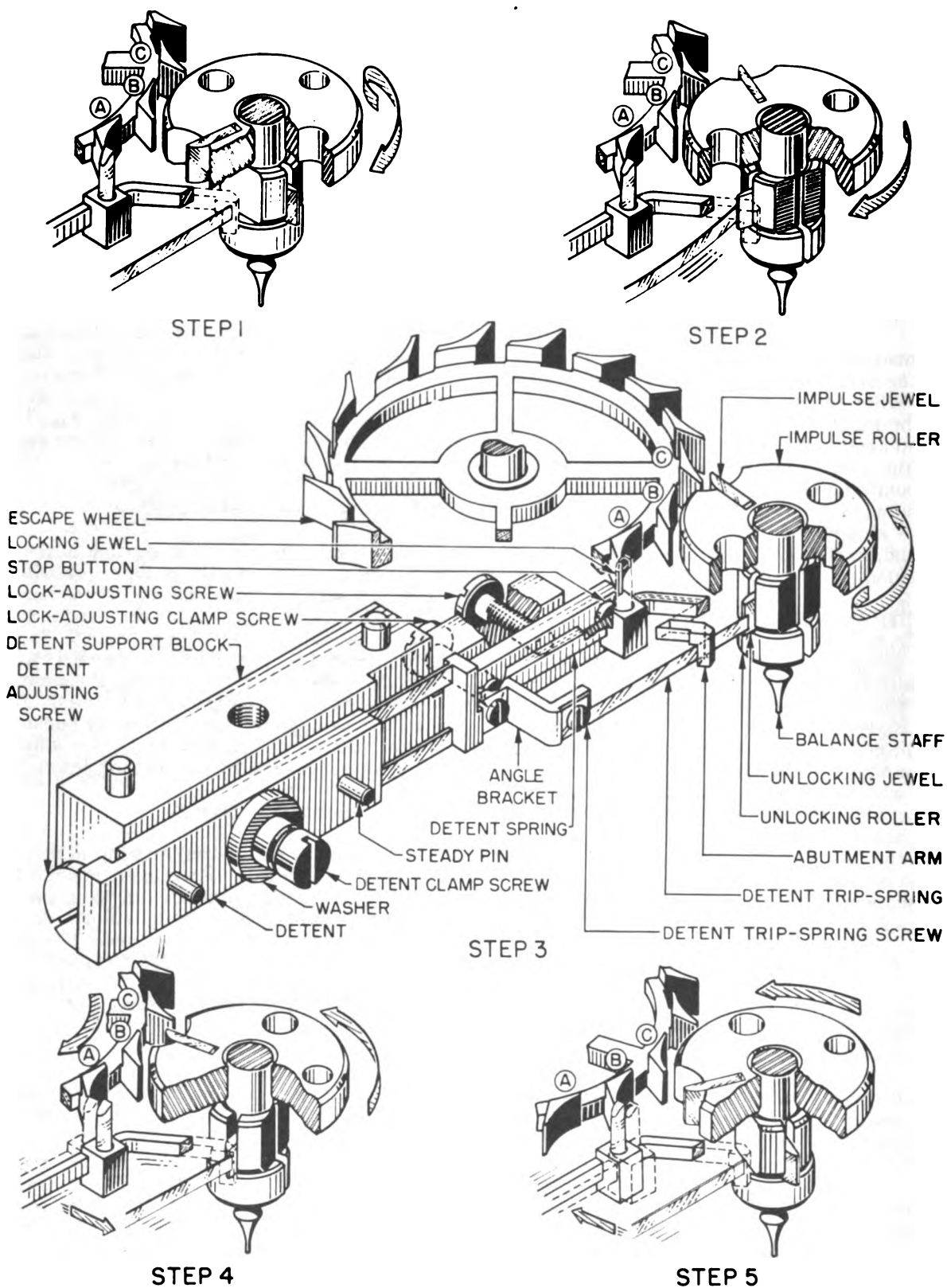


Figure 14

Detent Operation — Step 4

When the unlocking jewel strikes the detent trip spring in its counterclockwise swing, the trip spring is not free to be pushed aside as in the case of the unlocking jewel's clockwise swing, see Step 2. As shown in Step 4, Fig. 14, the detent trip spring is held by the abutment arm of the detent. When the unlocking jewel strikes the trip spring, the trip spring and the abutment arm are moved aside as a unit. Movement of the abutment arm displaces the locking jewel sufficiently to release escape wheel tooth A. Step 4 shows the locking jewel pulled aside and escape tooth A free to rotate; the unlocking jewel is riding over the tip of the detent trip spring, and mainspring power is beginning to turn the escape wheel clockwise.

As the unlocking jewel rides over and disengages the tip of the trip spring, escape wheel tooth C of the freely turning escape wheel will catch up to the impulse jewel and give it an impulse. This impulse imparts enough power through the impulse jewel to maintain motion of the balance wheel.

Detent Operation — Step 5

As shown in Step 5, Fig. 14, escape wheel tooth C has just delivered an impulse to the impulse jewel and escape wheel tooth A has rotated past the locking jewel. The detent trip spring, having been released by the unlocking jewel, allows the entire detent assembly to spring back into its former position, the locking position, in time to permit the locking

jewel to intercept tooth B of the escape wheel, thus again locking the escape wheel after an advance of one tooth.

The balance wheel and rollers will continue to move in their counterclockwise rotation and the sequence will repeat itself as described in Step 1.

Adjustment Provisions

Two adjustments are provided in the detent escapement. As shown in Fig. 14, a detent adjusting screw with clamp screw is provided to change the relationship of the detent to the detent support block. This adjustment allows correction of roller shake - see Operation 84 in Section VIII "Reassembly."

The detent escapement is also provided with a lock-adjusting screw and clamp screw. The lock-adjusting screw, working through the locking jewel button, adjusts the lock, or amount of engagement between locking jewel and escape wheel tooth. See Operation 85 in Section VIII "Reassembly."

INDICATING MECHANISM

As shown in Fig. 13, the center wheel staff mounts the hour and minute hands while the fourth wheel staff mounts the second hand. In addition to these conventional indicators, the chronometer is provided with a wind indicator hand to show the condition of wind. The wind indicator hand is actuated through its wind indicator wheel by a pinion on the dial end of the fusee arbor.

NOTE

By reference to the exploded views in Section XI "Maintenance Parts Catalog," the individual parts composing the operational schematics, Figs. 12, 13 and 14, may be completely identified and further light shed upon the driving, transmitting, controlling and indicating mechanisms of the Hamilton make of Ship Chronometer. Rugged simplicity characterizes the overall design, though the essential delicacy of certain parts and the fine tolerances required in part relationships speak highly of the engineering skills through which these instruments were mass-produced during the war.

The performance records of over 10,000 of these navigational timepieces, delivered to the Navy in three and one-half years, bear witness to the dependability and accuracy of Hamilton chronometers in meeting the precise requirements of modern warfare.

When in need of overhaul, a careful following of the procedures set forth in this book will restore them to their original mechanical perfection. In succeeding sections will be found a presentation of factory techniques developed for the purpose of conveniently achieving this objective.

SECTION V

DISASSEMBLY OF CHRONOMETER

The chronometer is removed from its gimbal mount by loosening the locknut on the case rear pivot screw in the gimbal ring. Then this screw is backed off sufficiently to permit the tilting of the instrument out of its ring and into the hand. The following instructions provide for the further disassembly of the movement for overhaul purposes.

Service tools, where required, are indicated in the text. These may also be identified in Section XIII of this book, "Service Tools."

Disassembly Inspection

Inspection for purposes of repair cannot be performed effectively at the time of disassembly. Except in the case of parts obviously damaged or out of alignment, such inspection may best be reserved until after cleaning. See Section VII "Inspection and Repair." However, to accumulate information relative to design,

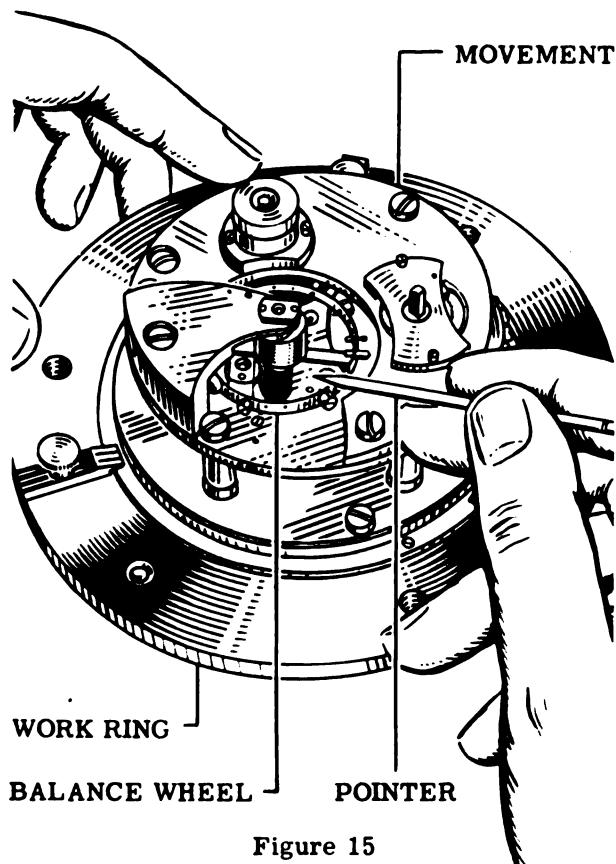


Figure 15

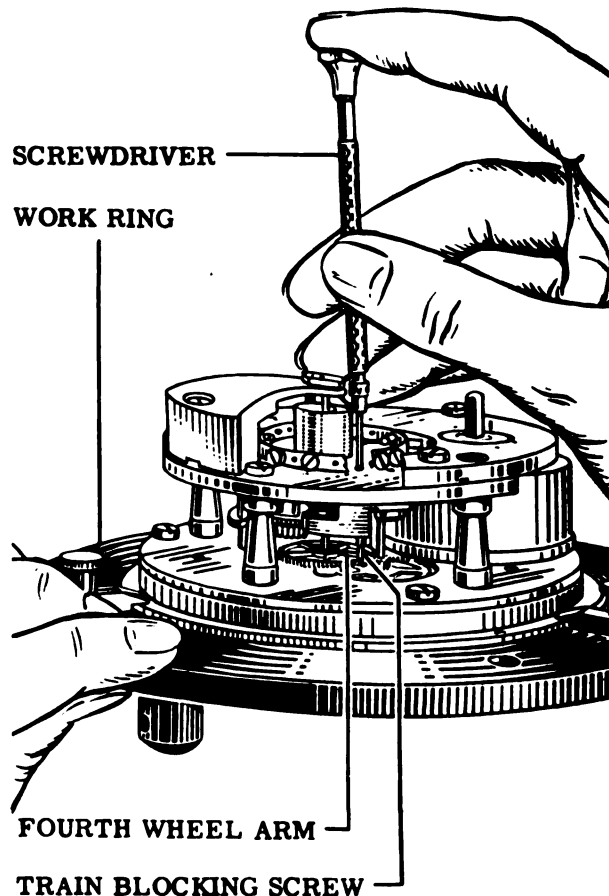


Figure 16

performance and usage, it is desirable to observe and record the condition of the chronometer during disassembly. The "Navigational Timepiece Record," NavShips 3587, which accompanies the instrument, may itself indicate that there are items which should be checked and entered into the chronometer history. Also see the section at the end of this book, "Ship Chronometer Service Bulletins."

To be of maximum utility, the disassembly inspection record must be specific, covering not only any apparent damage but also such items as the following:

- a. Fit of wheels on posts
- b. Fit of wheels in jewels
- c. Freedom of time and dial trains

- d. Functioning of barrel and fusee
- e. Functioning of balance and escapement mechanisms

Proper notations made during disassembly will be useful to the repairman during reassembly, as well as serving the larger purpose of collecting statistical data.

CAUTION

Disassembly operations should be performed only by qualified, experienced personnel skilled in the watchmaker's trade. The usual precautions in watch work are also applicable in the handling of chronometer parts.

DISASSEMBLY, MAIN MOVEMENT

1. Remove the bezel by turning in counter-clockwise direction.
2. Lower the movement out of the bowl, pushing gently with winding key if necessary. See description of this operation under "Taking Movement from Case," page 10.

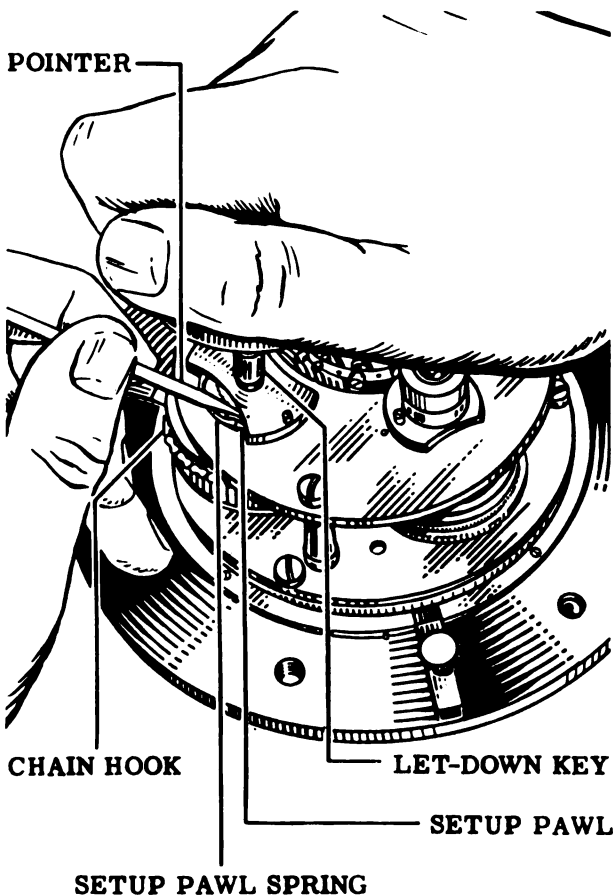


Figure 17

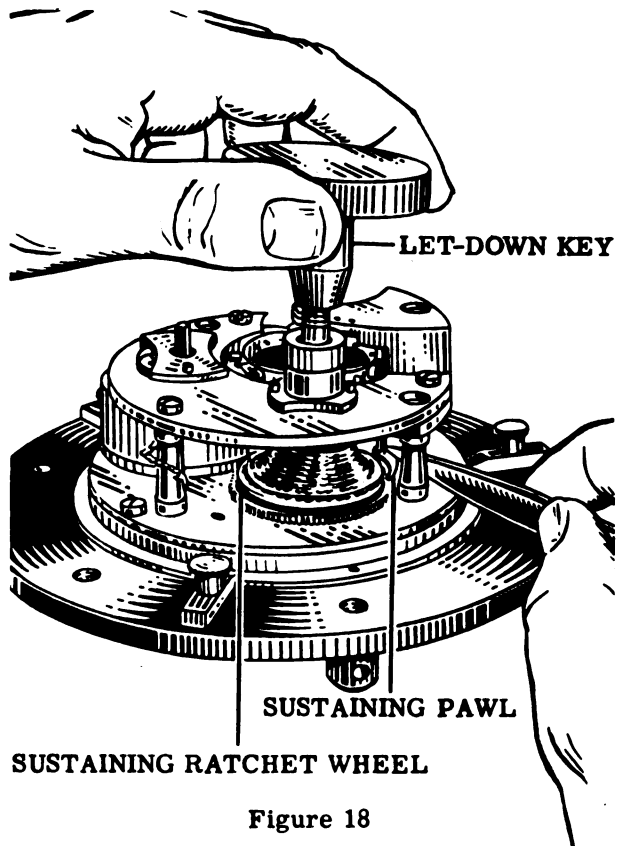


Figure 18

3. Remove legs from work ring and, keeping the movement train side up, install movement on work ring. Use Tool No. 98.

NOTE

When working on the train side, the three long legs of Tool No. 98 can be removed. If working on the dial side, replace the legs.

4. If running, stop the balance. See Fig. 15. Use only non-metallic pointer or dry, soft material. See Tool No. 87.
5. Carefully lower the train blocking screw, using Tool No. 89, making sure that this screw does not bear upon the top of one of the arms of the fourth wheel. See Fig. 16. If early model chronometer, without train blocking screw, place wire loop around rim of fourth wheel and nearest pillar. The use of this wire loop is plainly shown in Section IX. See Fig. 100.

CAUTION

Use care in performing the next operation that the balance does not increase its swing to the extent that there is damage to escape and balance pivots.

6. Oscillate the balance with Tool No. 87, or with pegwood or brush to allow the escapement to function until the train is stopped by the blocking screw bearing against one of the arms of the fourth wheel (or by the wire loop). This operation, in illustration, appears similar to Operation 4. See Fig. 15.
7. Using winding key (Tool No. 47) wind fusee chain fully onto the fusee.

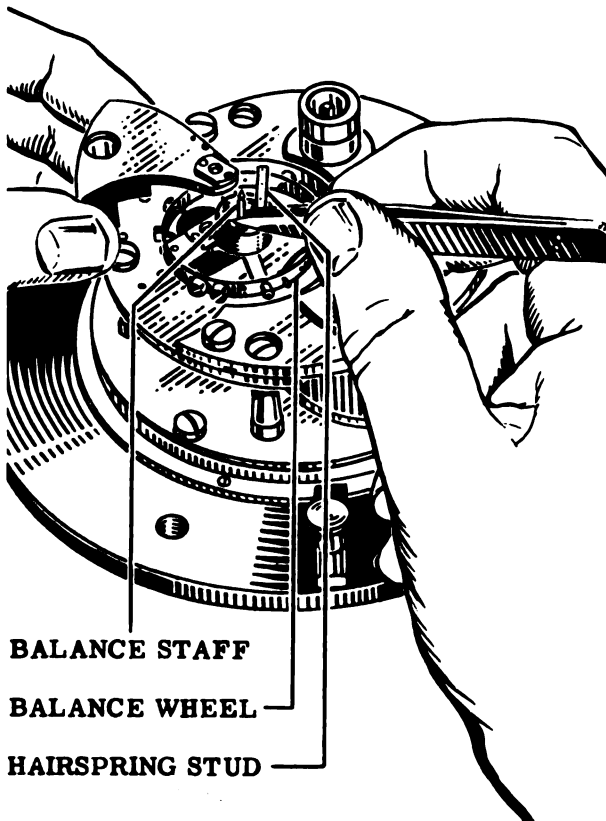
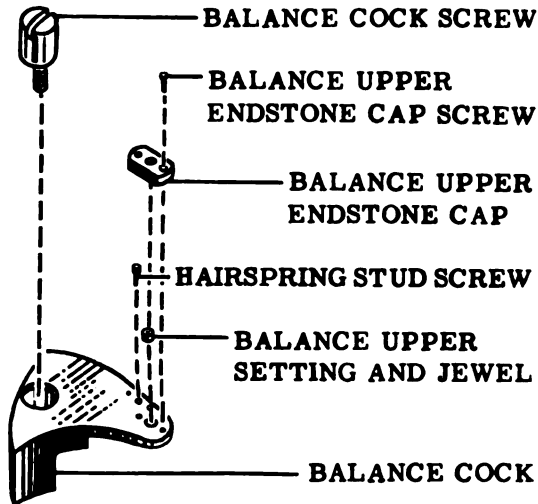


Figure 19

CAUTION

Do not use winding key for Operation 8.

8. Using let-down key (Tool No. 18) on the barrel arbor, disarm mainspring by allowing the key to rotate slowly in the fingers while the setup pawl is held away from the setup ratchet wheel. See Fig. 17. Unhook chain from barrel.
9. Using let-down key (Tool No. 18) on fusee arbor, turn slightly clockwise to release the sustaining pawl from the sustaining ratchet wheel. Hold the sustaining pawl away from the sustaining ratchet wheel and allow the key to rotate counterclockwise until the sustaining power is dissipated. See Fig. 18.
10. With gentle downward pressure on the screwdriver (Tool No. 89), unscrew the hairspring stud screw sufficiently to release the stud. See relative position of parts in Fig. 19.
11. Remove the balance cock screw, using large screwdriver (Tool No. 95). See the explosion in Fig. 19.
12. Remove the balance cock, holding the balance staff in an upright position. See Fig. 19, bottom portion.

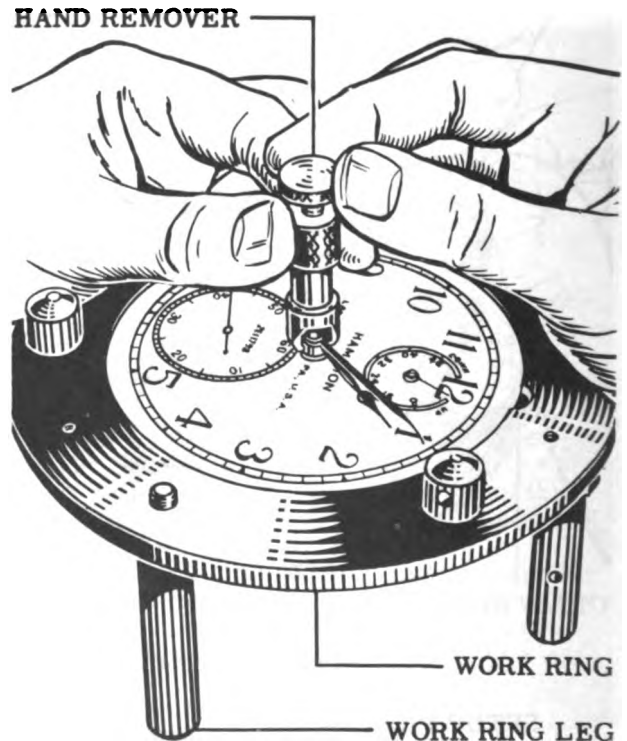


Figure 20

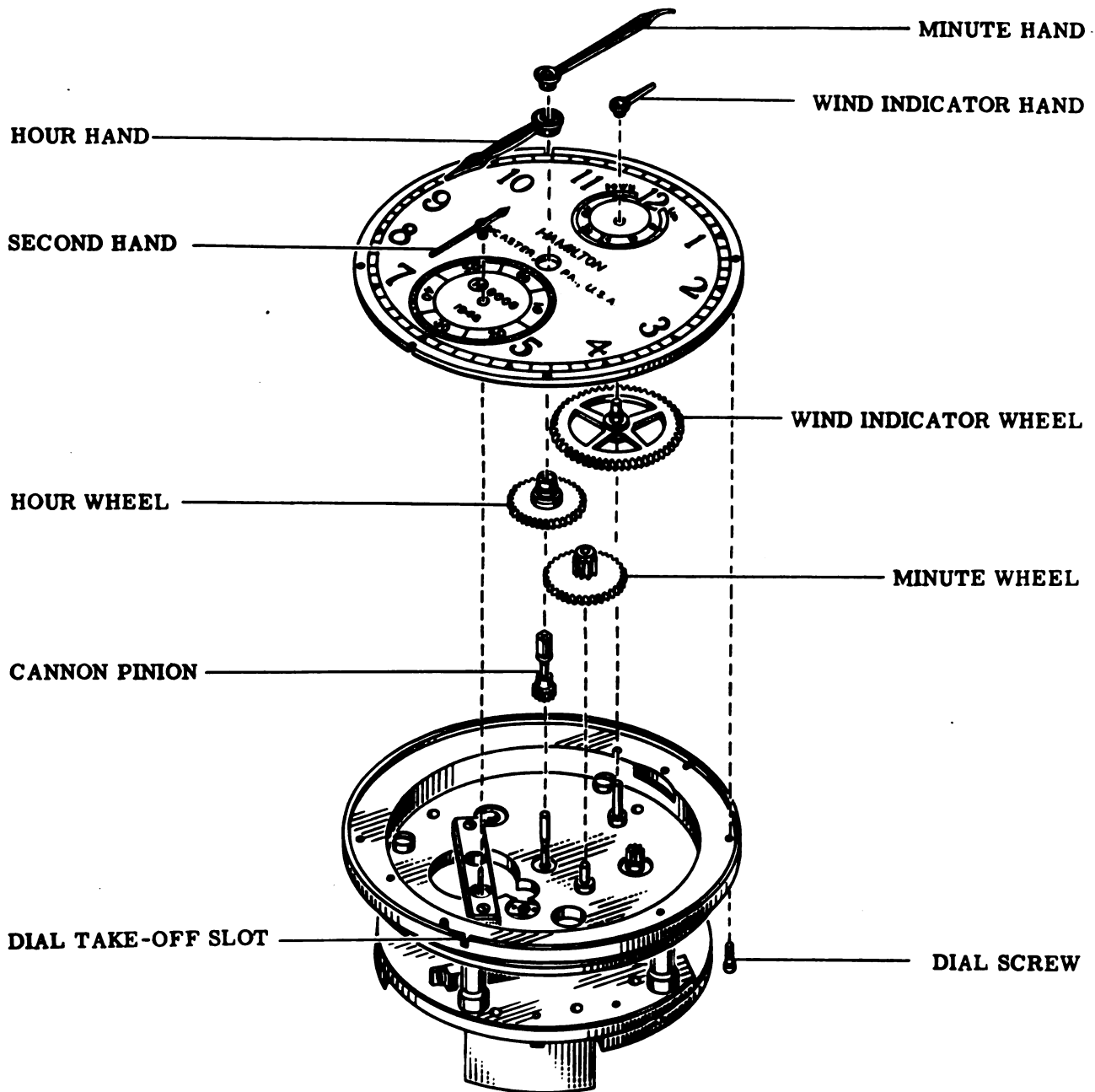


Figure 21

13. Lift the balance unit, moving slightly in the direction away from the detent. Place balance on work block. Use Tool No. 52.

NOTE

For cleaning purposes, the balance unit is not disassembled unless the balance pivots are in poor condition or there is some other reason for doing so. For balance disassembly, see Section VII, "Inspection and Repair."

14. Remove balance upper endstone cap screws, using Tool No. 90. See Fig. 19. Place cock, cap and screws in tray.

NOTE

Endstone caps are to be placed in the same tray compartments as their respective bridges. See Section VI "Cleaning."

15. Remove hairspring stud screw (see Fig. 19) and place in the screw compartment

of the cleaning tray. Refer to Section VI "Cleaning." See Fig. 33.

16. Turn movement over (after replacing work ring legs).
17. Remove hour hand and minute hand with Tools Nos. 19 and 20. See Fig. 20.
18. Remove second hand and wind indicator hand (see Fig. 21 for position of parts), using hand removers (Tools Nos. 79 and 80) with shield between tools and dial. The shield is known as Tool No. 49.
19. Turn movement, grasping the dial side with the left hand, using tissue paper between the palm of the hand and the dial. Remove four dial screws, using Tool No. 91. See Fig. 21 for position of parts.

NOTE

Place these dial screws in screw compartment of the cleaning tray, where all small screws from succeeding operations are placed. See Section VI "Cleaning."

20. Turn movement over. Place on bench, dial side up. Remove dial, using dial take-off slot. Refer to Fig. 21.

ESCAPE UPPER SETTING AND JEWEL

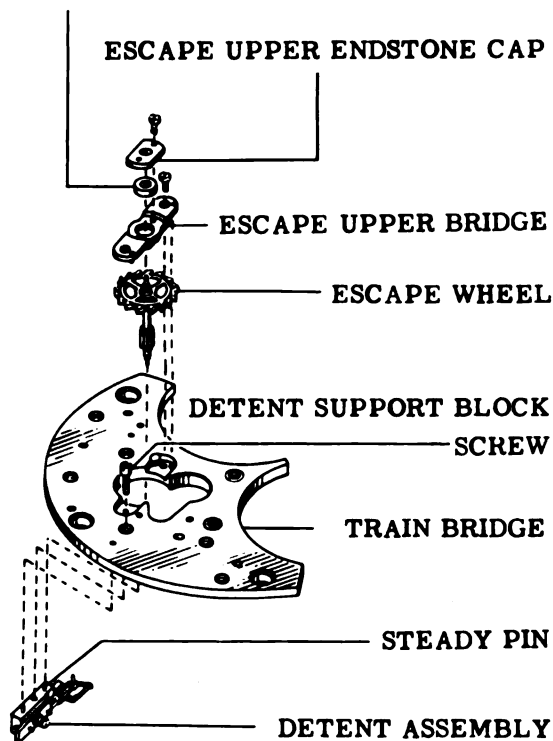


Figure 22

BALANCE LOWER BRIDGE

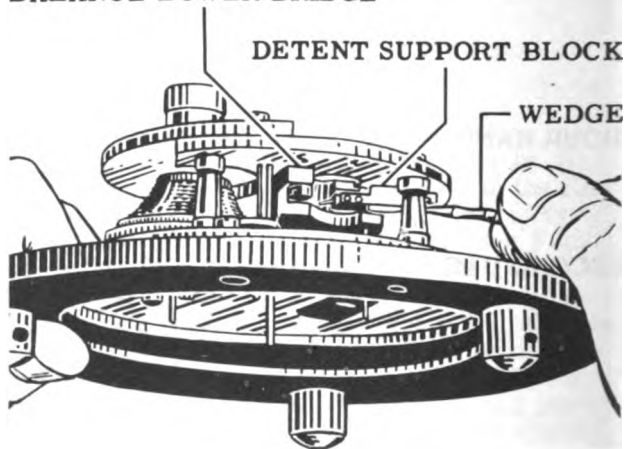


Figure 23

21. Remove wind indicator wheel, hour wheel and minute wheel. Remove cannon pinion, using Tool No. 21. See Fig. 21.
22. Turn movement over (removing legs on work ring). Remove two screws of escape upper bridge and place screws in tray. Lift out escape upper bridge. See the explosion in Fig. 22.
23. Remove escape wheel and place with the balance unit. See Fig. 22.
24. Remove two endstone cap screws from escape upper bridge. Remove endstone cap from bridge. See Fig. 22. If not already done, mark the numeral "1" in the milled recess of the escape upper endstone cap. Place bridge, endstone cap and cap screws in the cleaning tray with bridge screws.
25. Place wedge (Tool No. 11) between balance lower bridge and detent support block. See Fig. 23. Remove detent support block screw and place screw in tray.
26. Carefully remove wedge, and remove detent assembly (see Fig. 22) by straddling detent assembly with the tweezers. If detent is tight, remove by pushing nickel wire against steady pins. Place detent assembly with escape wheel and balance.
27. Remove two setup coverplate screws. Remove plate. See Fig. 24.
28. Remove setup pawl and setup ratchet wheel. See Fig. 24.
29. Remove two dust-seal screws. Remove dust seal and from it take off the packing ring unit. See Fig. 24.

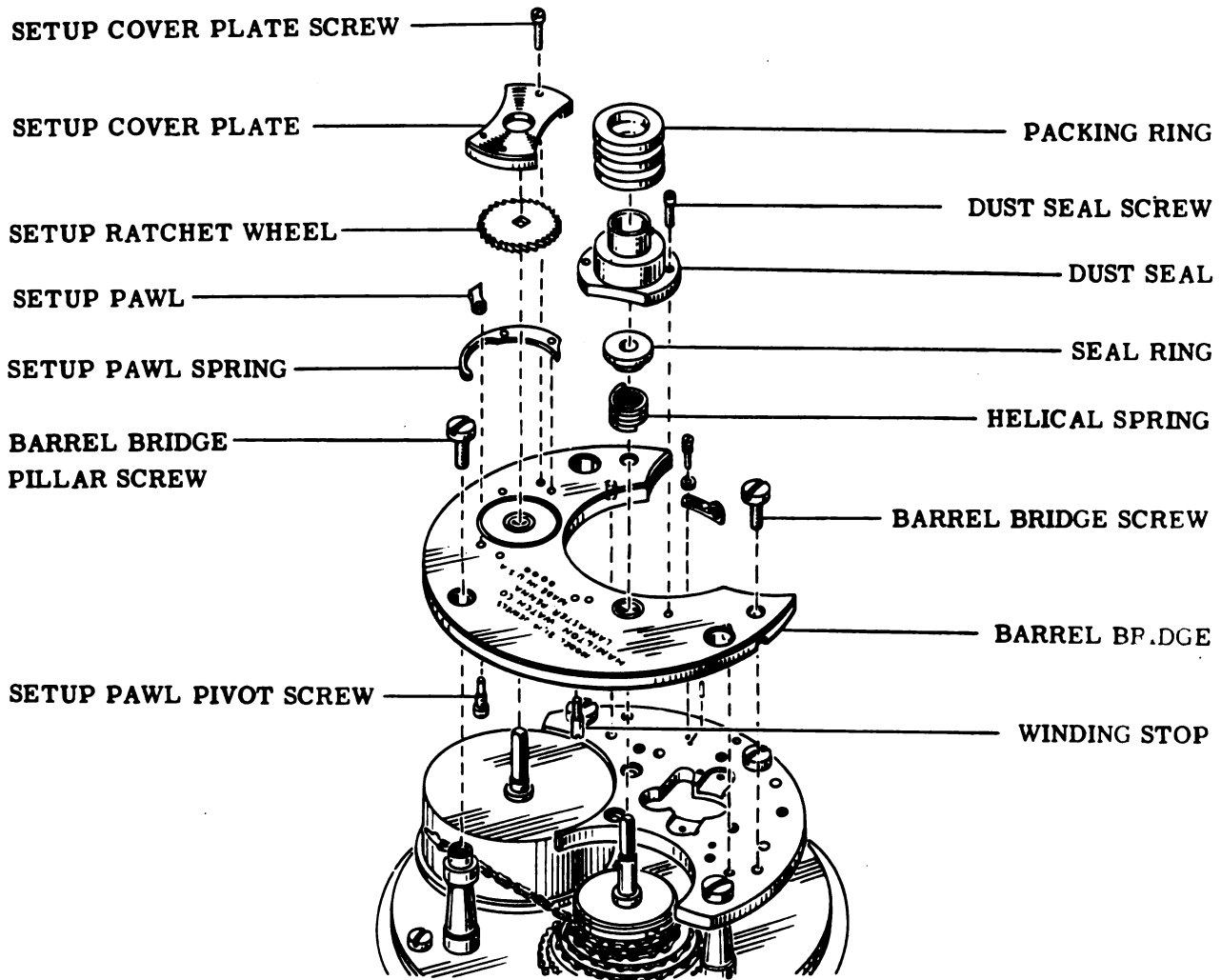


Figure 24

CAUTION

The packing rings should be replaced whenever the chronometer is overhauled. Never allow the packing rings to contact cleaning solution.

NOTE

The following Operations, Nos. 34 through 37, are customarily performed after the last disassembly step. For clarity, they are placed here immediately after the barrel is lifted from the movement.

30. Remove seal ring and its helical seal ring spring. See Fig. 24.
31. Using special shell screwdriver (Tool No. 96), remove three screws fastening the barrel bridge. See Fig. 24. Remove barrel bridge. See Fig. 25.
32. Push off setup pawl spring, shown in Fig. 24, by pressure on steady pins from the lower side of barrel bridge. Remove setup pawl pivot screw. Unscrew winding stop (left hand thread). See Fig. 24.
33. Lift out mainspring barrel, indicated in Fig. 25 on the next page.

34. Remove barrel cap screws. See Fig. 26.
35. Remove barrel cap, tapping the end of the barrel arbor, if necessary. See Fig. 26.
36. Remove barrel arbor. See Fig. 26.

CAUTION

In performing Operation 37, there is always danger of physical injury unless the spring is tightly held. Use ring to contain spring as it is removed from the barrel. Refer to Section VIII, Fig. 76.

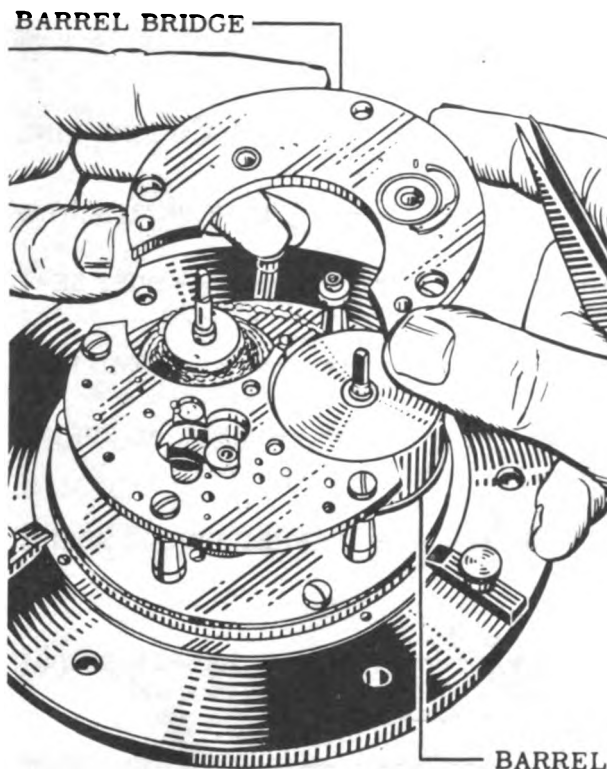


Figure 25

37. Remove mainspring (see Fig. 26), using the mainspring winder and restraining ring. For convenience and safety, use Tools Nos. 56 and 57.
38. Release sustaining pawl and remove fusee. See Fig. 27.

NOTE

The following Operations, Nos. 39 through 45, are customarily performed after Operation 58. For clarity, they are set forth here, immediately after the fusee is lifted from the movement.

39. Remove chain from fusee and hang over rod to prevent kinking. Fig. 28 shows how the fusee chain is fastened to the fusee by means of a pin. For cleaning chain, see Section VI.
40. Remove taper pin and fusee end plate, both shown in Fig. 28.

NOTE

If not already marked, place slight mark on the fusee end plate on the side from which the pin is removed.

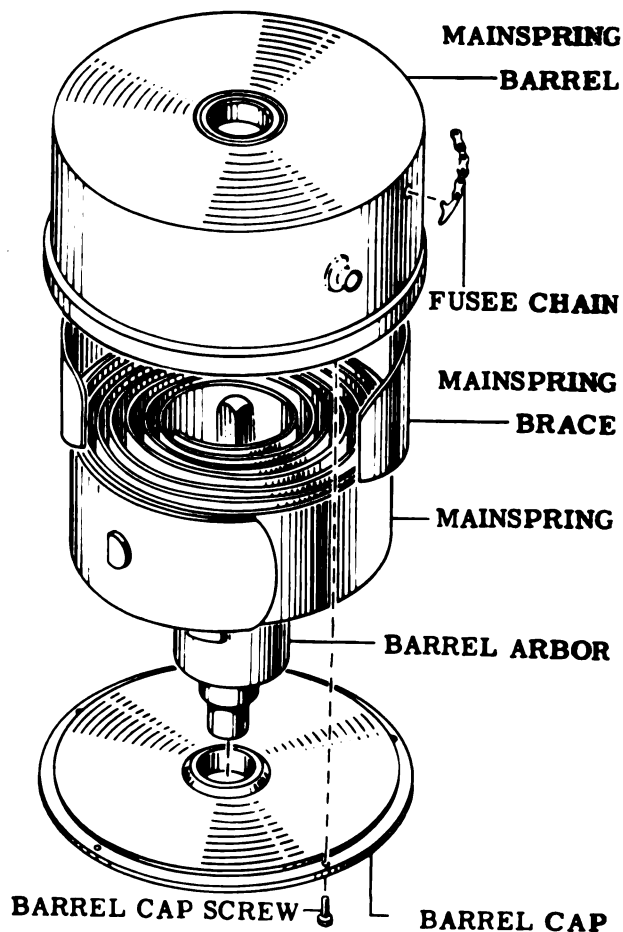


Figure 26

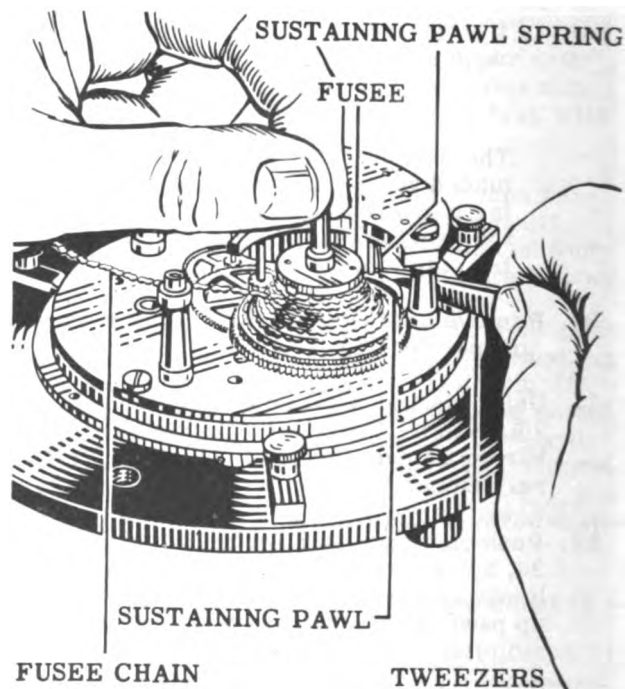


Figure 27

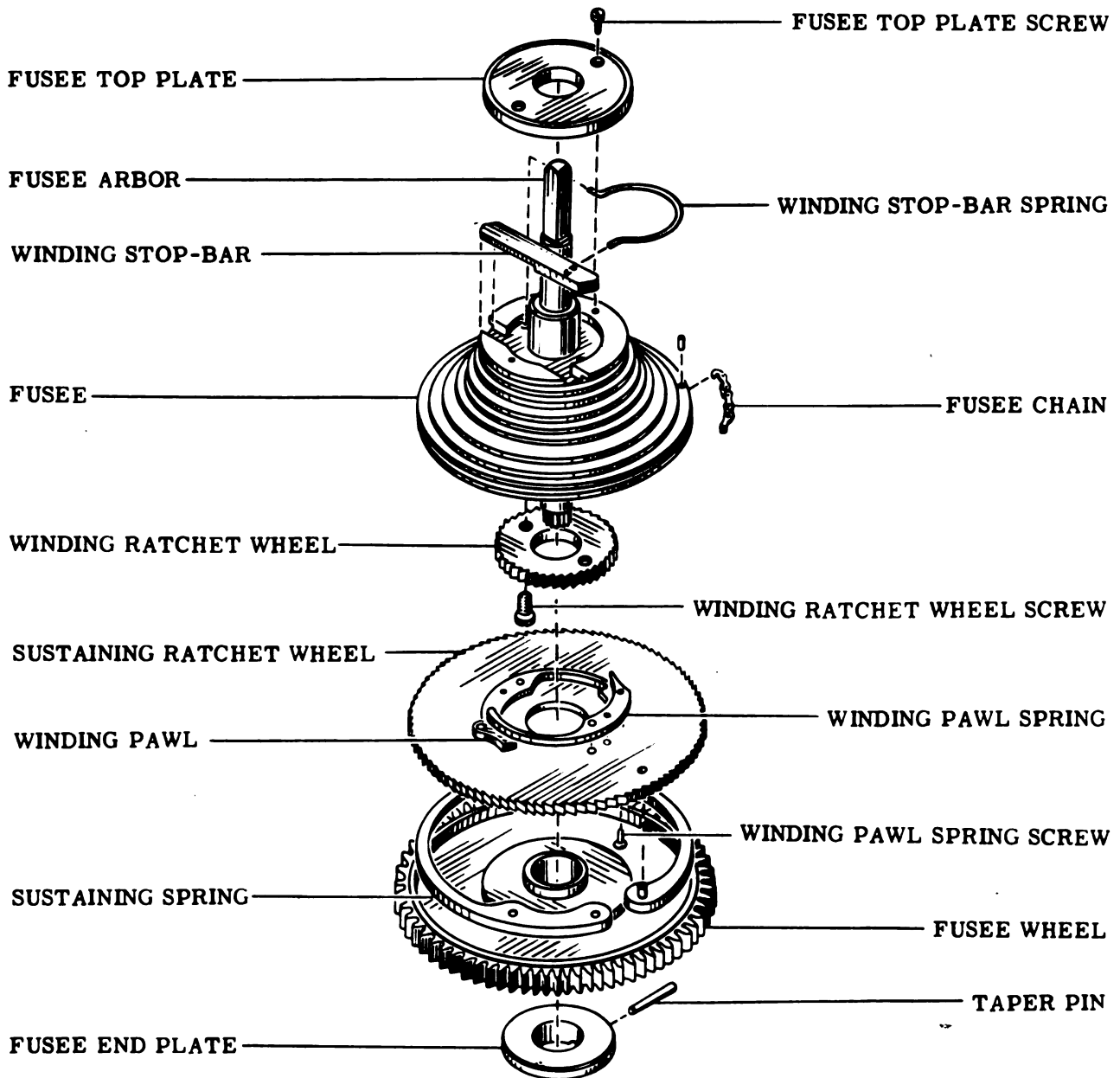


Figure 28

41. Remove fusee wheel, sustaining spring and sustaining ratchet wheel. See these parts exploded in Fig. 28.
42. Remove four screws from sustaining ratchet wheel, releasing winding pawl springs. See Fig. 28.
43. Remove two winding ratchet wheel screws from fusee, using Tool No. 93, releasing winding ratchet wheel. See Fig. 28.
44. Remove two fusee top plate screws, releasing fusee top plate. See Fig. 28.
45. Remove winding stop-bar spring and winding stop-bar. See Fig. 28.
46. Free train by backing up the train blocking screw. See Fig. 16 and Operation 5.
47. Remove three upper train bridge screws, indicated in Fig. 29, using Tool No. 96.
48. Lift the upper train bridge straight up off the pillars. See Fig. 29.
49. Remove fourth wheel, center wheel, third wheel and sustaining pawl. See Fig. 29.

HAMILTON SHIP CHRONOMETER

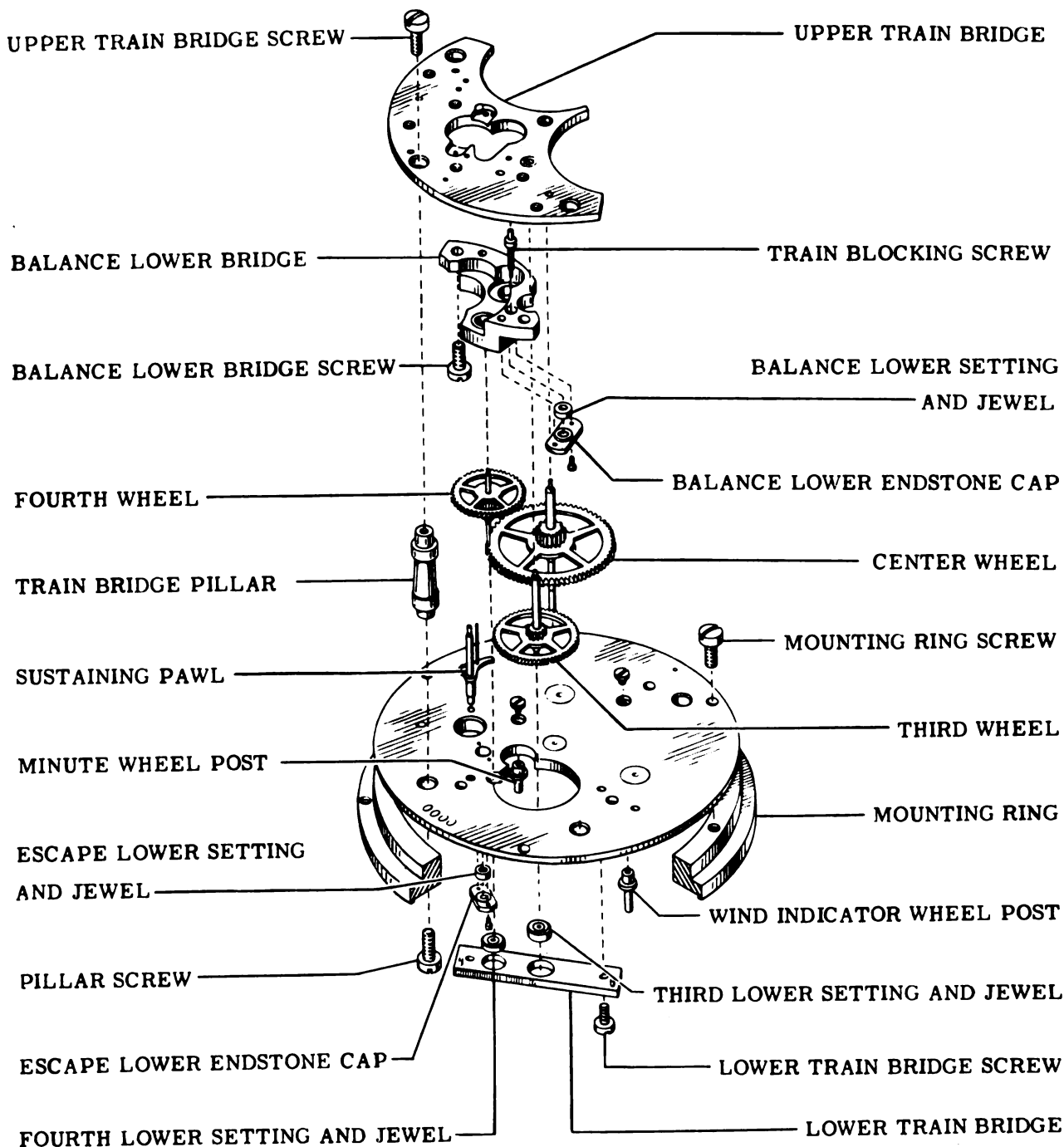


Figure 29

50. Remove two balance lower bridge screws from the upper train bridge, using the special shell screwdriver, Tool No. 96. Remove balance lower bridge. See Figs. 29 and 30. The upper part of Fig. 30 shows condition of the pillar plate after removal of the upper train bridge.
51. Remove train blocking screw. See Figs. 29 and 30. Remove two balance lower endstone cap screws. Remove endstone

cap from bridge. See Figs. 29 and 30. If not already done, mark the numeral "2" in the milled recess of the balance lower endstone cap.

NOTE

For the following operations, the work ring will be turned back and forth, and held in the hand, as is found convenient.

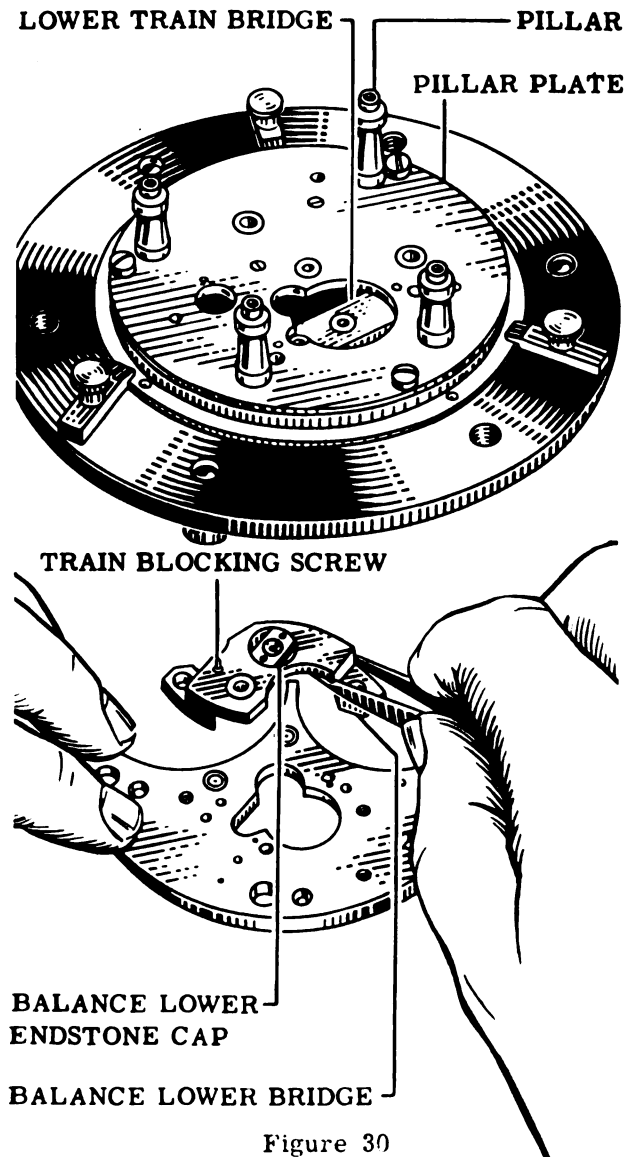


Figure 30

CAUTION

Pillars, upon reassembly, should be placed in the same location from which removed. They should therefore be marked upon removal.

52. Remove four pillar screws from the dial side, allowing the pillars to drop out, using for this purpose the special shell screwdriver, Tool No. 96. See Fig. 29.
53. Remove two screws from the lower train bridge and remove lower train bridge. Use Tool No. 94. See Fig. 31. Refer to Fig. 29 for exploded view.
54. Remove two escape lower endstone cap screws and remove endstone cap. See Fig. 29. If not already done, mark the

numeral "3" in the milled recess of the escape lower endstone cap.

55. Using special shell screwdriver (Tool No. 96), remove mounting ring screws. Lift pillar plate from mounting ring. See Fig. 29.
56. Remove wind indicator post screw and remove post. See Fig. 29.
57. Remove minute wheel post screw and remove post. See Fig. 29.
58. Remove mounting ring from work ring.

CAUTION

Since the mounting ring is lacquered, contact with alcohol should be avoided.

Further disassembly procedures, which may be required for purposes of repair, are given in Section VII "Inspection and Repair."

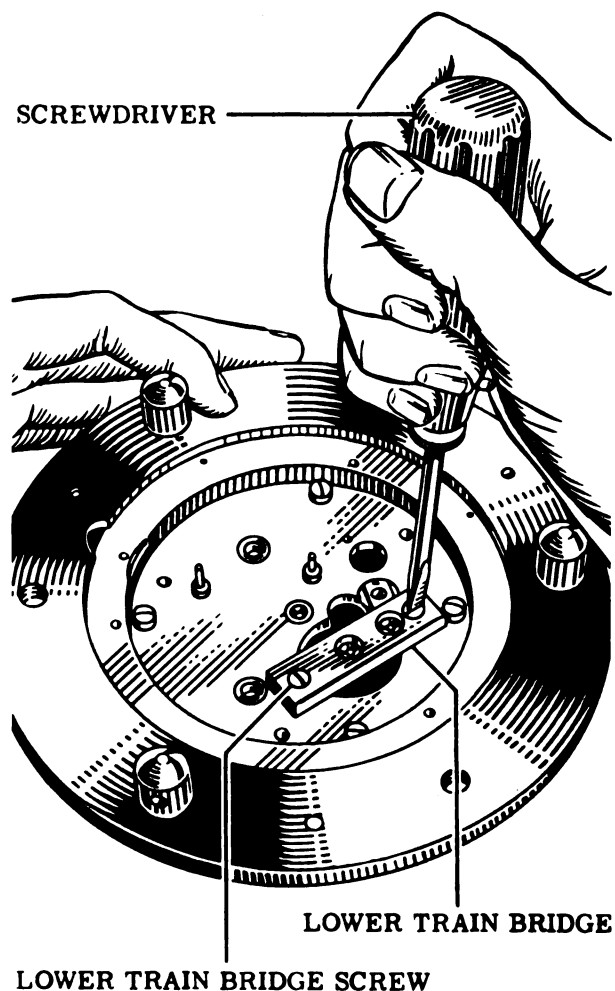


Figure 31

SECTION VI

CLEANING

Cleaning practice, as outlined in this section, indicates the employment of a standard cleaning machine as shown in Fig. 32, and a hand spinner as shown in Fig. 35. However, manual methods may be used, and any of the commonly accepted watch cleaning materials available to the repairman will prove satisfactory if thorough manual procedures are employed. See the note on importance of cleaning at the end of this section.

CAUTION

Some chronometer parts are not cleaned in accordance with the general procedure as outlined. Refer to special paragraph headings in this section for the following:

Dial and Hands
Balance and Hairspring
Escape Wheel
Packing Rings
Fusee Chain
Mainspring and Brace
Detent Assembly
Mounting Ring

All parts other than those listed are cleaned by agitation in various solutions followed by thorough drying in warm air.

PRELIMINARY

Parts obviously defective need not be cleaned. These should be replaced in the cleaning operation with perfect parts. Then, upon reassembly, the repairman will have available whatever parts are needed to complete each operation.

If old oil has become thick and adheres to the jewels or bushings, carefully remove this residue with pegwood before proceeding. If excessively dirty, the less fragile parts should be brushed with a medium-stiff watch brush dipped in naphtha. This should be followed by an alcohol rinse and drying over the warm air blower.

Parts not placed in the proper compartments of the cleaning baskets (Tools Nos. 109, a, b, c) during disassembly, should be arranged therein as illustrated in Figs. 33 and 34. Care should be exercised that pieces are

not left in contact with each other in such a manner as to impede cleaning.

Before proceeding with cleaning instructions, it is necessary to identify the materials to be used incidental to the routine cleaning operations.

MATERIALS USED

The instructions appearing in this section refer to the use of various materials and solutions which, for purposes of accuracy, must be defined.

Naphtha

Where naphtha is mentioned, what is meant is naphtha of 70 to 72 degrees Baumé, API.

Alcohol

By alcohol is meant alcohol, S.D. 1, or S.D. 3A, or S.D. 30 or isopropyl alcohol, 99 percent.

Cleaning Solution

The following ingredients and proportions are used:

30 liters	distilled water
300 grams	oxalic acid (Tech. 98-100 percent, Gen. Chemical Company)
3 liters	denatured alcohol (S.D. 3A)
800 cc	ammonium hydroxide (C. P.)
200 cc	alkamine (alkyl sulphate, Amalgamated Chemical Corporation)

Place distilled water in carboy. Dissolve oxalic acid in separate quantity of water and add to carboy. Add alcohol, ammonium hydroxide and alkamine in order named. Mix thoroughly.

The solution should be replenished during the day and renewed each working day. It should always be handled and used in glass containers, which should be covered when not in use. It may be used hot or cold. If hot, the temperature should not exceed 180 degrees F.

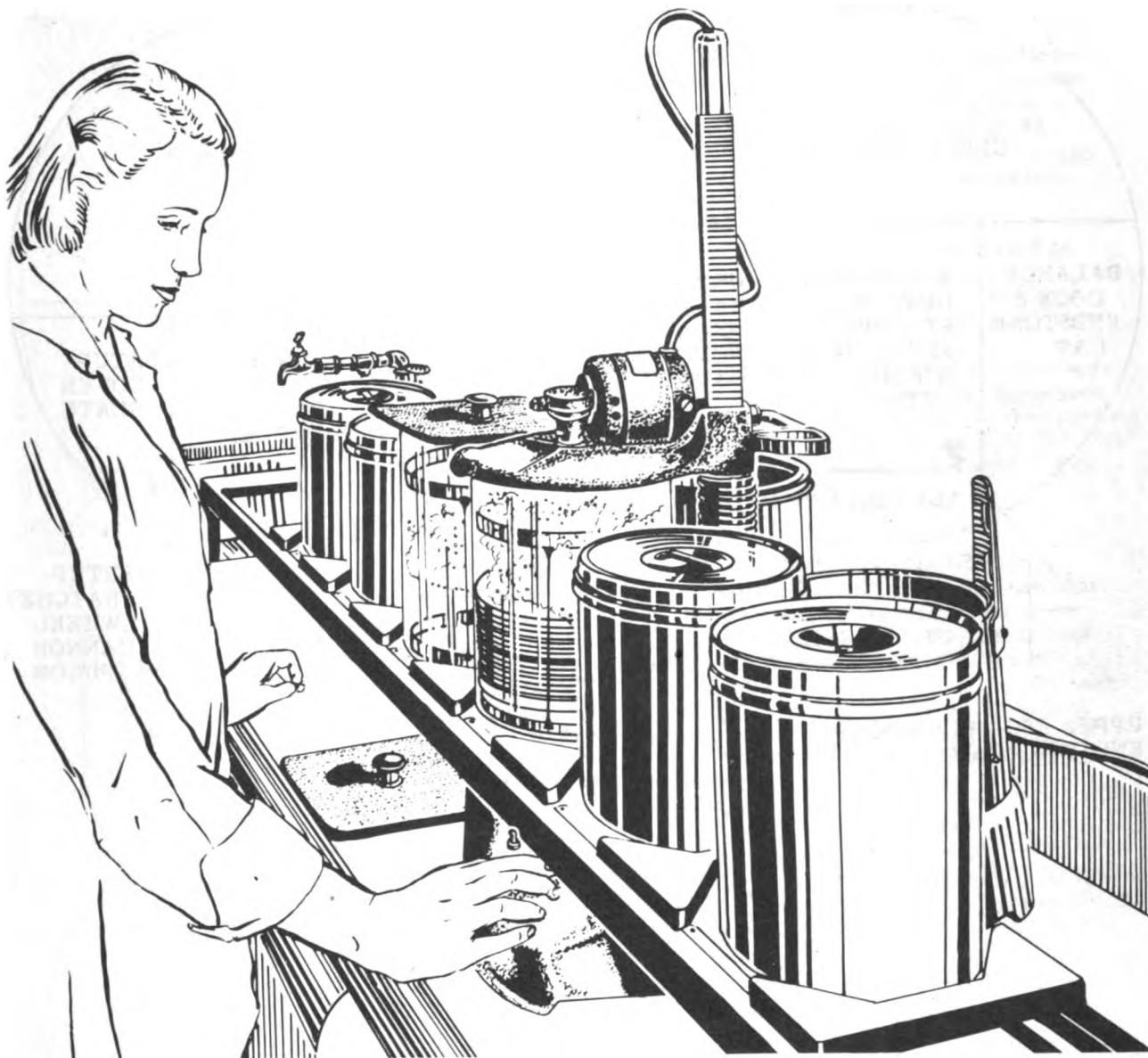


Figure 32

Rinse Solution

The following ingredients and proportions are used:

2400 cc	distilled water
48 grams	orvus flakes or granules (alkyl sulphate, Proc- tor and Gamble)

Measure the desired volume of water into the container and add the weighed amount of orvus flakes.

Gun Oil

A commercial product of the Alox Corporation, known as L-1165M Gun Oil.

CLEANING IN SOLUTIONS

The wire baskets containing parts are stacked in the cleaning machine holder according to its capacity. Immerse the baskets in the following, as indicated:

1. For two minutes in naphtha

CAUTION

The vapors of naphtha are explosive and poisonous. This product is an excellent solvent but it should be used with ventilation and with proper caution as to fire hazard.

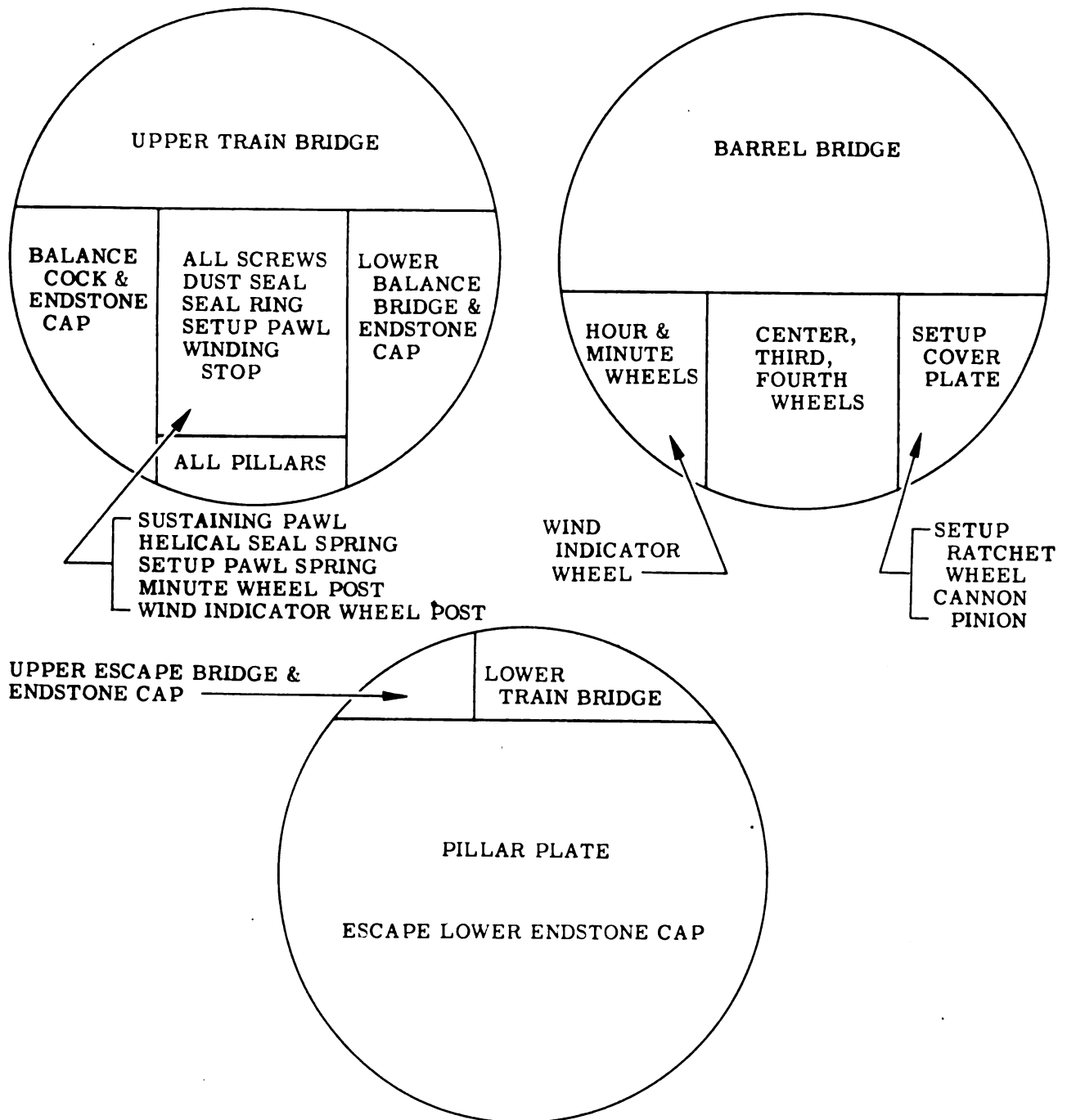


Figure 33 - Arrangement of Parts in Cleaning Baskets

2. For one minute in first alcohol bath

CAUTION

Alcohol should be replaced as soon as it becomes cloudy in use. Failure to do this may cause noticeable rusting or staining of steel parts.

3. For two minutes in cleaning solution

4. For one minute in rinse solution

5. For two minutes, or longer if required, in distilled water

6. For one minute in second alcohol bath

7. For one minute in third alcohol bath

8. Until completely dry in hot air blower chamber

CAUTION

During each of the immersion operations, spin the machine, adjusting speed with the control knob as the operator is doing in Fig. 32. Stop the rotation two or three times in each solution so that entrapped air may escape.

Parts may appear dry and still be damp in their hidden recesses. Minimum drying time is five minutes under average conditions.

After cleaning, all parts should be thoroughly examined to be sure they have been completely cleaned. When inspection reveals that a perfect cleaning job has been done, carefully peg all bushings and jewels with clean, soft pegwood.

CAUTION

Great care should be exercised that none of the pivots or jewels be touched with fingers or greasy buff stick after cleaning. Traces of lint and dust can usually be removed with a blast of air. Reclean, if touched. To avoid leaving fingermarks, keep the handling of parts to a minimum. Use tweezers.

SPECIAL CLEANING

Certain parts are not cleaned in the manner outlined in the foregoing paragraphs. These parts will be individually mentioned, and instructions given for each.

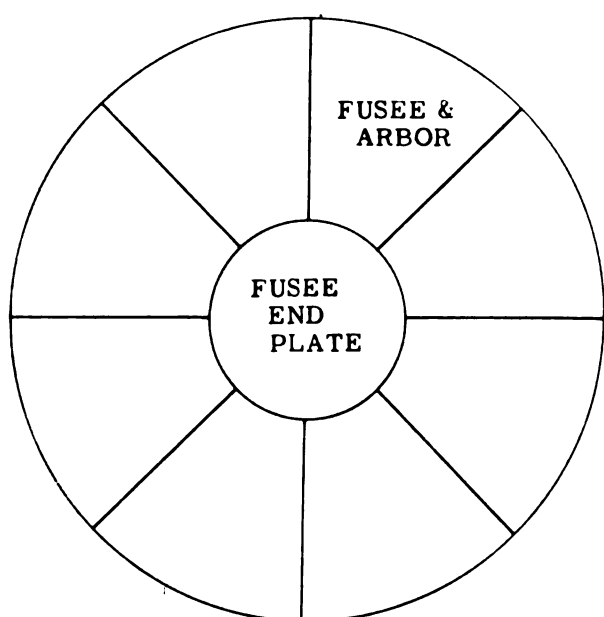
Dial and Hands

The dial and hands should be cleaned with a soft, dry camel's hair brush. To remove fingermarks use a buff stick moistened with naptha. Do not allow the dial and hands to come in contact with any other liquid.

Balance and Hairspring

For cleaning purposes, the balance unit is not disassembled unless the balance pivots are in poor condition or there is some other reason for doing so. For balance disassembly and reassembly, see Section VII "Inspection and Repair."

If the hairspring is to be cleaned separately, this should be done with the collet and stud in place ready for use. Clean by stringing on dip wire. Use Tool No. 76. Dip in naptha and follow with an alcohol rinse. Dry over the warm air blower of the cleaning machine.



WINDING PAWL SPRINGS
WINDING PAWL SPRING SCREWS
WINDING RATCHET WHEEL
WINDING RATCHET WHEEL SCREWS
WINDING STOP-BAR
FUSEE TOP PLATE SCREW

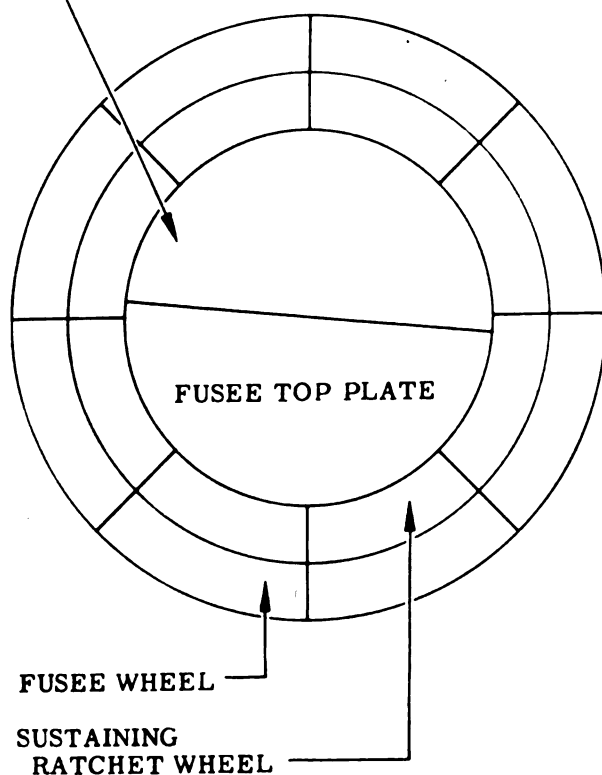


Figure 34

If the balance and hairspring assembly or the balance alone is to be cleaned, the assembly or the balance wheel should be set in the wire tripod. Use Tool No. 111. This unit can then be placed in the cleaning basket of the hand spinner (see Fig. 35) and passed through the various cleaning solutions and dried as for other parts.

Where several chronometer balances are to be cleaned, Tool No. 66 may be used with the large cleaning machine.

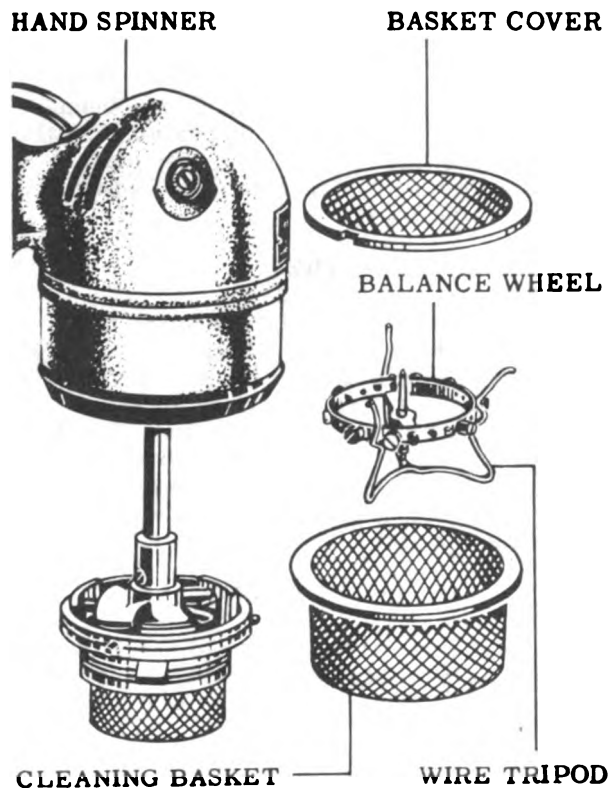


Figure 35

If a dark deposit is present on the impulse jewel due to the action of the escape wheel teeth, it can be removed by careful work with clean, hard pegwood.

CAUTION

Shellacked parts must not be placed in alcohol for longer than the time specified. Check for loose jewels after cleaning.

Escape Wheel

The escape wheel is cleaned by stringing on dip wire (Tool No. 76), dipping in the various solutions and drying as for other parts. If a dark deposit is present on the impulse face of the escape teeth, clean with hard pegwood.

Packing Rings

The packing rings are discarded and replaced with new rings each time the chronometer is overhauled. No cleaning solution should ever touch the packing rings.

Fusee Chain

The fusee chain is not immersed in solutions. Simply wiping with a lintless cloth moistened with gun oil usually suffices.

Mainspring and Brace

With the exception of the mainspring and brace, the parts comprising this assembly should be cleaned through the solutions as previously described. The mainspring and brace should be cleaned by wiping with tissue. Use care in wiping to keep fingers off the spring and to avoid distorting its shape. If the grease on the spring is very dry, it should be cleaned by dipping in naphtha only, and dried with tissue. Naphtha previously employed for other cleaning may be used to remove the bulk of the dry grease.

Detent Assembly

The detent may be cleaned as a unit by stringing on dip wire (Tool No. 76), dipping in the various solutions and drying over warm air as previously described. If a dark deposit appears on the locking jewel, this may be removed by a clean pegwood stick.

CAUTION

Shellacked parts must not be placed in alcohol for longer than the time specified. Check for loose jewels after cleaning.

Mounting Ring

The mounting ring should be cleaned by dipping in naphtha. Since the ring is lacquered, contact with alcohol is not permitted.

NOTE

No precision instrument will operate at its best with faulty lubrication. Proper lubrication is impossible unless parts are clean. The slightest film or deposit causes oil to spread, leaving dry the area where its presence is essential and, very likely, causing difficulty elsewhere. Thus, satisfactory service after overhaul is entirely dependent upon the care and attention given to cleaning.

SECTION VII

INSPECTION AND REPAIR

After the disassembled parts are perfectly cleaned, they should be completely inspected for defects. The following check list contemplates replacement of worn, damaged or rusted parts. It mentions the points to check particularly. Other obvious reasons for replacement may well be found.

Items of significance should be entered into the chronometer's "Navigational Timepiece Record," NavShips 3587.

INSPECTION CHECK LIST

1. Check all bushings for wear, scoring and tightness. Check jewels for wear, tightness, freedom from cracks, chips and scoring. Check endstones for pitting. Replace jewels or bushings, if necessary, using a jeweling press. See Fig. 36. Adjust endshake after replacement as required, with endshaking stand. Use Tools Nos. 1, 2 or 5.

NOTE

Worn jewels are usually noticeably out of round.

2. Replace one-piece rollers with two-piece rollers, if necessary by reason of breakage or defect. Otherwise, leave the one-piece roller in place on the balance staff.

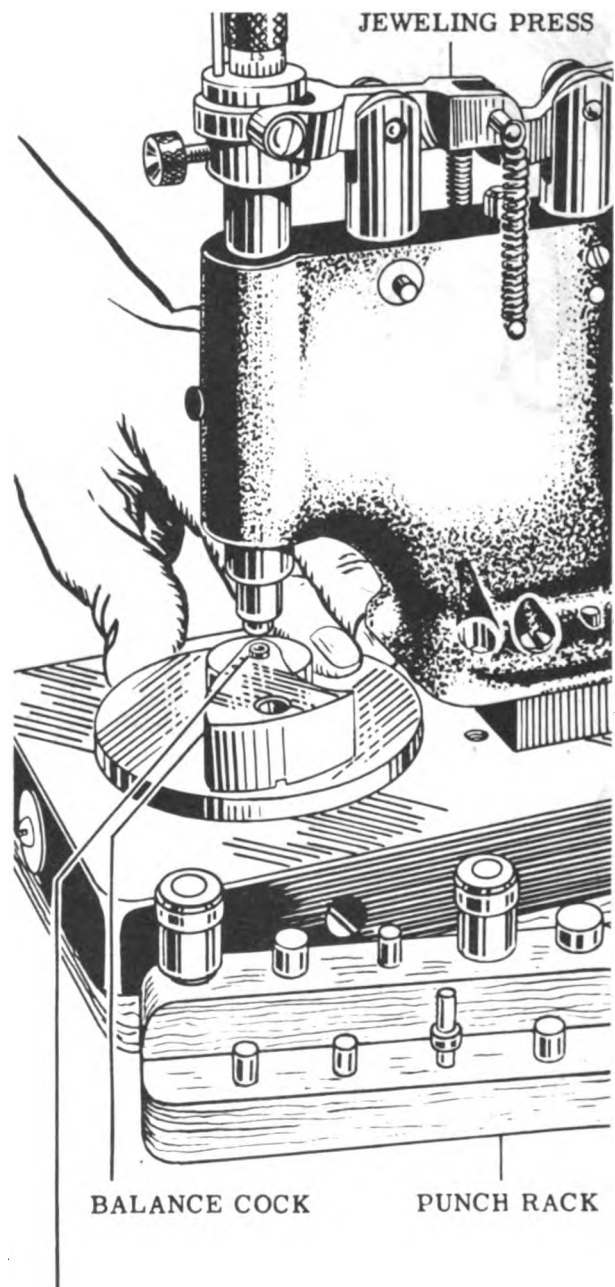
(Nearly all chronometers up to No. 6408 were fitted with one-piece rollers. If one of these needs to be replaced, it should be removed and two-piece rollers substituted on the same staff. See instructions given later in this section.)

3. Check all train pivots with eyeglass. If scored, replace the unit.
4. Check cone pivots with 18 power microscope using special fixture. See Fig. 37. If there is any evidence of damage or wear, replace the part. Fig. 38 shows the appearance of a perfect cone pivot, together with one type of defective pivot.

CAUTION

It is extremely important that balance and escape pivots be in perfect form and condition.

5. Check escape wheel teeth for wear. Appreciable wear indicates that replacement should be made. See Fig. 38.



BALANCE UPPER JEWEL SETTING

Figure 36

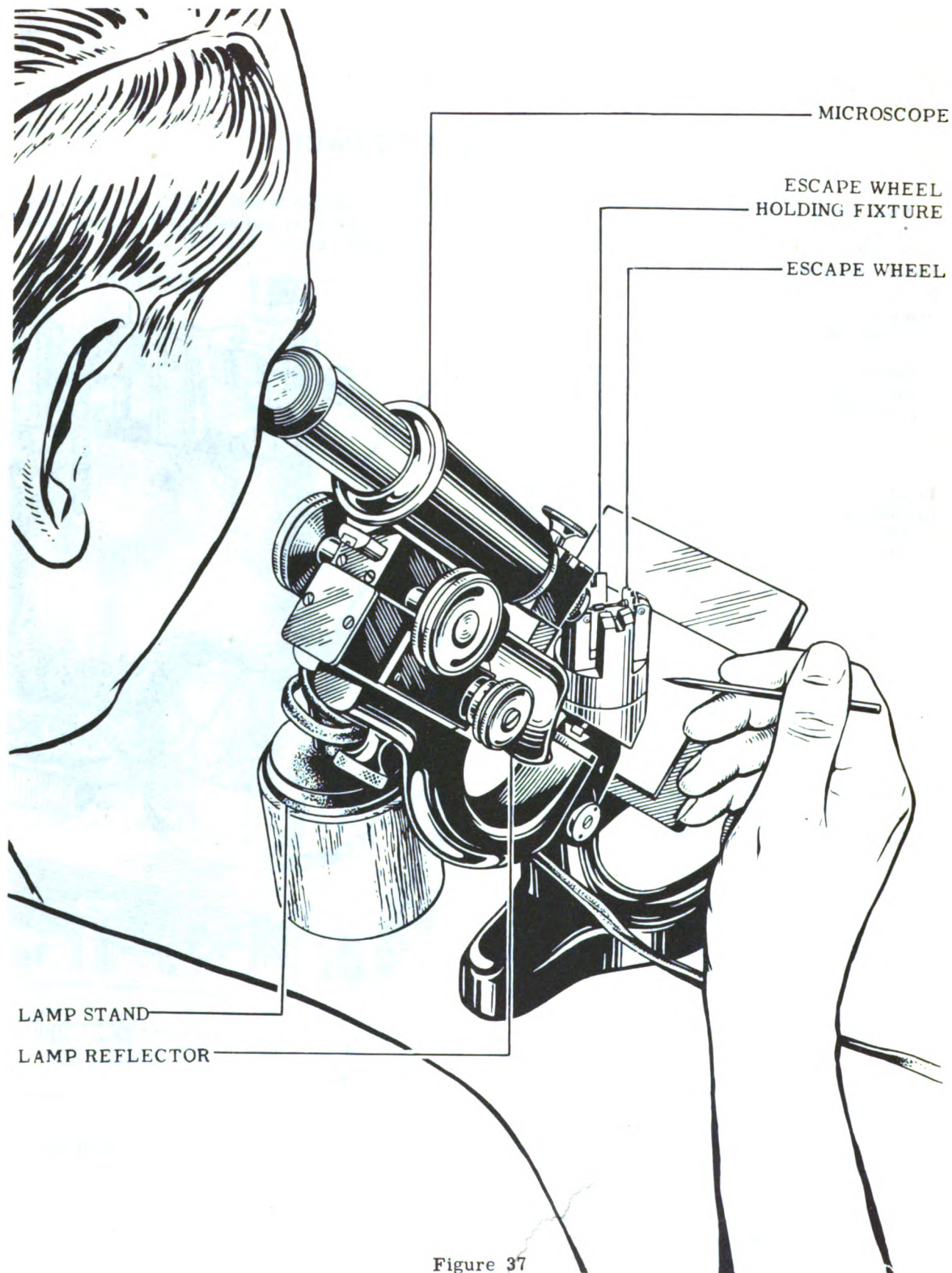


Figure 37

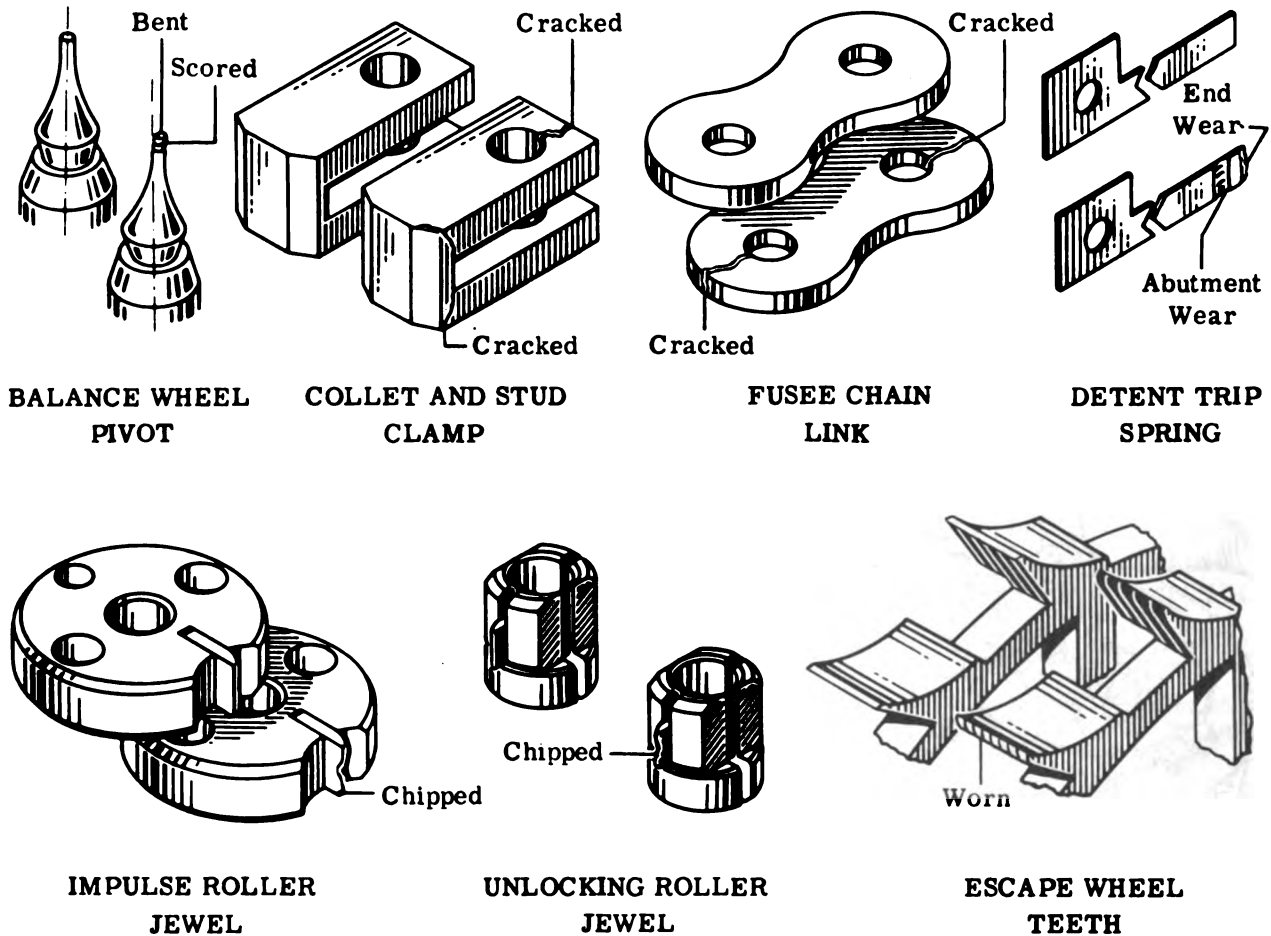


Figure 38

6. Check rollers for loose or chipped jewels. See Fig. 38.
7. Check hairspring stud and collet clamps for cracks. See Fig. 38. If cracks are found, follow instructions for replacement of clamp given later in this section.
8. Check fusee chain for worn and damaged links. See Fig. 38.
9. Check detent assembly for damaged locking jewel, damaged or worn trip spring, damaged or broken detent spring. If defective parts are found, replace by following disassembly and reassembly instructions given later in this section.

NOTE

Two types of trip spring wear are shown in Fig. 38. The trip spring may be worn where it touches the abutment. It may be worn at the end where it is struck

by the unlocking jewel, especially if the contact faces of that jewel have become rough.

10. Check timing weights, vernier timing weights and balance screws for tightness. Use Tools Nos. 60, 61 and 63.
11. Check fusee parts for wear, especially noting the bearing surfaces of the arbor, and the condition of the stop-bar spring.
12. Check barrel assembly, especially noting bearing surfaces of barrel and arbor.
13. Check condition and set of mainspring.

The procedures outlined in subsequent paragraphs of this section cover routines often required for repair.

DISASSEMBLY OF BALANCE UNIT

1. Place balance in balance wheel work block, hairspring up. Use Tool No. 52.

CAUTION

It is important to record exact relation of the stud to a balance screw hole in order to facilitate easy adjustment upon reassembly.

- Place the wedge of the hairspring wedge tool (Tool No. 64), into the collet slot, remove the hairspring unit with stud and collet, and place in work tray. See Fig. 39.

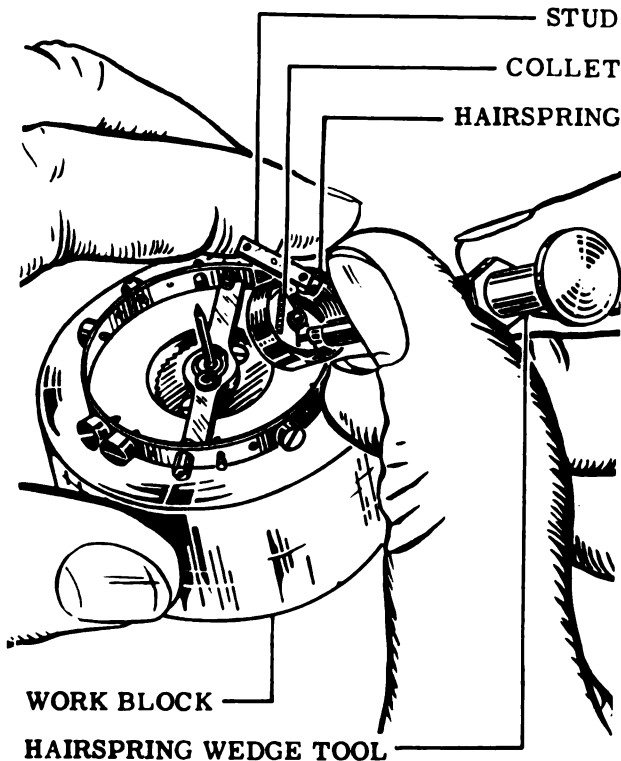


Figure 39

- Lift balance from work block and remove unlocking roller with special pliers. Use Tool No. 48.

NOTE

If one piece double roller is to be removed, this can be accomplished with the tool indicated in Operation 4.

- Remove impulse roller, using roller removing stand, Tool No. 4. See Fig. 40.
- Remove two balance hold-down screws using balance cap screwdriver (Tool No. 88) and remove balance wheel cap. See Fig. 41.
- Remove staff. If tight, it can be placed in work block and removed with gentle pressure by a punch. Use Tool No. 16.

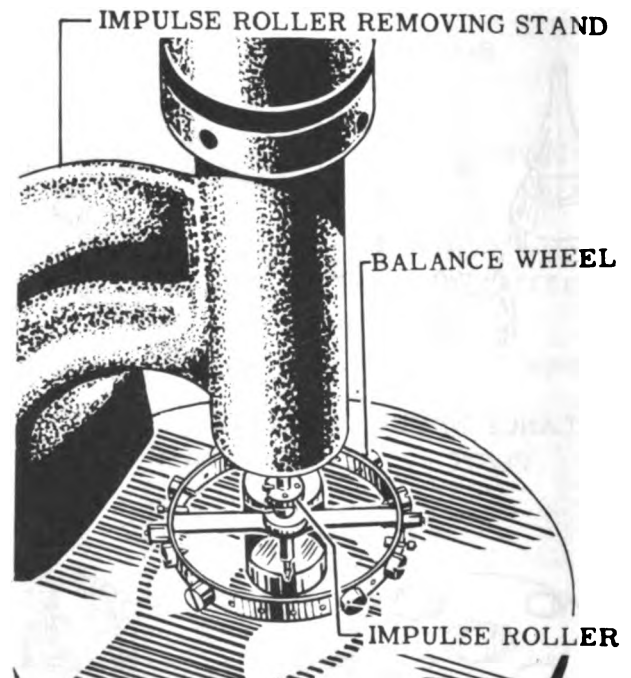


Figure 40

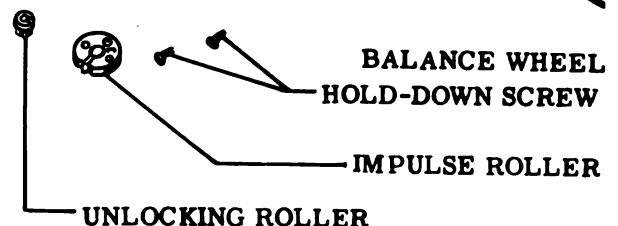
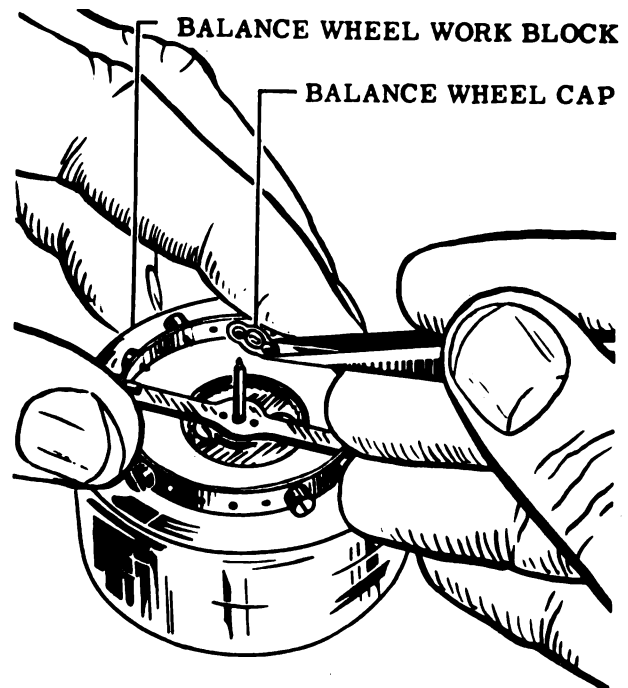


Figure 41

ASSEMBLY OF BALANCE UNIT

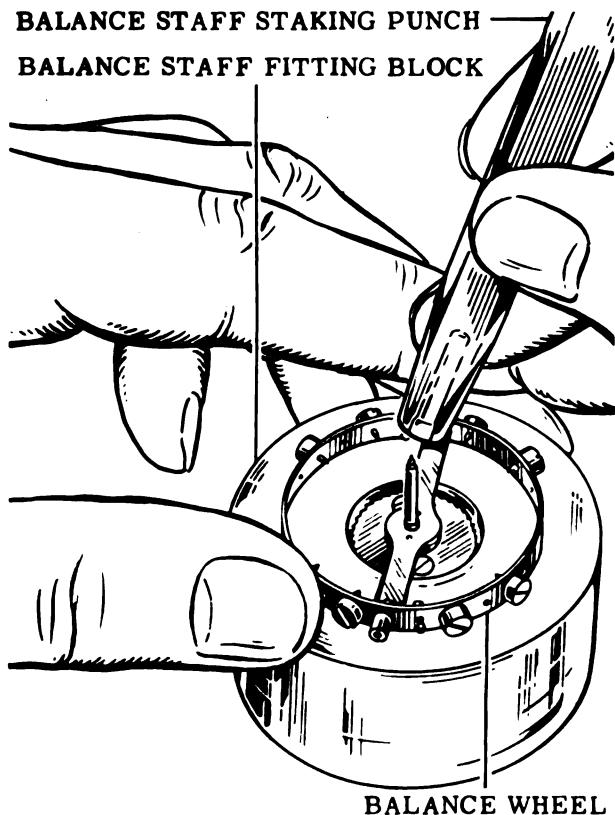
1. Check fit of impulse roller on staff, holding roller with tweezer in work holes and with the numbered side of the roller facing the balance hub. The roller should go up to within 0.010 in. of the roller shoulder on the staff. If correct, leave the roller on the staff. If too small, ream. Use Tools Nos. 65 and 81. If large, replace.

NOTE

If one piece double roller is to be reinstalled, this can be accomplished by following the same procedure as for the single roller.

BALANCE STAFF STAKING PUNCH

BALANCE STAFF FITTING BLOCK



BALANCE WHEEL

BALANCE WHEEL CAP

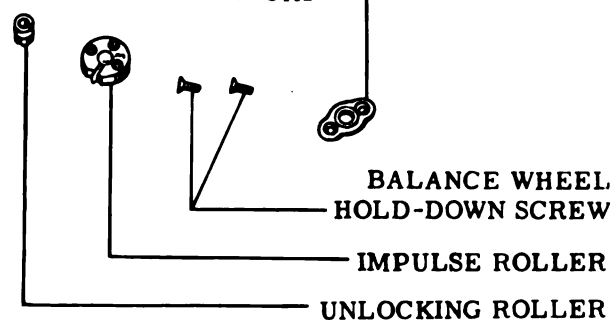
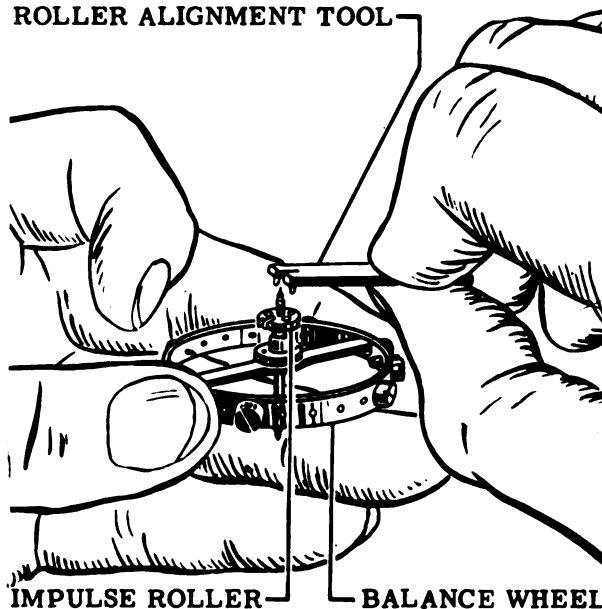


Figure 42

ROLLER ALIGNMENT TOOL

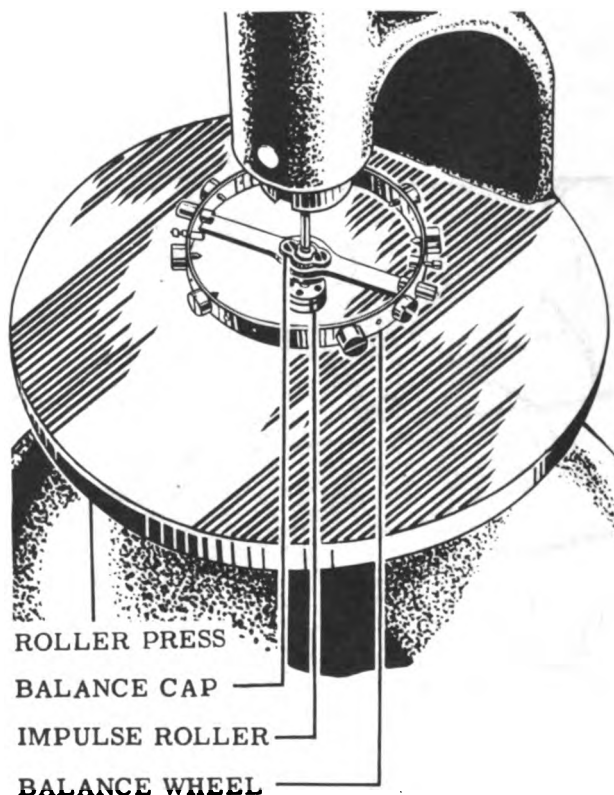


IMPULSE ROLLER

BALANCE WHEEL

Figure 43

2. Place staff with impulse roller on balance staff fitting block. Use Tool No. 50. Install wheel on staff with special plastic punch (Tool No. 59), making sure that the screw holes in the balance wheel line up with those in the hub. See Fig. 42.



ROLLER PRESS

BALANCE CAP

IMPULSE ROLLER

BALANCE WHEEL

Figure 44

3. Locate balance wheel cap in place (see Fig. 41), and screw in two cap screws tightly. Use balance cap screwdriver, Tool No. 88.
4. Using roller alignment tool (Tool No. 43), line up impulse face of impulse jewel with center of balance hold-down screw hole. See Fig. 43.
5. Press impulse roller tightly against roller shoulder of staff in roller press, using Tool No. 3. See Fig. 44.
6. Check and fit unlocking roller, using same standard and method as in Operation 1.
7. Using Tool No. 42, locate unlocking roller with unlocking jewel approximately 90 degrees ahead of the impulse jewel (clockwise of the unlocking jewel when facing it). See Fig. 45. Press unlocking roller tightly against the impulse roller on the roller press. Use Tool No. 3.
8. Check trueness of balance in the round and in the flat with balance-truing caliper. Use Tool No. 23.

NOTE

The balance wheel may be corrected if slightly out of true in the flat, but

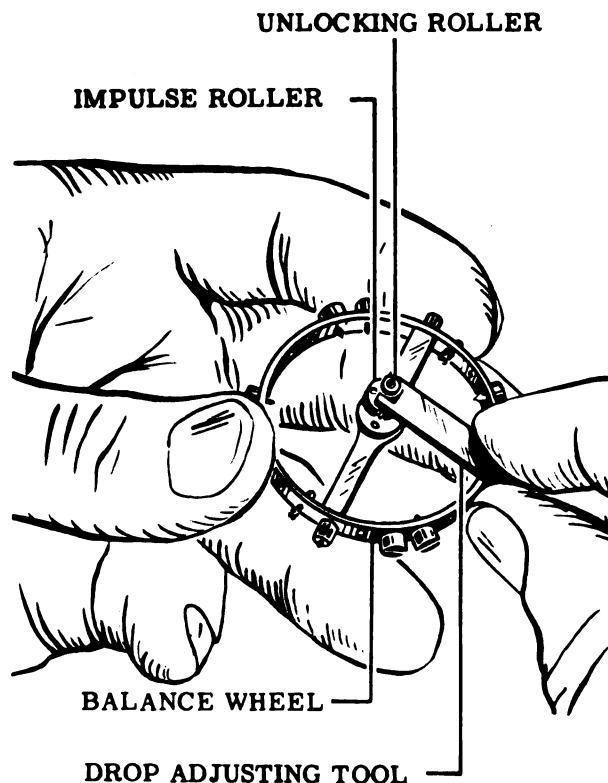


Figure 45

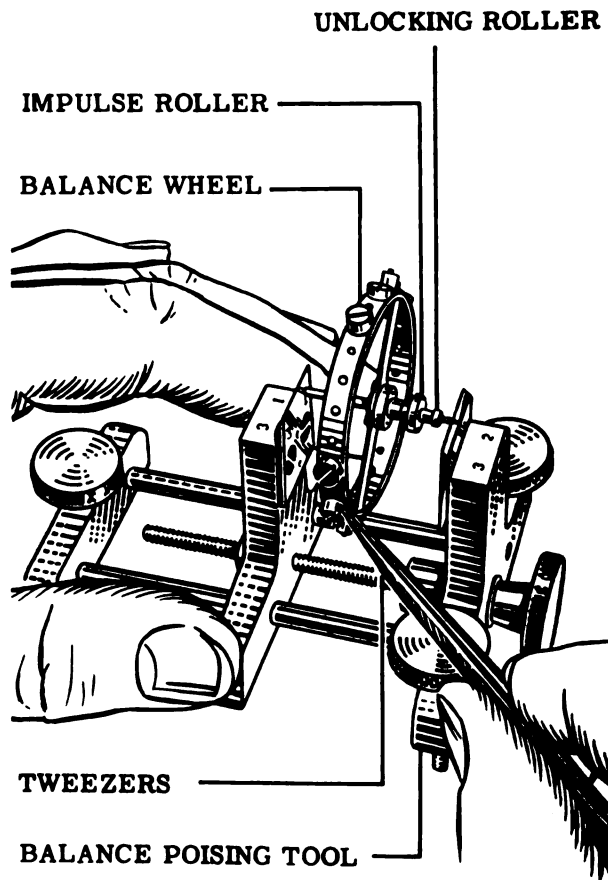


Figure 46

if appreciably out, use new part. No attempt should be made to true the wheel in the round. If necessary to use a new balance wheel, set it up with weights the same as the old one.

9. Poise balance wheel in balance poising tool, Tool No. 24. See Fig. 46. Place balance in numbered block (Tool No. 54), if screw changes are needed. Undercut screws, if necessary. See Fig. 47. Use balance screw undercutter, Tool No. 30. Install washers, if required.

CAUTION

In performing the foregoing operation on a balance wheel originally in the unit, the same static weight of the balance should be closely maintained. If a new balance wheel is used, further static weight adjustment will be required. See Section IX "Test and Adjustment."

10. Place balance in balance wheel work block (Tool No. 52) and apply a drop of Hamilton No. 47 Red Oil to that portion of the staff where the hairspring collet will be located.

11. Using the hairspring wedge tool (Tool No. 64) in the collet slot, locate the hairspring unit in the proper position on the staff.

CAUTION

The position of the hairspring unit, when relocated, should be the same as that recorded when it was removed. See "Disassembly of the Balance Unit."

REMOVAL AND REPLACEMENT OF HAIRSPRING COLLET OR STUD

While removal and replacement of the hairspring stud involves operations practically identical to the corresponding work on the collet, it is well to read the following instructions completely before proceeding. Colletting work should be performed first.

1. Drive out clamp wedge pin of hairspring collet or stud on wedge pin staking tool (Tools Nos. 9 and 17), and withdraw hairspring. See Fig. 48.
2. Place collet or stud on colletting block (Tool No. 44) or in studding tool (Tool No. 45). Locating pin of collet should be in work hole of block; stud should be upside down on studding tool. Colletting block and studding tools are illustrated in Fig. 51.
3. Place collet clamp on tongue of collet, or place stud clamp on tongue of stud. Fig. 49 shows relation of clamp to collet, which is similar in arrangement to the stud.
4. If a new hairspring is being used, cut off the fabrication terminal at about one wire width beyond the bend. Use Tool No. 78. See Fig. 50.
5. Place the cut off piece of hairspring between the clamp and tongue of the collet or stud to facilitate fitting of the wedge pin.
6. Insert wedge pin, drive in lightly, and mark for correct length. See Fig. 51. Allow approximately 0.050 in. for driving. Remove pin; using pin vise (Tool No. 83), cut to correct drive length with cutting pliers (Tool No. 46), and smooth the cut end. Use Tool No. 97 or 10, for finishing.
7. Remove wedge pin and short piece of hairspring. Insert end of hairspring sideways

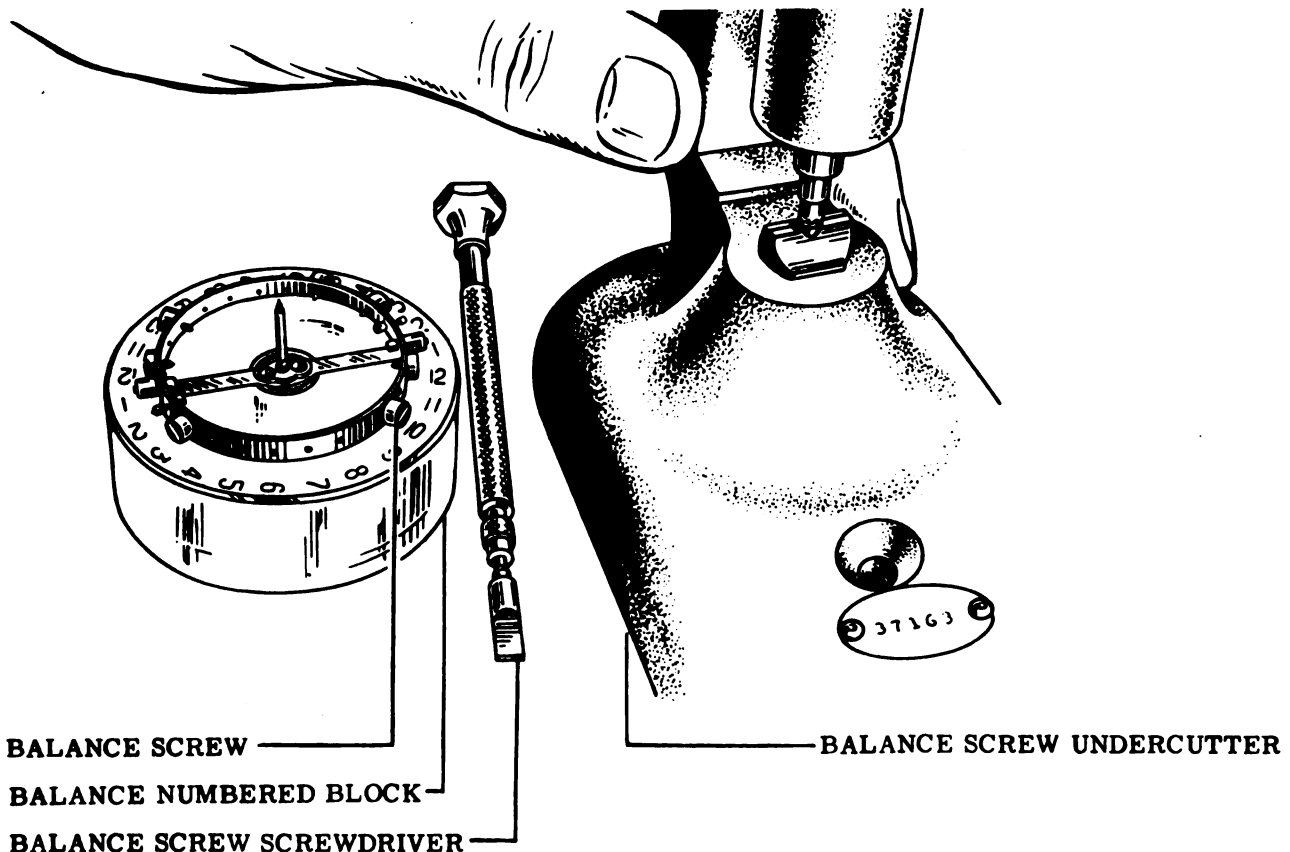


Figure 47

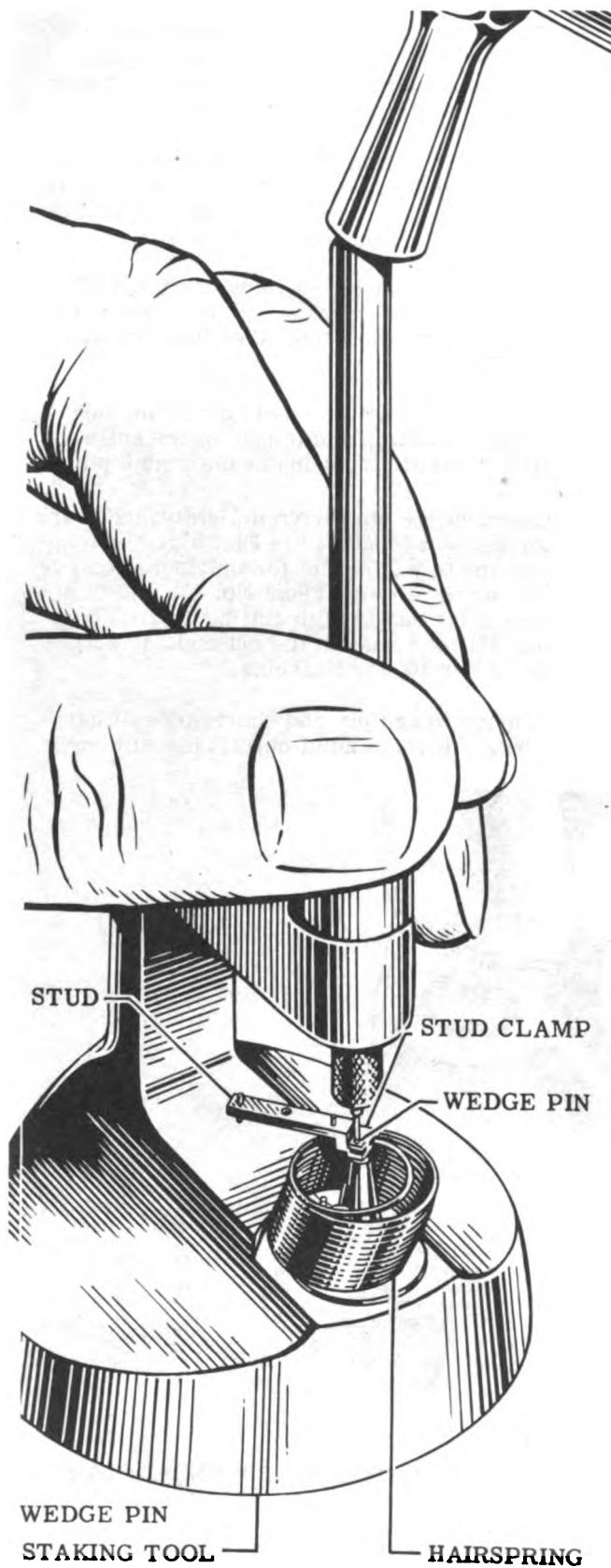


Figure 48

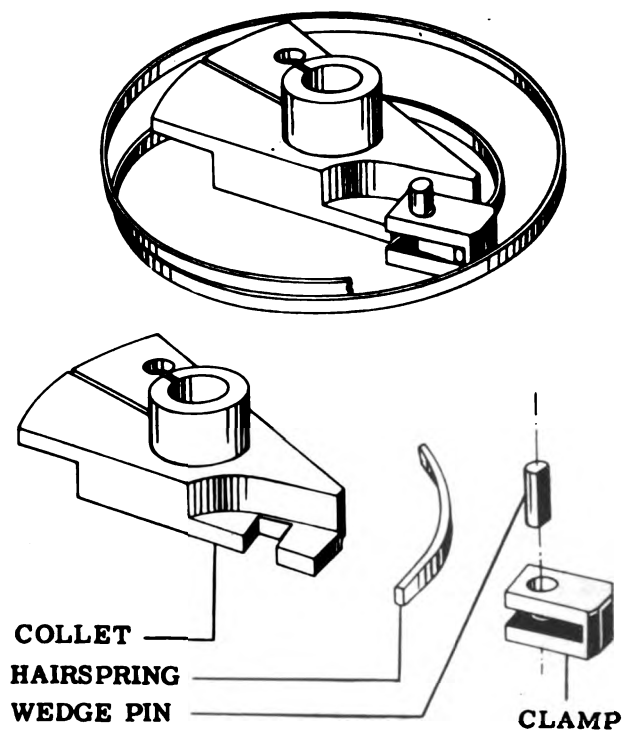


Figure 49

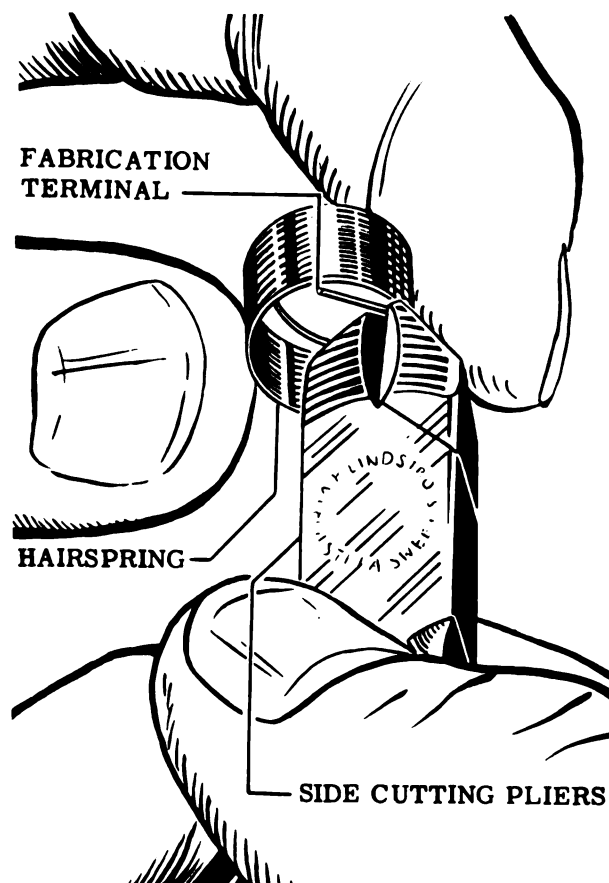


Figure 50

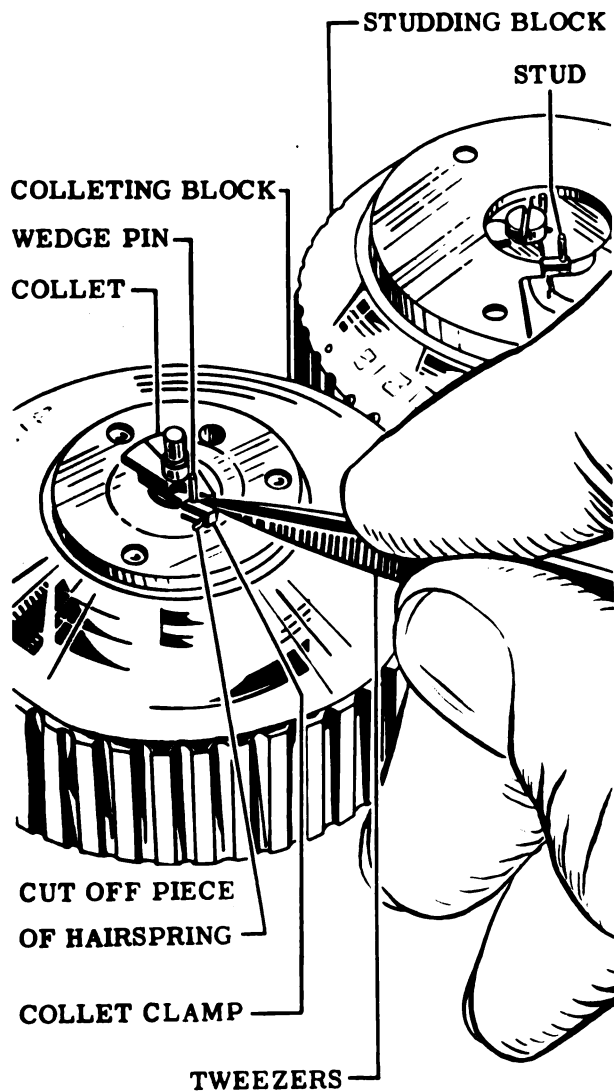


Figure 51

between the clamp and the tongue of the collet or stud, as shown in Fig. 52. Tip hairspring up straight.

CAUTION

Make sure that the hairspring collet or stud tongue face and the collet or stud clamp face are perfectly aligned before the next operation.

8. Insert wedge pin and drive it tight with special concave punch. Use Tool No. 15.
9. Remove assembly from colleting block or studding tool. Place collet on collet taper pin filing block, Tool No. 58. Place stud in pin vise, Tool No. 83. Cut wedge pin off with pliers, using care not to strike the hairspring coil or touch the clamp with the pliers. Use Tool No. 46.

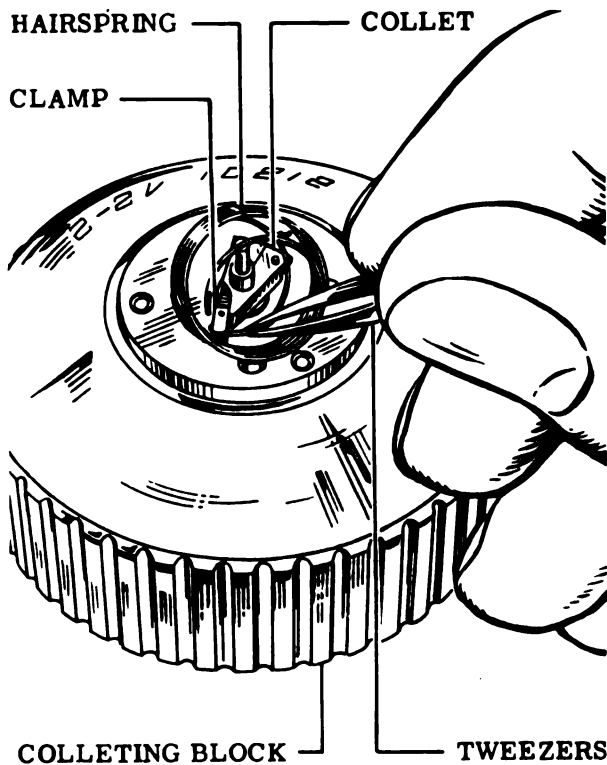


Figure 52

10. File wedge pin flush with collet clamp or, in case of stud, file pin below the top surface of the stud, using care not to strike the hairspring coil or clamp with the file.
11. True spring in the truing calipers (Tool No. 22), after colleting and before studding. See Fig. 53.

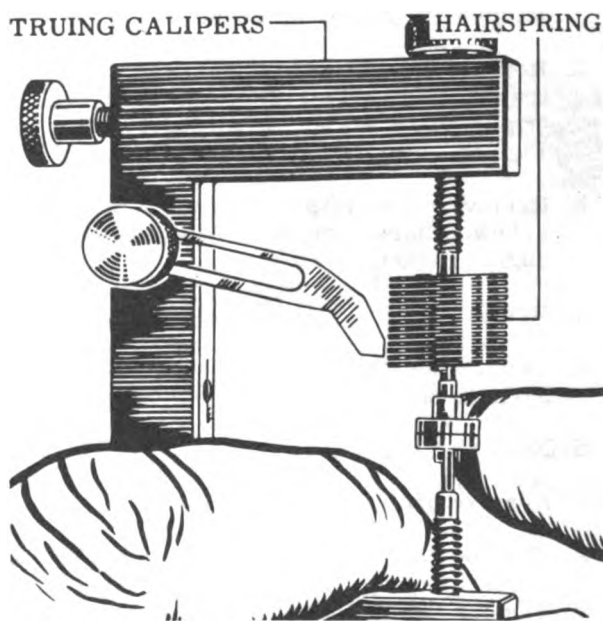


Figure 53

CAUTION

Bending of the hairspring should be kept to a minimum.

12. Clean assembly by stringing on dip wire (Tool No. 76), dipping in naphtha, followed by alcohol rinse. Dry thoroughly over warm air blower.

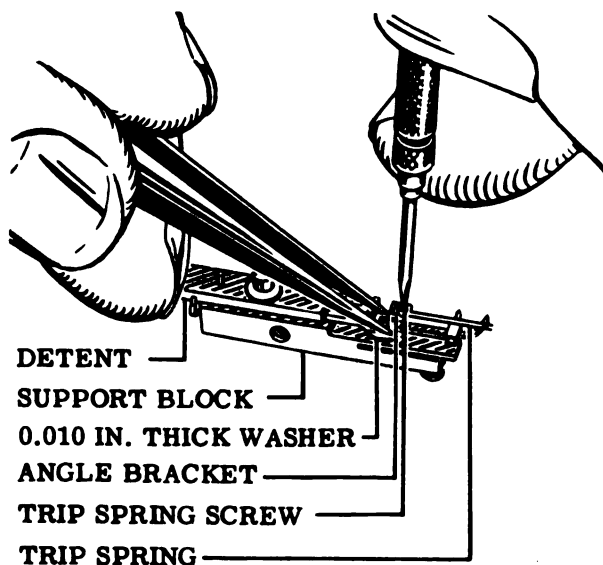


Figure 54

DISASSEMBLY OF DETENT

1. Place 0.010 in. thick washer (Tool No. 14) under the detent trip spring support angle bracket to sustain the load while removing the detent trip spring. Timing washer 42256 may be used.
2. Remove detent trip spring screw, releasing trip spring. See Fig. 54. Check trip spring for wear or damage. Refer to Fig. 38. Use new spring, if necessary.
3. Remove detent trip spring support bracket screw. Remove detent trip spring support angle bracket.
4. Remove supporting washer.
5. Remove detent clamp screw and washer. See Fig. 55.
6. Remove detent.
7. Remove detent adjusting screw.
8. Remove detent lock adjusting clamp screw.
9. Remove detent lock adjusting screw. See Fig. 56 for final disassembly.

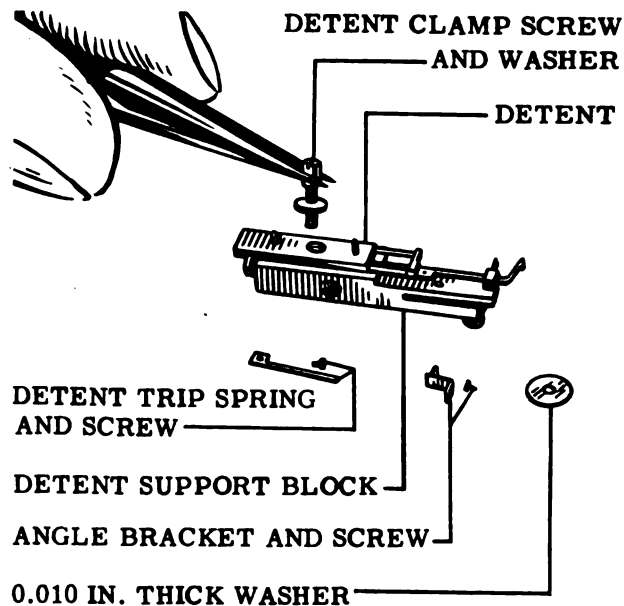


Figure 55

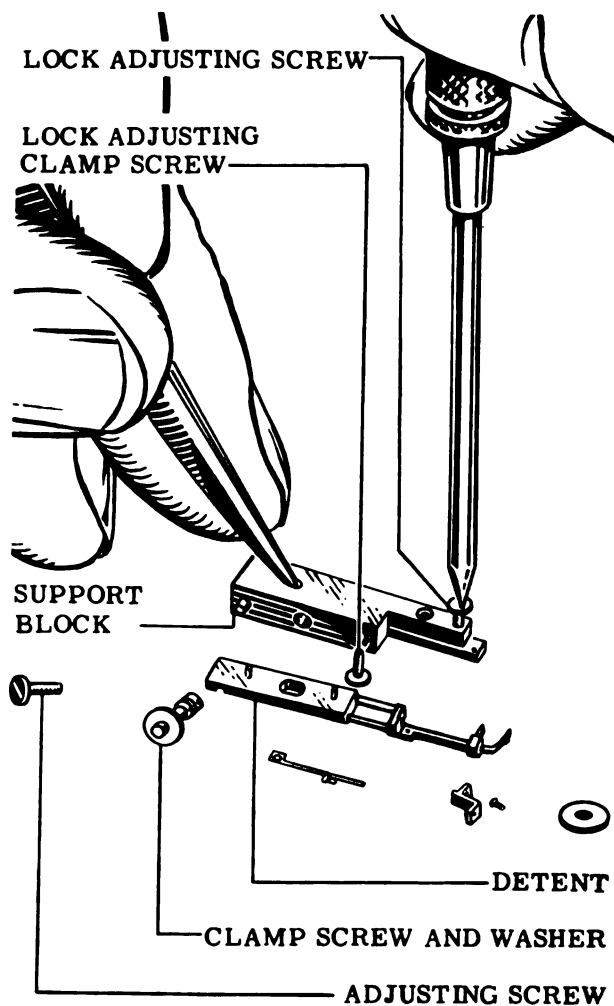


Figure 56

REASSEMBLY OF DETENT

1. Replace detent lock adjusting screw. See Fig. 56 for beginning of reassembly.
2. Replace detent lock adjusting clamp screw.
3. Replace detent adjusting screw.
4. Place detent on detent support block.
5. Replace detent clamp screw and washer. See Fig. 55.

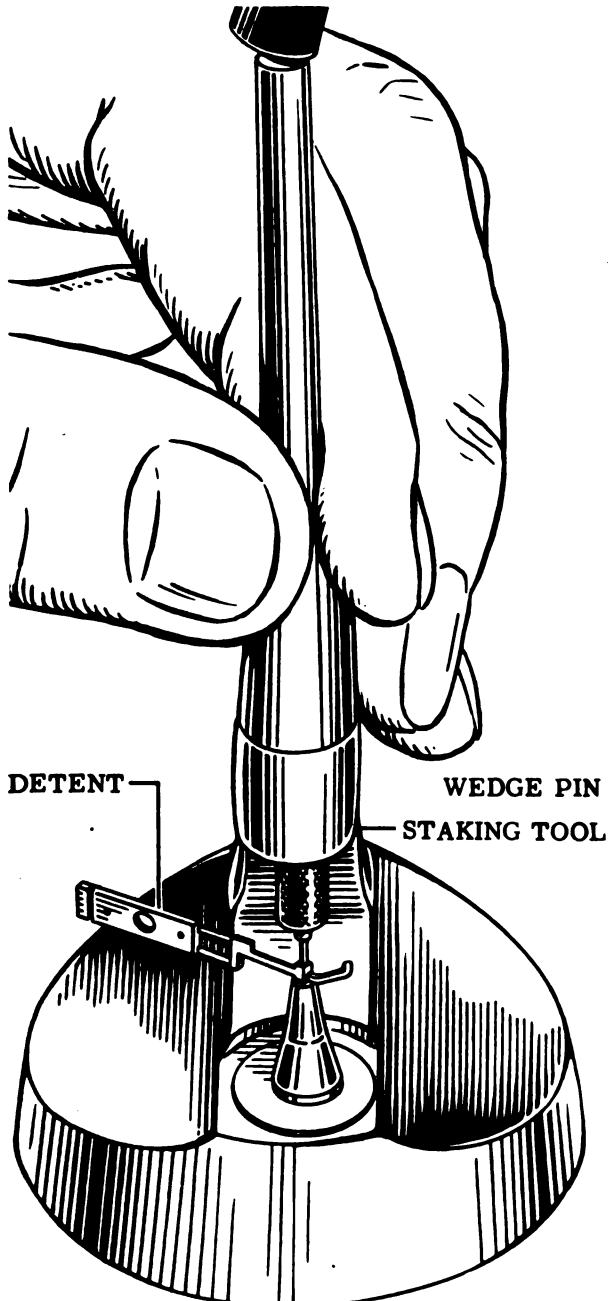


Figure 57

6. Place 0.010 in. thick washer (Tool No. 14) under the detent trip spring support angle bracket, centered on the screw hole for the trip spring support bracket screw.
7. Replace detent trip spring support angle bracket, and screw in the detent trip spring support bracket screw.
8. Place trip spring screw in the hole of the trip spring. Hold trip spring in position and assemble the screw as shown in Fig. 54. Remove washer.

CAUTION

Be sure all parts are in alignment and all screws tight.

RE-JEWELING THE DETENT

1. Drive out old jewel pin, working on the bottom with wedge pin staking tool (Tool No. 9). See Fig. 57.
2. Remove old shellac by scraping and dipping in alcohol.
3. Place detent on locking jewel locating tool. Use Tool No. 41, which is shown in Figs. 58 and 59.

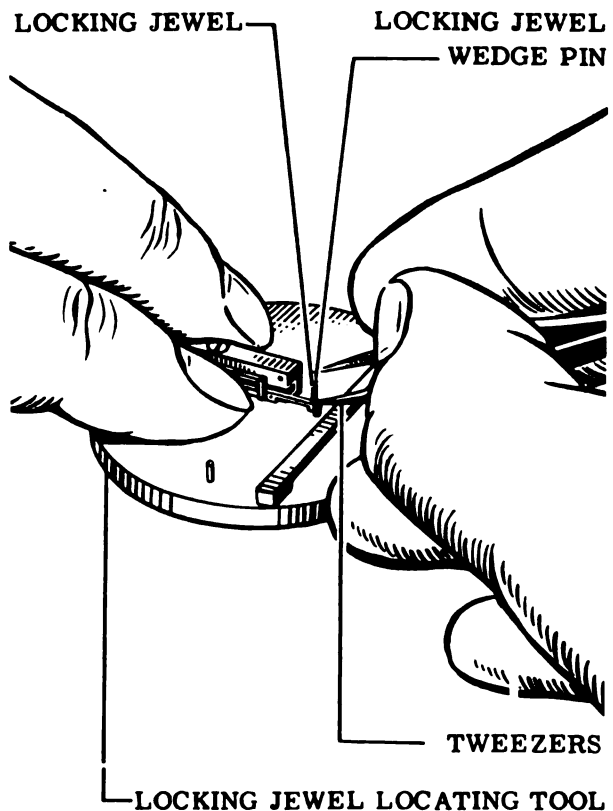


Figure 58

HAMILTON SHIP CHRONOMETER

4. Insert locking jewel, with the large flat facing the locating bar of the fixture.
5. Swing the locating arm of the fixture to line up the jewel at the correct angle.
6. Insert locking jewel wedge pin lightly. With tweezers, mark pin just above the detent arm. See Fig. 58. Place pin in pin vise. With end-cutting pliers (Tool No. 77), cut off pin about 0.020 in. above the mark. Stone cut end smooth. Remove pin.
7. Line up locking jewel as in Operation 5.
8. Replace wedge pin and push in with tweezers so pin is flush with top of detent arm. Check the alignment of the large flat. Adjust, if necessary.
9. Remove detent from locating fixture. Turn over. With end-cutting pliers (Tool No.

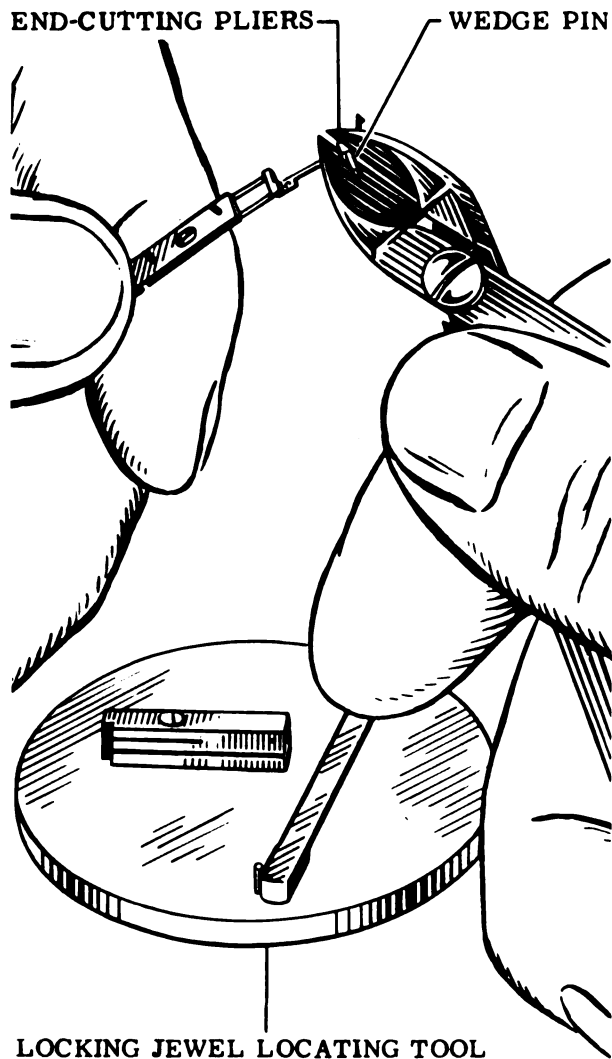


Figure 59

77), cut off excess of wedge pin. See Fig. 59. Stone smooth.

CAUTION

No more heat should be used in Operation 10 than necessary to flow the shellac.

10. Heat detent by holding on heating fixture, Tool No. 67. See Fig. 60. Insert thread of jewel-setting shellac at the bottom and top of the detent so it flows around the locking jewel.

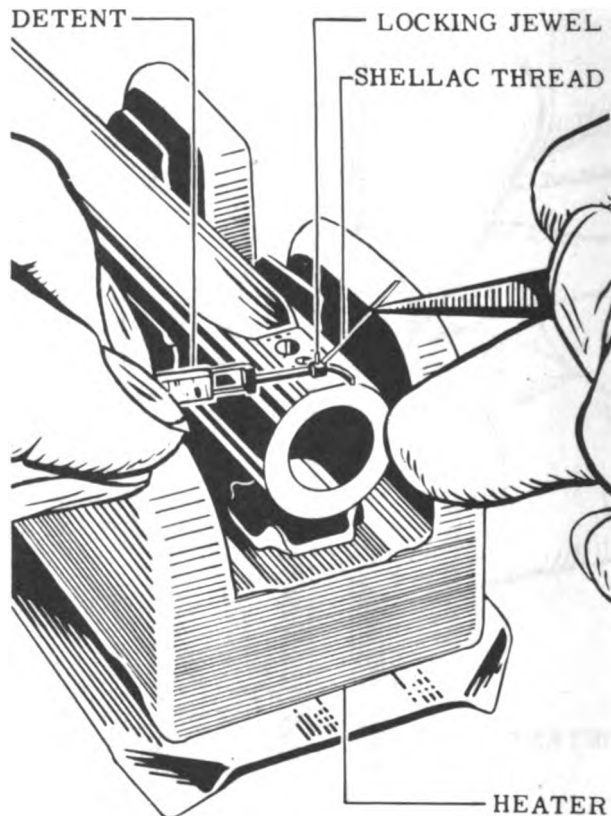


Figure 60

11. Remove excess shellac by stoning the bottom and picking with pointed nickel wire (Tool No. 92) at the top.

CAUTION

All exposed portions of the locking jewel must be free from shellac or its function will be impaired.

RE-JEWELING IMPULSE ROLLERS

To protect the roller and jewel during re-jeweling operations, tweezers should be used for handling.

1. Clean impulse roller with naphtha and dry in warm air.
2. After drying thoroughly, place roller on rubber mat (Tool No. 28) with the numbered side down, slot to the right.
3. Insert a jewel in the slot, curved side of the jewel toward the operator and flush on the bottom. See Fig. 61. Select the proper size jewel for light push fit with no side shake, by trying several jewels in the slot. Leave the selected jewel in the slot of the roller.

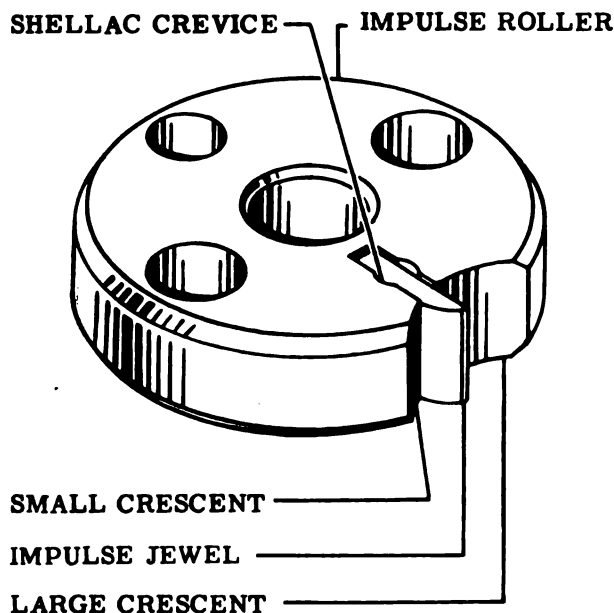


Figure 61

4. Pull the jewel a little to the right to permit shellac to enter behind the jewel. Pick up the roller, with jewel in place, and lightly press it on the roller holding tool. Use Tool No. 100. Place the roller holding tool across the impulse roller jewel heater unit so that the roller is barely touching the heater. The heater is Tool No. 6.
5. Apply a shellac thread to the warmed jewel, back and forth across its upper surface.
6. Take the roller holding tool from the heater unit and tilt up the outer end of the jewel to permit shellac to flow down the sides of the jewel.
7. Again, place the tool across the heater unit (Tool No. 6) so that the roller barely touches the heater, with the jewel up and away from the heater.

8. Apply shellac in the shellac crevice on each side of the jewel (see Fig. 61 which shows these crevices before the jewel was tipped up); when the shellac bubbles, remove the roller from the heater and push shellac in at the top and back of the jewel. Be sure there is sufficient shellac on the top and back.
9. Cool in air and remove the roller from the roller holding tool.
10. Set the roller in the impulse roller jewel setting tool (Tool No. 29), keeping the numbered side down. This tool is shown in Fig. 62. Hold the tool with the left hand and tighten the clamping screw so that the roller jewel is toward the operator.
11. Place the setting tool on the heater unit until the shellac softens. Remove from heater and push tipped-up jewel back into normal position. Pull the jewel out slightly. Push downward lightly on each end of the jewel to insure that the bottom of the jewel is flush with the numbered side of the roller.
12. Push the jewel locating slide of the setting tool forward gently as far as it will go to

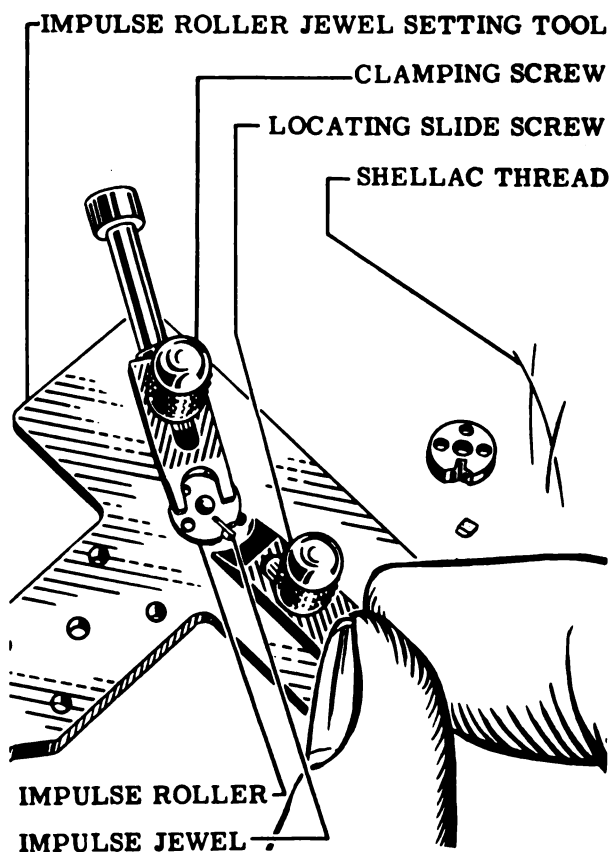
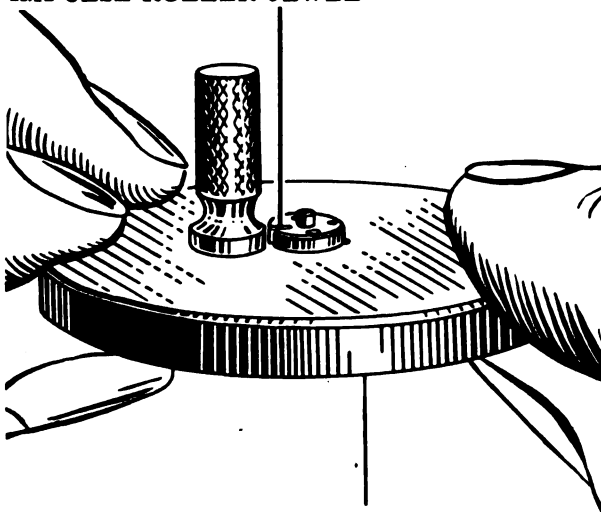


Figure 62

set the jewel in its exact position. See Fig. 62. Quickly tighten the slide screw to hold the jewel.

13. Cool in air. Remove all excess shellac with nickel wire pointer (Tool No. 92), being especially careful to remove shellac from large and small crescents. See Fig. 61. Clean in naphtha with stiff brush; dry in warm air.
14. Place the roller on the impulse roller jewel gage. Use Tool No. 25. See Fig. 63. Check to see that the jewel is perfectly upright with the edge of the roller, that it is flush with the numbered side of the roller, and that it is in perfect concentricity with the outside diameter of the roller. Chipped jewels should be replaced and and cocked jewels repositioned.

IMPULSE ROLLER JEWEL



IMPULSE ROLLER JEWEL GAGE

Figure 63

RE-JEWELING UNLOCKING ROLLERS

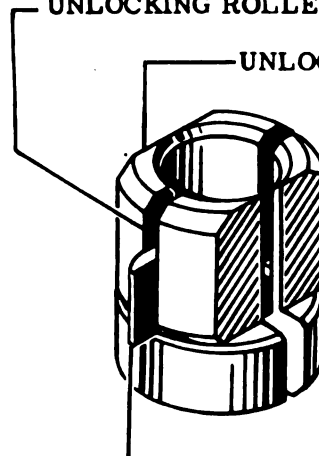
To protect the roller and jewel during re-jeweling operations, tweezers should be used for handling.

1. Clean unlocking roller with naphtha and dry in warm air.
2. After drying thoroughly, place roller on rubber mat (Tool No. 28) with the wider slot facing the operator.
3. By trial, select a jewel with light push fit and no side shake, and insert it in the slot at the left of the operator. See Fig. 64. Use unlocking roller jewel inserting tool (Tool No. 99), to push the jewel down all the way in the slot.

4. Lightly press the roller on the roller holding tool. Use Tool No. 100.
5. In the slot above the jewel, place a shellac thread as long as the jewel at each side of the jewel.
6. Place the tool on the heater unit (Tool No. 100), so that the roller barely touches the heater. Heat until the shellac bubbles.
7. Cool the roller in air and clean the excess shellac from the inside of the roller with the nickel wire pointer. Use Tool No. 92.

UNLOCKING ROLLER JEWEL SLOT

UNLOCKING ROLLER



UNLOCKING ROLLER JEWEL

Figure 64

8. Place roller in unlocking roller jewel setting tool (Tool No. 27) with the wider slot of the roller toward the operator. Tool No. 27 is shown in the operator's fingers in Fig. 65. Insert the tool centering pin in the roller. While holding down the roller with the centering pin, push the clamp on the right side of the tool up to the center. Tighten the clamp nut with the clamp nut wrench supplied with the setting tool.
9. Place the setting tool and roller on the heater unit and heat until the shellac softens. With the finger, push the handle of the jewel adjusting slide forward gently as far as it will go to set the jewel in its proper position. See Fig. 65.
10. Still keeping the finger on the adjusting slide handle, take the setting tool off the heater unit and cool in air.
11. Remove the roller from the setting tool and place it on the unlocking roller jewel gage (Tool No. 26). This gage is shown in the background in Fig. 65. Place the gage centering pin through the roller. Push the

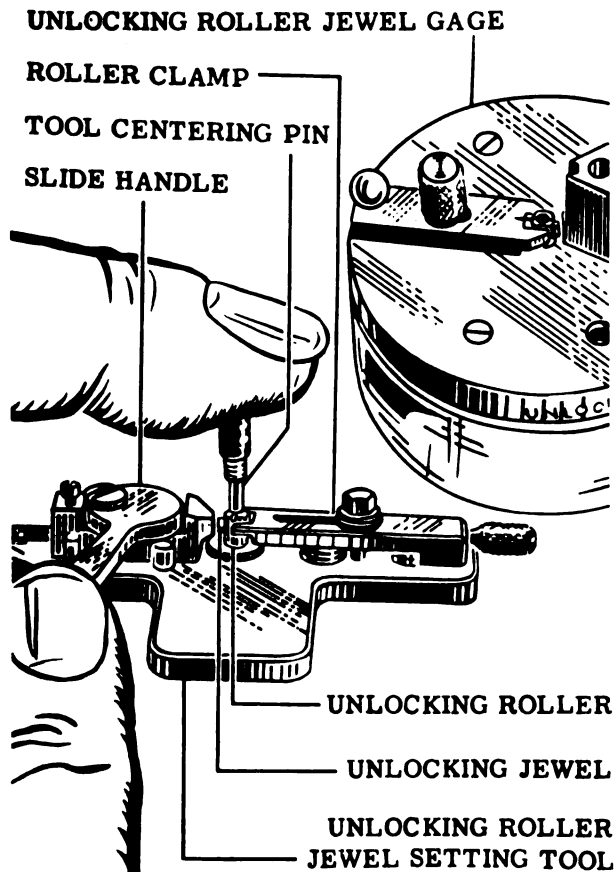


Figure 65

clamp on the gage forward, up to the center; while holding down the roller with the centering pin, tighten the knurled clamp nut. Remove the centering pin. Check the jewel with the gage face. See Fig. 66. The gage face should pass the jewel freely without any drag or any light showing between the stone and the gage. Check to be certain that the jewel is perfectly upright.

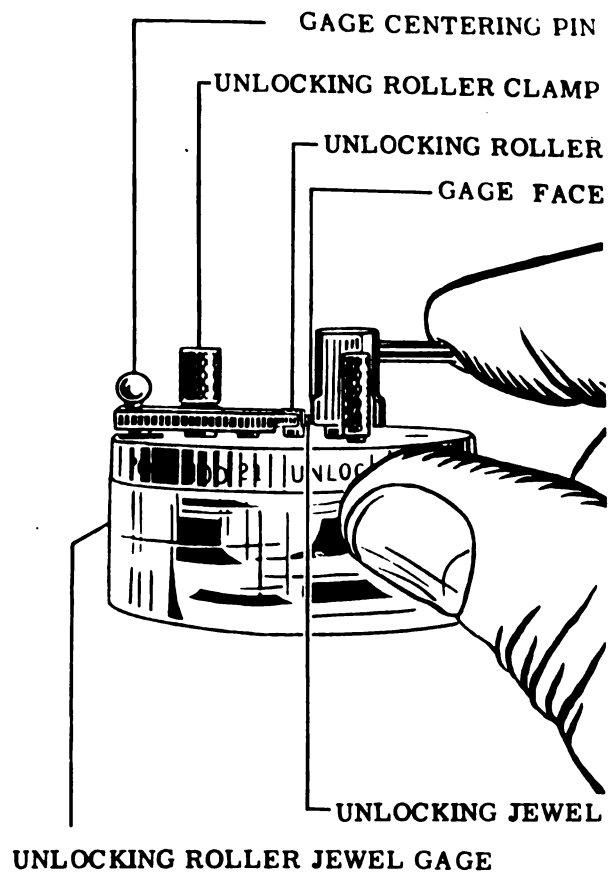


Figure 66

Chipped jewels should be replaced and cocked jewels repositioned.

12. Place the roller on the roller holding tool (Tool No. 100). With nickel wire pointer (Tool No. 92), clean off excess of shellac on each side of the jewel, leaving a little on each side and on top in the slot. Clean the roller in naphtha and dry in warm air.

SECTION VIII

REASSEMBLY OF CHRONOMETER

Using clean parts in perfect condition, re-assembly of the chronometer becomes a matter of carefully following the detailed instructions in this section. Special tools, where required, are indicated in the text. These may also be identified in Section XIII of this book, "Service Tools."

Lubrication

Specific instructions regarding lubrication, as to type and where needed, are noted in the reassembly operations. Do not use any lubricant unless specifically required by these instructions. Three types of lubricant are used.

- a. Red Oil
Hamilton No. 47 Red Oil
- b. Argon Oil
Atlantic Refining Argon "R" Mineral Oil
- c. Grease
Hamilton T-324 Grease

Do not wipe to remove oil. A smear from any cause will draw oil away from the lubrication point. Use a covered oil cup and be sure oil is free from dust, lint, metal chips or other foreign matter. The tip of the oiler should always be cleaned by inserting in clean pith. Be extremely careful of the amount of oil or grease used in all operations calling for lubrication.

CAUTION

Instructions in this section are adequate for qualified personnel, both as to assembly and lubrication. This work should be performed only by repairmen experienced in the watchmaker's trade. The usual precautions in such work are also applicable in chronometer work.

REASSEMBLY, MAIN MOVEMENT

1. (See notes following Operation 15 and Operation 34.) Place mounting ring on work ring, Tool No. 98. Place pillar plate on mounting ring. Screw in three mounting ring screws, using shell screwdriver, Tool No. 96. Fig. 67 shows the mounting ring in its position relative to the pillar plate.

CAUTION

All screws in a chronometer must be secured tightly in place.

2. Turn work ring over and insert the minute wheel post and the wind indicator wheel post, both shown in Fig. 67.
3. Hold posts in place with tissue, turn over and screw in two post screws. Screws are illustrated in Fig. 67.
4. Turn movement over and install barrel bridge pillar. Fasten screw from below. Use Tool No. 96. Fig. 67 shows a train bridge pillar, similar to and opposite in location to the barrel bridge pillar.
5. Install three pillars for upper train bridge. Fasten screws, using shell screwdriver. Use Tool No. 96. See Fig. 67.
6. Place the escape lower endstone cap (number 3) in its recess and screw in the two endstone cap screws with screwdriver, Tool No. 90. Refer to Fig. 67 for proper location of parts.
7. Install the lower train bridge. Screw in the two lower train bridge screws. Tighten screws with large screwdriver. Use Tool No. 94. See Fig. 67 and Fig. 68.
8. Turn movement over and place third wheel, center wheel, and fourth wheel in their pivot holes. See Fig. 67.
9. Install the sustaining pawl. Make reference to Fig. 67.
10. Screw the train blocking screw into the balance lower bridge with screwdriver (Tool No. 84), so that the cone part of the screw is below flush with the top surface of the balance lower bridge. Refer to explosion of this screw in Fig. 67.

NOTE

The use of the train blocking screw started with chronometer No. 3500. All chronometers after 4002 were equipped with train blocking screws.

REASSEMBLY OF CHRONOMETER

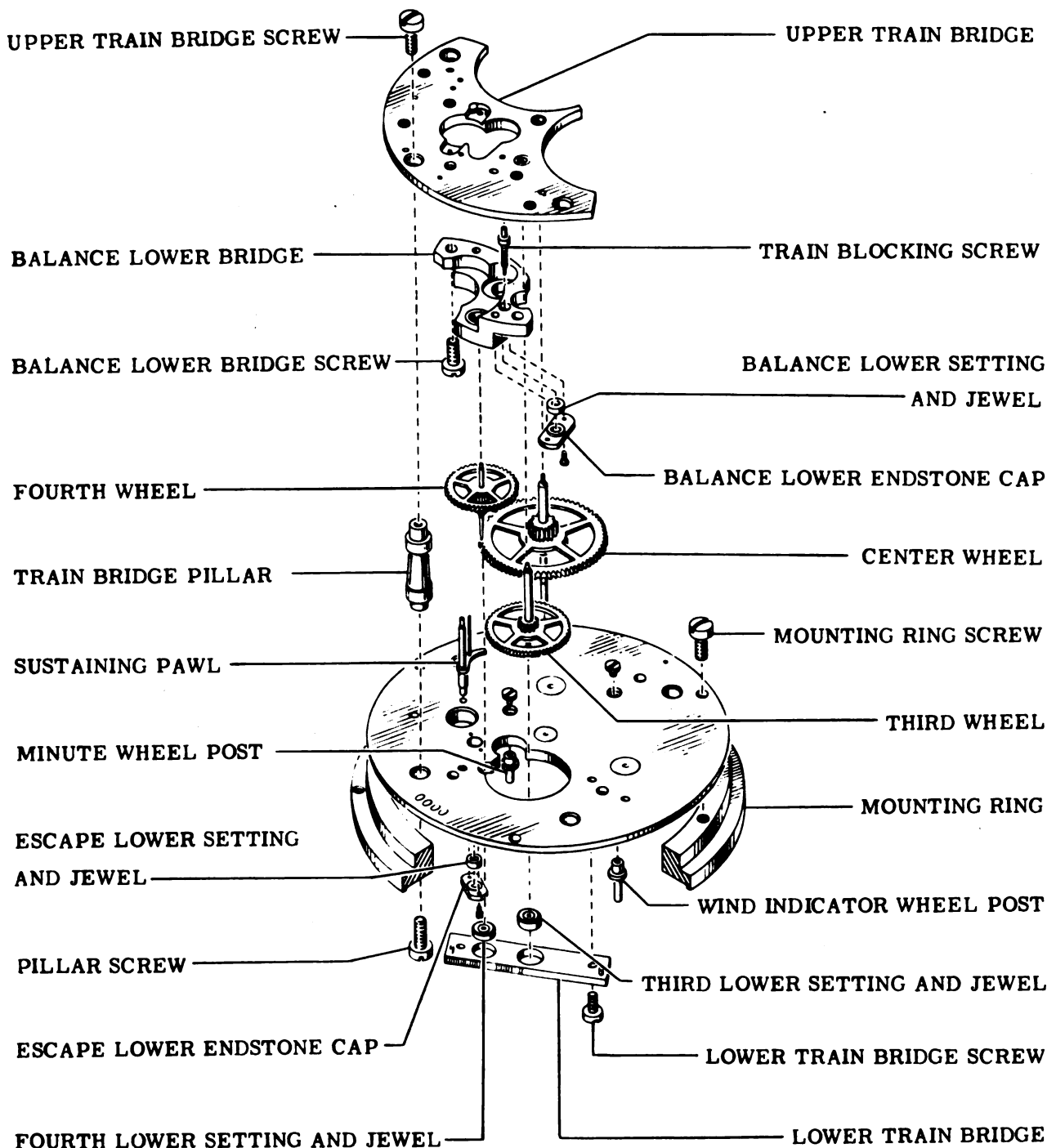


Figure 67

11. Turn balance lower bridge over and install balance lower endstone cap (number 2) with two cap screws. Use Tool No. 90. Refer to Fig. 67.

12. Subassemble the balance lower bridge to the upper train bridge with two screws, using shell screwdriver, Tool No. 96. See the operation being performed in the lower half of Fig. 30, Section V.

13. Turn the upper train bridge over and back-up the train blocking screw to provide installation clearance. Refer to Fig. 67.

CAUTION

In lowering the upper train bridge, Operation 14, be sure all pivots are properly in their holes and wheels and pinions are in mesh.

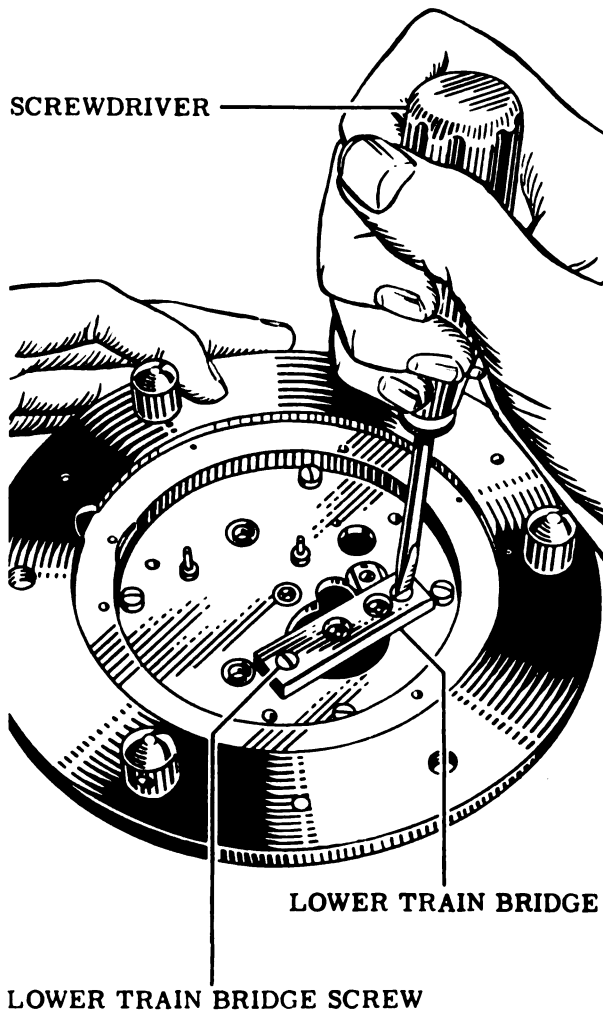


Figure 68

14. Place the upper train bridge over the pillar plate. Slowly and carefully lower it into position. Screw in three pillar screws for train bridge. See Fig. 67 for relative positions of the parts.

NOTE

The endshakes to be checked in train reassembly (see Operation 15) may be adjusted, if necessary, by pressing one of the bushings or jewels in or out, as required. Use Tools Nos. 1 and 2. The sustaining pawl is an exception, since it has no bushing, and must be replaced if endshake is unsatisfactory at the time of overhaul. The train bridge may have been undercut to provide more endshake for the sustaining pawl at the time of manufacture, as will be noted on some of the instruments returned from service.

15. Check endshake of fourth, third, and center wheels and sustaining pawl. These shakes should be from 0.001 to 0.003 in.

NOTE

The following Operations, 16 through 34, are customarily performed before Operation 1. To avoid confusion, they are set forth here in the reassembly at the point where the fusee assembly is installed in the movement.

16. Place fusee, complete with arbor, square end down on fusee assembly block, Tool No. 53. See Fig. 69 for proper position.

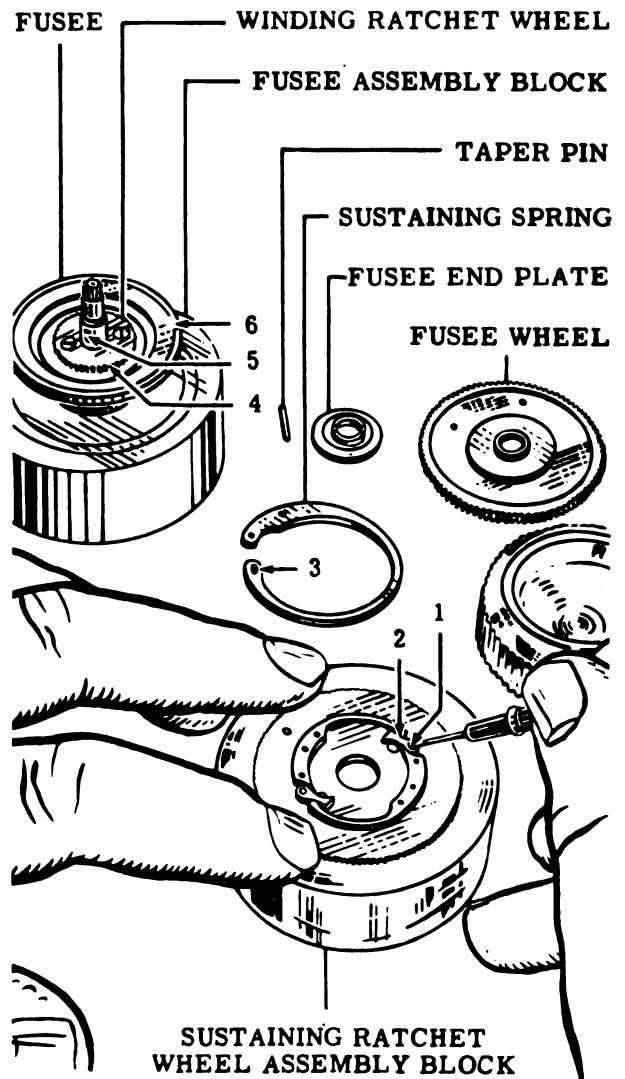


Figure 69

17. Attach fusee winding ratchet wheel to fusee with two screws. Fig. 69 shows the ratchet assembled. Also see Fig. 70.

REASSEMBLY OF CHRONOMETER

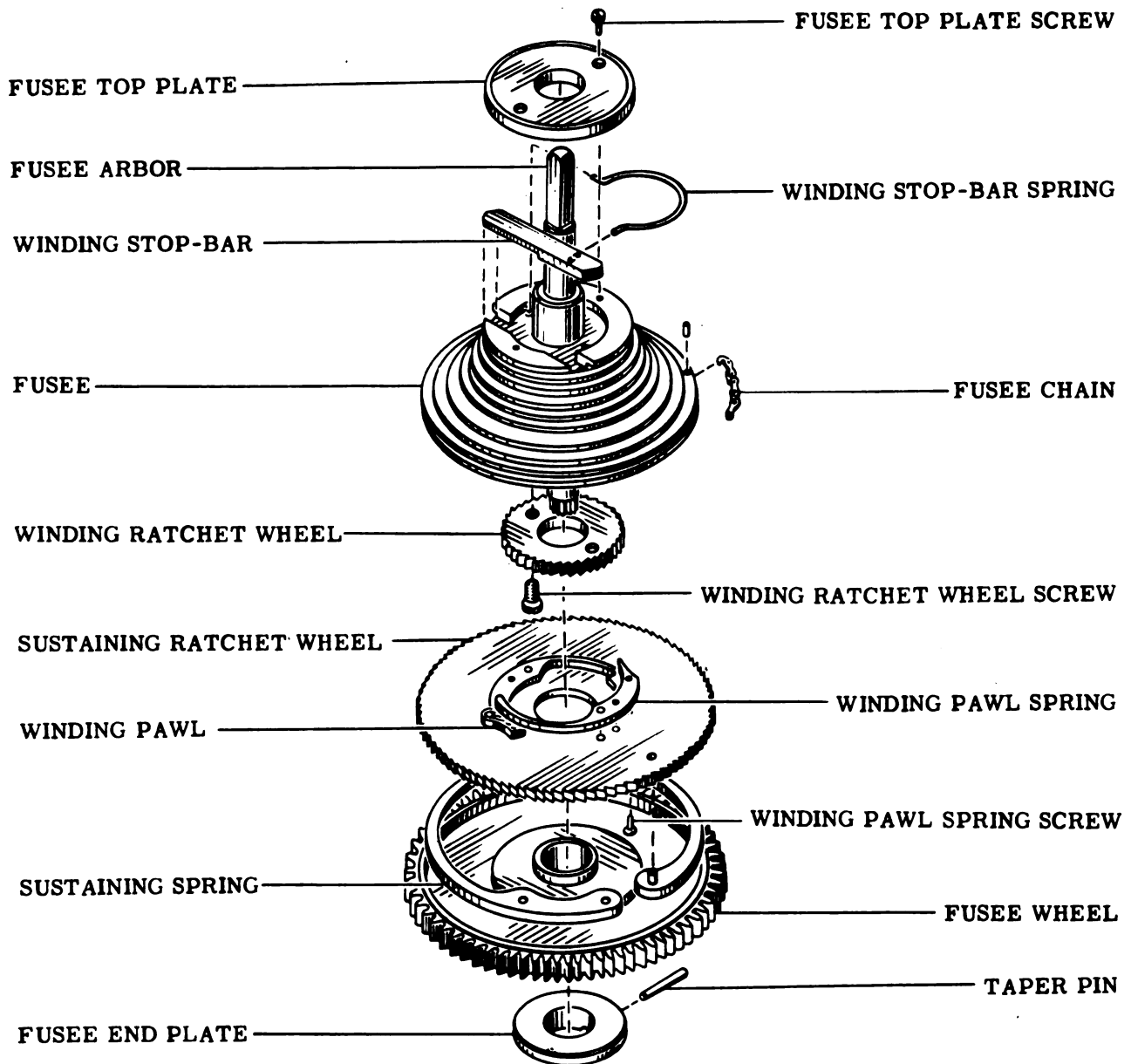


Figure 70

13. Assemble winding pawl springs to sustaining ratchet wheel with four winding pawl spring screws, using sustaining ratchet wheel assembly block, Tool No. 51. Apply drop of red oil to each post of winding pawls as shown in Fig. 69, arrow No. 1. Grease pawl springs where they contact the pawls. Arrow No. 2.
19. Place spot of grease on the bottom of the working end of the sustaining spring. See Fig. 69, arrow No. 3. Locate sustaining spring in fusee wheel. See Fig. 71 for proper position. Also refer to Fig. 70.
20. Put six drops of grease on the lip of the fusee wheel (see Fig. 71, arrow No. 1), and a small amount of grease on the outside of the hub (see arrow No. 2).
21. Lightly grease the teeth of the winding ratchet. See arrow No. 4 in Fig. 69.
22. Grease fusee arbor above ratchet wheel. See arrow No. 5 in Fig. 69.
23. Grease elevation at outer rim, on top of fusee (as viewed with fusee on the assembly block). See arrow No. 6 in Fig. 69.
24. Place the sustaining ratchet wheel on the fusee, using care to see that the pawls are properly positioned. See Fig. 70 for relationship of parts.

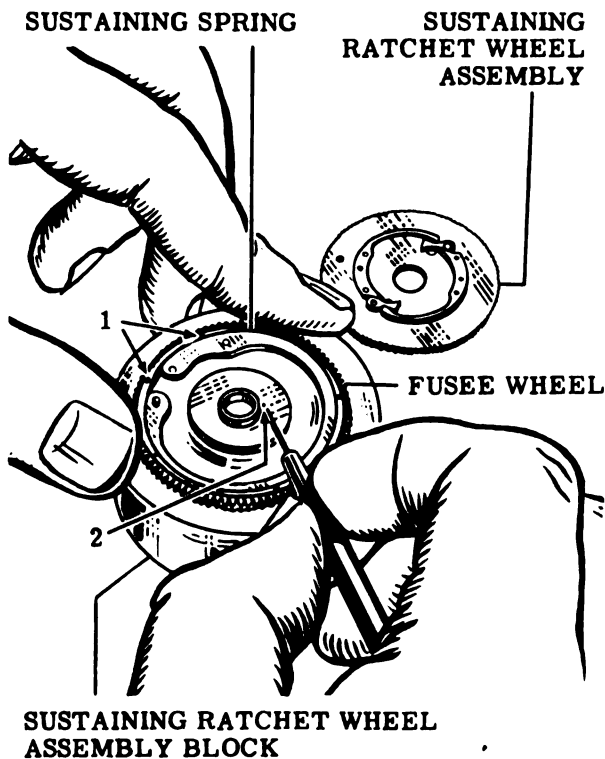


Figure 71

25. Put the fusee wheel in place on the fusee. See Fig. 70.
26. Grease both elevations in the recess of the fusee wheel. See Fig. 72 for location of grease spots.

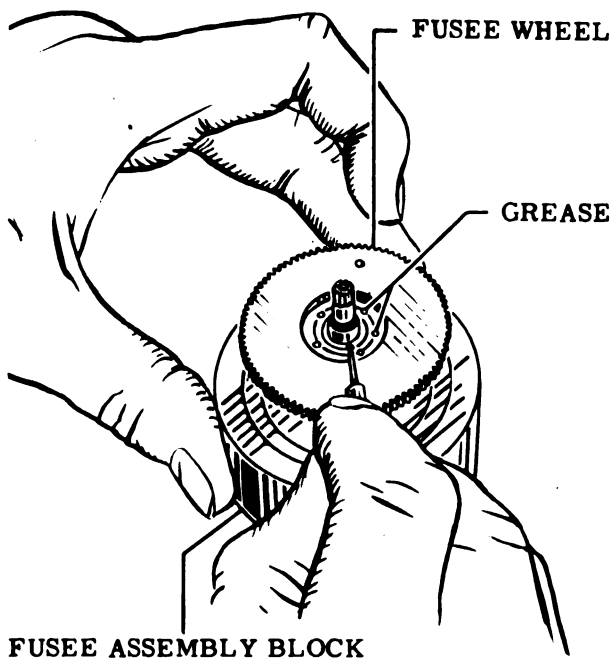


Figure 72

27. Install fusee end plate. Refer to Fig. 70.

NOTE

Make sure that the end plate notches correspond to the taper of the taper pin. Where new parts are used in the fusee assembly, it may be necessary to file these notches deeper. See Operation 29.

28. Install taper pin, inserting in the arbor hole at the end having larger diameter. Refer to Fig. 70. The end plate is marked on the side from which the taper pin should be inserted.
29. Using the assembly block (Tool No. 53) as a key, assure freedom of motion in a counterclockwise direction. If necessary, file slots in end plate with Tool No. 107. Observe that the sustaining spring operates properly.
30. Turn fusee over in assembly block. Grease bottom of the stop-bar slot.
31. Install winding stop-bar spring and winding stop-bar. Grease top of stop-bar. See grease spots in Fig. 73.

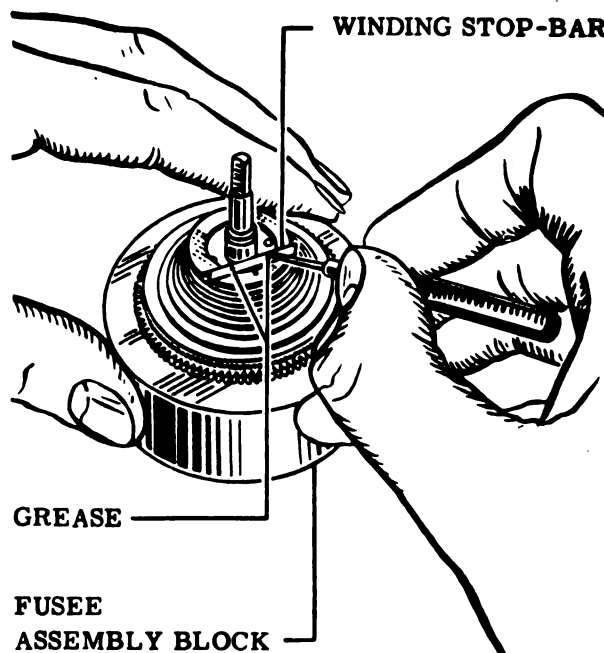


Figure 73

32. Install fusee top plate with two top plate screws. See Fig. 70 for proper position. Check freedom of the winding stop-bar.
33. Wind the chain carefully on the fusee, except for the last turn.

CAUTION

In performing Operation 33, care should be exercised to avoid bending the links of the chain in a direction opposite to that intended.

34. Hold sustaining pawl aside and place fusee assembly in position. See Fig. 74.

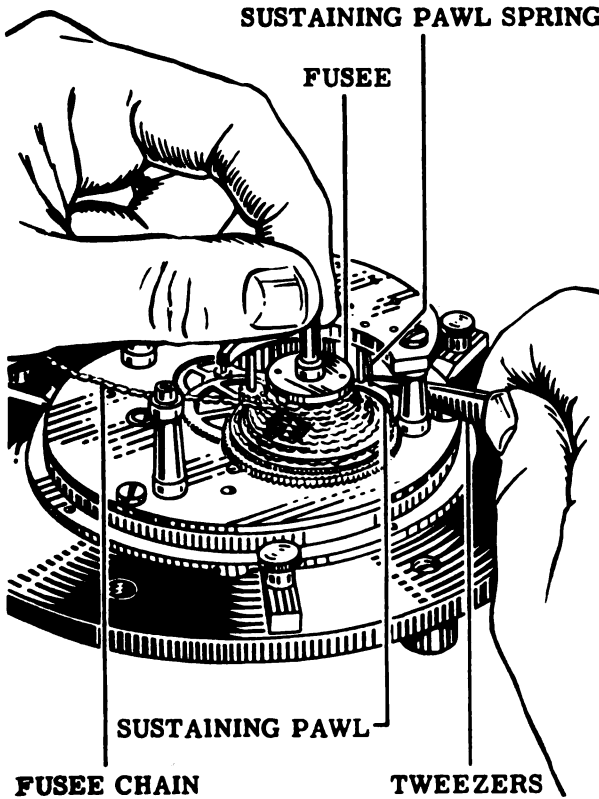


Figure 74

NOTE

The following Operations, 35 through 39, are customarily performed before the main movement assembly is started (usually first). To avoid confusion, these operations are set forth here at the point where the mainspring assembly is installed in the movement.

35. Grease and install mainspring brace in mainspring barrel, making sure that the angle of the ends is correct to prevent slipping. See Fig. 75 for proper relative positions.

CAUTION

In carrying out Operation 36, there is always danger of physical

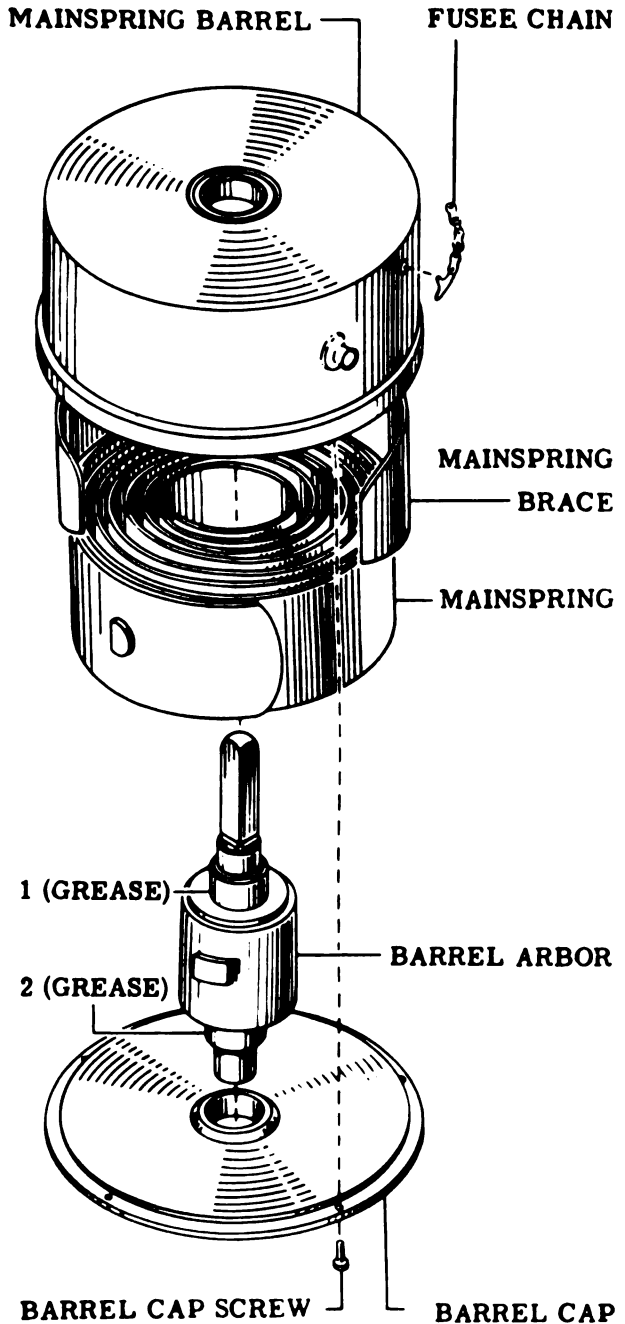


Figure 75

injury unless the spring is tightly held. Use ring to contain spring as it is transferred from mainspring winder to barrel. Use Tools Nos. 56 and 57. See Fig. 76.

36. Install mainspring in barrel, using Tools Nos. 56 and 57, shown in Fig. 76, and applying a thin layer of grease along the length of the spring with syringe, Tool No. 108. Make sure the grease covers the entire flat surface of the spring.

HAMILTON SHIP CHRONOMETER

37. Place a small amount of grease on the two bearing surfaces of the barrel arbor. See arrows Nos. 1 and 2 in Fig. 75.
38. Insert barrel arbor. Make sure the mainspring is hooked on the arbor. Check position with Fig. 75.
39. Place in barrel assembly block, Tool No. 55. Install barrel cap in the position shown in Fig. 75. The five barrel cap screws must be tightly screwed in place.
40. Place barrel assembly in its pivot hole in the pillar plate, winding square up.
41. Screw in setup pawl pivot screw into barrel bridge. See Fig. 80 for relative position of parts.
42. Screw in winding stop (left-hand thread). Refer to Fig. 80.
43. Oil upper sustaining pawl pivot with argon oil. See arrow No. 1 in Fig. 80.

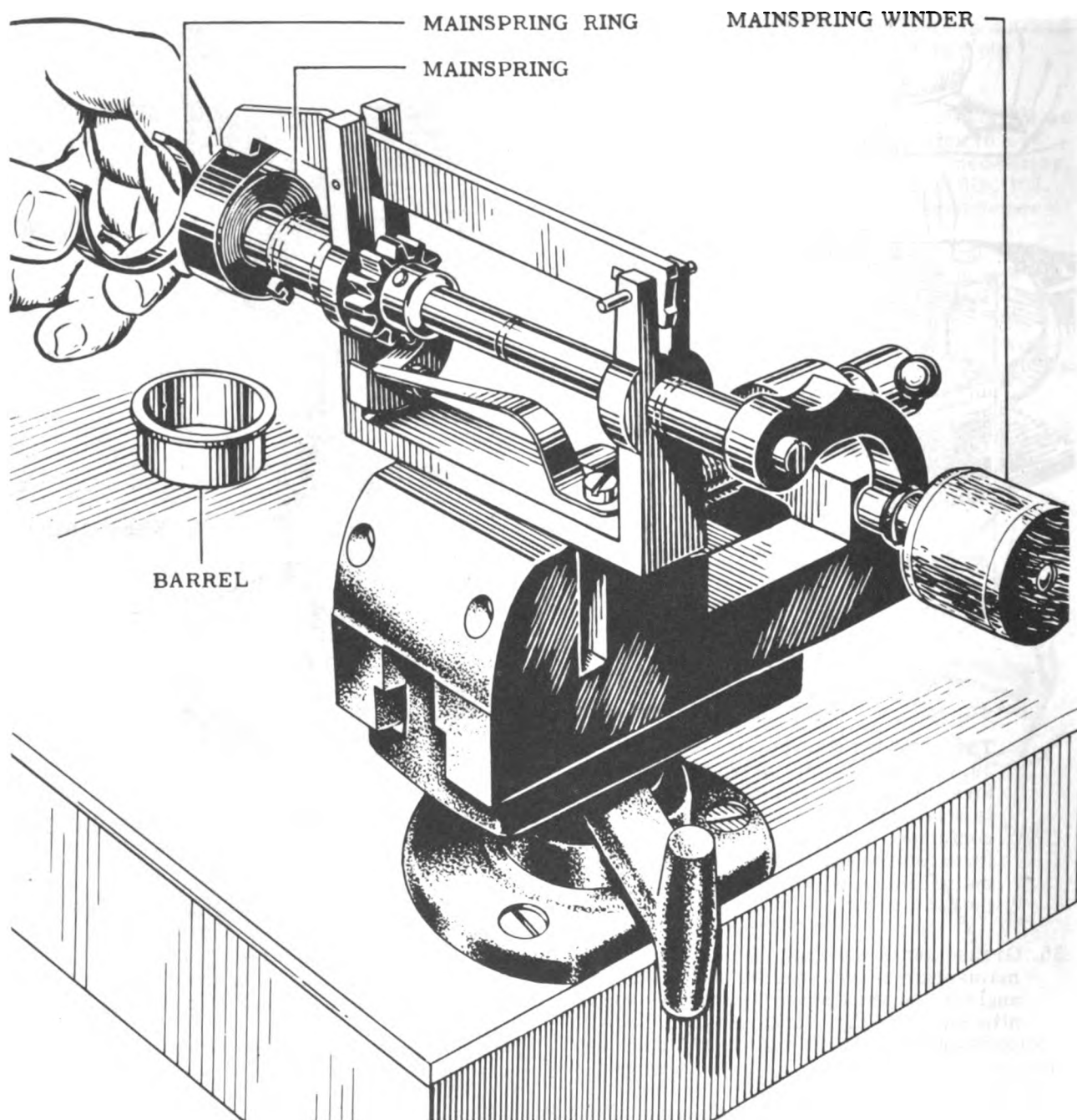


Figure 76

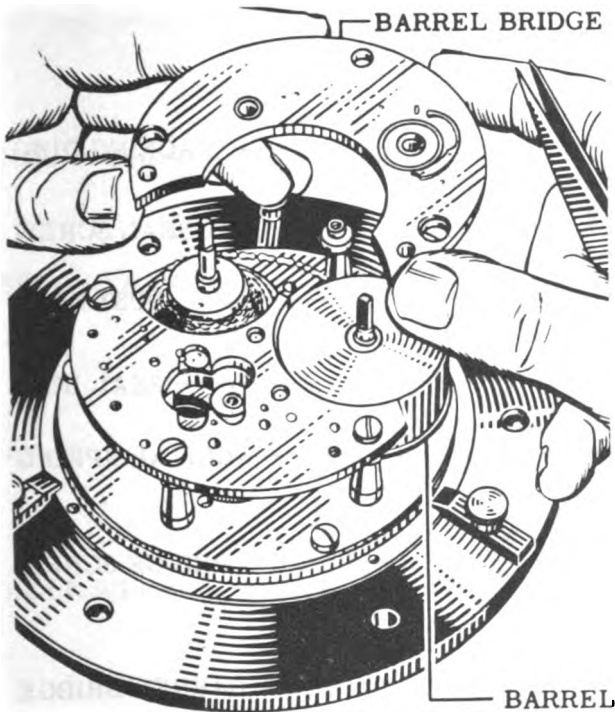


Figure 77

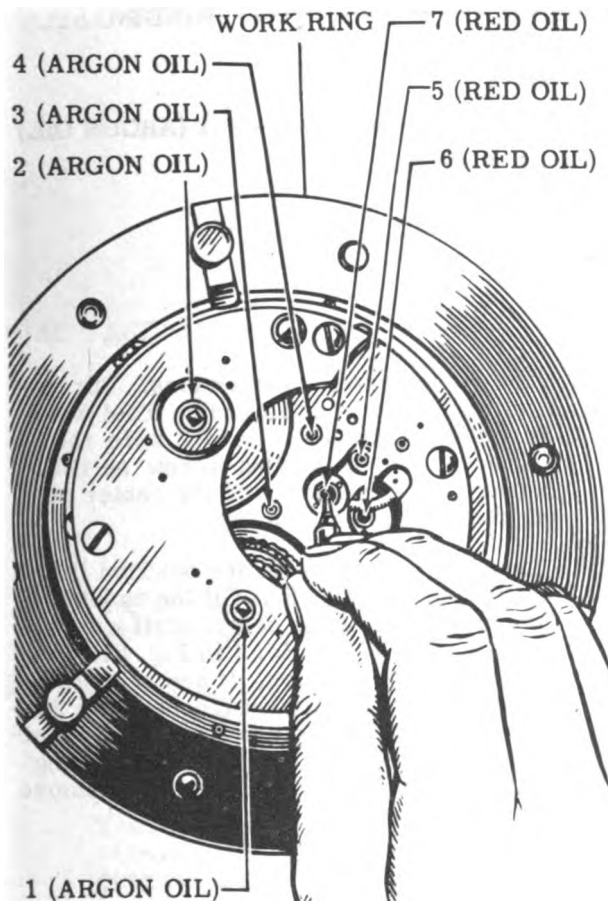


Figure 78

44. Install barrel bridge (see Fig. 77) and fasten three barrel bridge screws, using shell screwdriver, Tool No. 96.
45. Lower the train blocking screw between the arms of the fourth wheel. Refer to Section V, Fig. 16, Operation 5.
46. Using argon oil, oil the fusee upper bushing (arrow No. 1), the barrel upper bushing (arrow No. 2), the center upper bushing (arrow No. 3) and the third upper bushing (arrow No. 4). See Fig. 78.
47. Using red oil, oil the fourth upper jewel (arrow No. 5), the escape lower jewel (arrow No. 6) and the balance lower jewel (arrow No. 7). See Fig. 78.
48. Turn the movement over. Using argon oil, oil the fusee lower bushing (arrow No. 1), barrel lower bushing (arrow No. 2), center lower bushing (arrow No. 3) and sustaining pawl pivot (arrow No. 4). See Fig. 79.

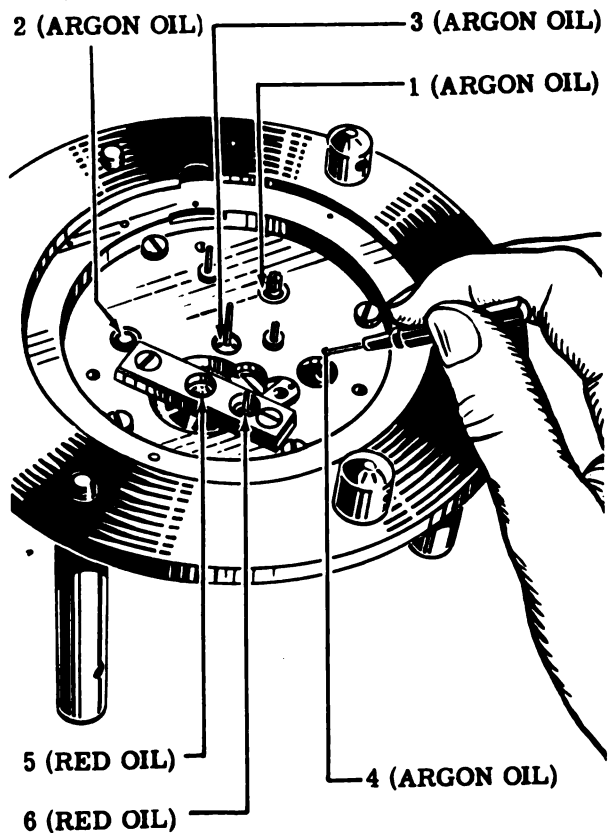


Figure 79

49. Using red oil, oil the third and fourth lower jewels (arrows Nos. 5 and 6). See Fig. 79.
50. Turn movement over. Install the setup ratchet wheel, pawl and pawl spring. See Fig. 80 for relative positions of the parts.

HAMILTON SHIP CHRONOMETER

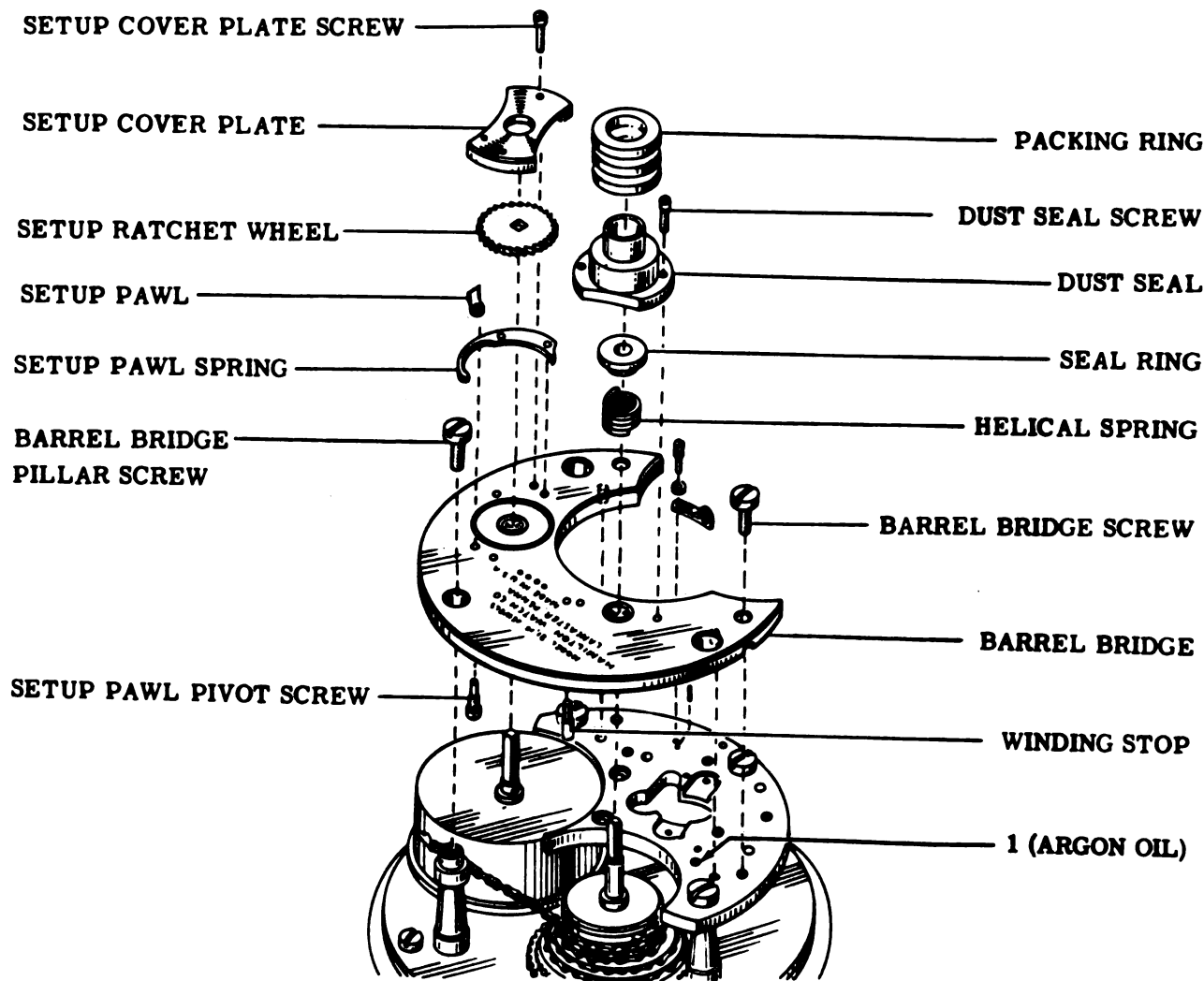


Figure 80

51. Install the setup cover plate with two screws. See Fig. 80.
52. Subassemble new packing rings to dust seal. Install helical seal rings, seal ring and dust seal with two dust seal screws. Refer to Fig. 80 for location.
53. Hook fusee chain on barrel and wind barrel about one turn. Method of fastening is shown in Fig. 75.
54. Wind the chain on the fusee to check operation of the winding stop-bar and the winding stop.
55. Check the chain as installed to make sure it is straight and true, and fits properly on all turns of the fusee.
56. Wind mainspring fully and let down approximately $1/4$ turn.
57. Take movement from work ring. Turn over and place in dialing ring, Tool No. 7. Turn the movement over. Be sure the center pinion supporting screw on the dialing ring fits down onto the center pivot. See position in Fig. 83.
58. Turn movement over. See position in dialing ring in Fig. 82. Oil the cannon pinion bearing of the center staff with argon oil. See arrow No. 1 in Fig. 81. Install the cannon pinion. If new pinion, indent using cannon pinion closer, Tool 33. Check friction of the cannon pinion by putting the minute hand in place and moving it in counterclockwise direction. Remove the minute hand.

NOTE

The cannon pinion should be tight on the staff of the center wheel, but

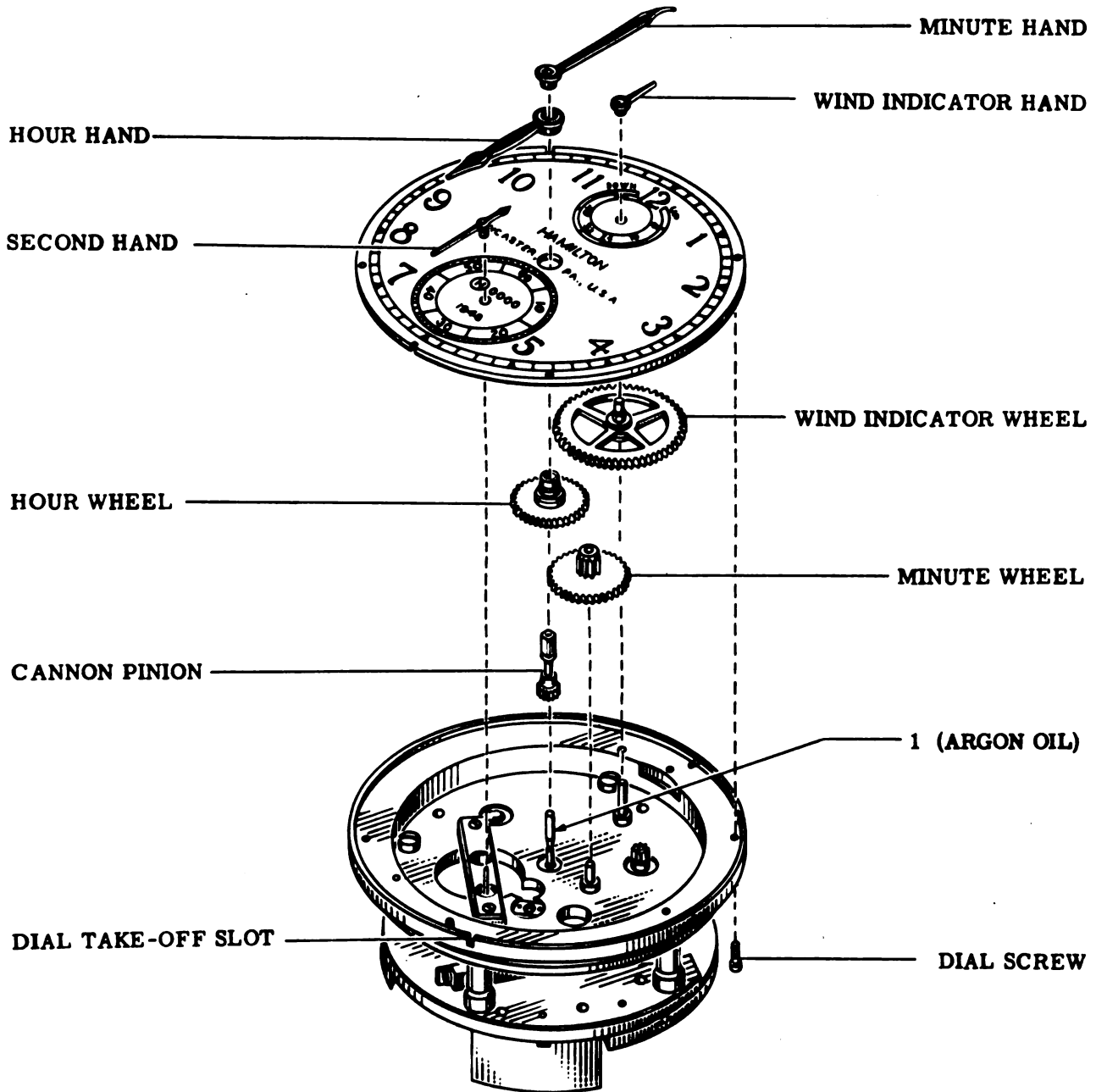


Figure 81

not so tight that it moves the train when being checked in Operation 58.

59. Install the minute wheel, hour wheel, wind indicator wheel as shown in Fig. 81.

NOTE

The hour wheel must be free on the cannon pinion. New wheels may need broaching with Tool No. 39.

60. Place the dial in position. See Fig. 82.

61. Turn the movement over, holding the dial in position with the hand protected by tissue. Insert and fasten four dial screws, using Tool No. 91, as shown in Fig. 83.

62. Turn the movement over. Install the wind indicator hand with pointer in "UP" position, using small punch, Tool No. 37. Refer to Fig. 81.

63. Use the minute hand to position the cannon pinion so that the minute hand points to "12". Remove the minute hand.

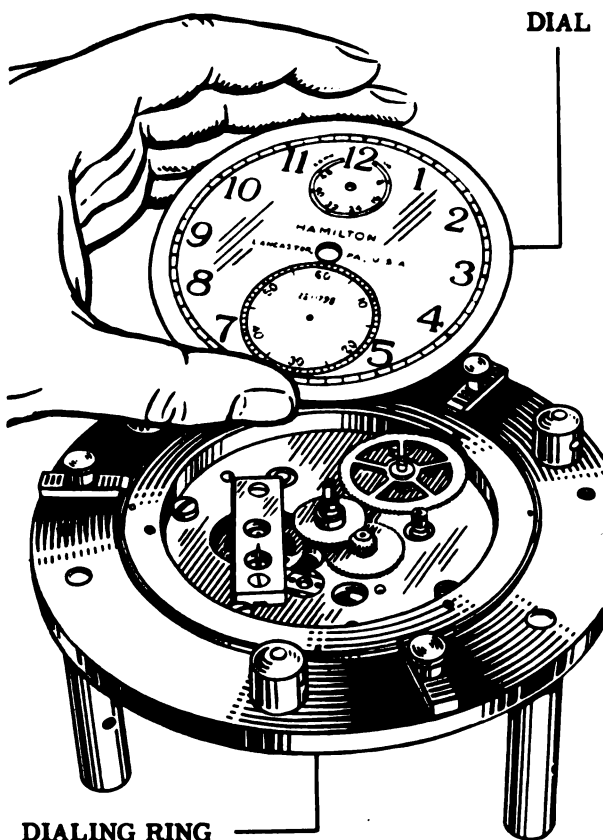


Figure 82

64. Install the hour hand, pointing to "3" on the dial. Use Tool No. 35. Install the minute hand pointing to the "12" position on the dial. Use Tool No. 36.

NOTE

Where a new hour hand is installed, if the fit is too tight, broach with Tool No. 40. The minute hand can be broached with a square file. If either of these hands is too loose, use Tool No. 106 (Chuck No. 1 for the minute hand, and Chuck No. 2 for the hour hand).

65. Remove dialing ring, holding dial with tissue, and place movement on work ring, train side up.
66. Subassemble escape upper endstone cap (number 1) to escape upper bridge with two screws. See Fig. 84 for proper positions.
67. Oil escape upper jewel with red oil. See arrow No. 1 in Fig. 84.
68. Install escape wheel.

CAUTION

Do not use lubricant on escape wheel teeth.

69. Lower escape upper bridge into place and install two escape upper bridge screws. Refer to Fig. 84 for position of parts. Check endshake of the escape which should be 0.001 to 0.003 in.

NOTE

If endshake is too much, adjust by milling off surface of endstone cap. Use Tool No. 31. This operation is like that illustrated in Fig. 86. Press the endstone down on the bridge so that it seats properly, using for this purpose the endstone press, Tool No. 32.

If endshake is too little, select a better fitting endstone cap and press the hole jewel into contact with the endstone. Use press like Tool No. 1.

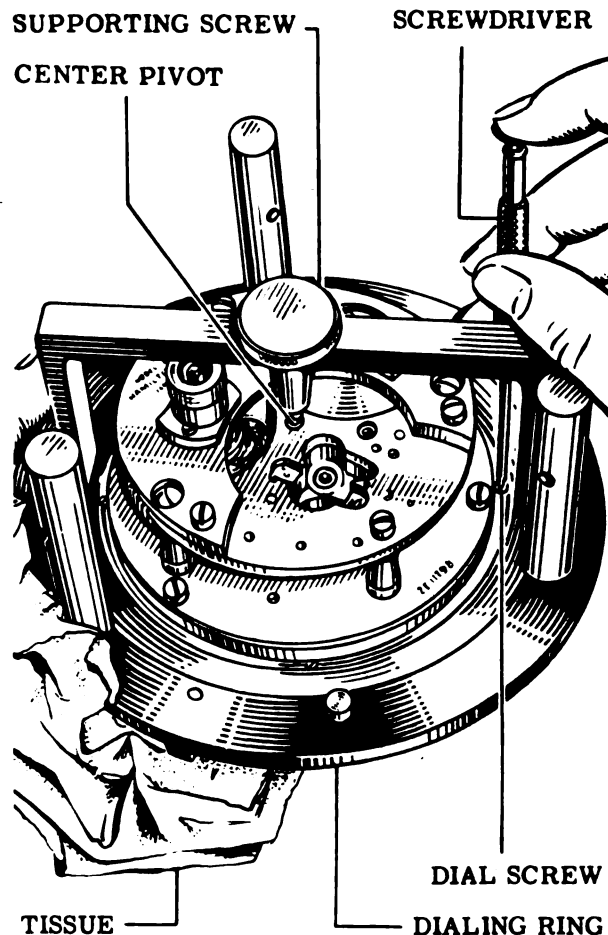


Figure 83

ESCAPE UPPER SETTING AND JEWEL

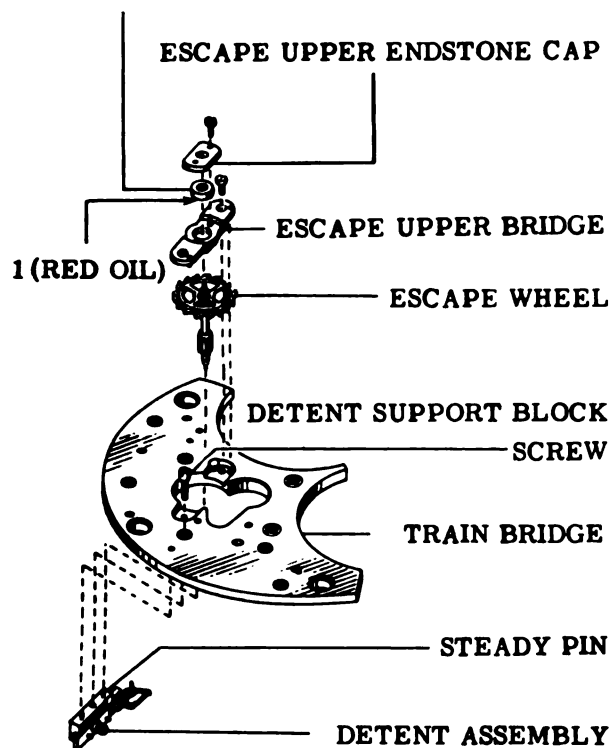


Figure 84

70. Subassemble balance upper endstone cap to the balance cock with two screws. (This is a diamond endstone cap, identified by having both sides flat.) See Fig. 85 for relative positions.
71. Oil the balance upper jewel with red oil. See arrow No. 1 in Fig. 85.
72. Install the balance assembly as a unit. For subassembly of the Balance Unit, see Section VII "Inspection and Repair."

CAUTION

Do not use lubricant on the locking and unlocking jewels.

73. Carefully lower the balance cock into place. Make sure the staff pivot freely enters the jewel hole while guiding the balance staff with tweezers. See Fig. 85.
74. Hold balance cock in position with the fingers, insert and tighten the balance cock screw, making sure that the endshake of the balance staff is sufficient before applying pressure to the balance cock screw. When the balance cock screw is in place, again check the endshake of the balance staff, which should be 0.001 to 0.003 in.

NOTE

If endshake is too much, adjust by milling off surface of endstone cap. Use Tool No. 31. See Fig. 86. Press the endstone down on the bridge so that it seats properly. Use Tool No. 32. If endshake is too little, select a better fitting endstone cap and press the hole jewel into contact with the endstone. Use press like Tool No. 1.

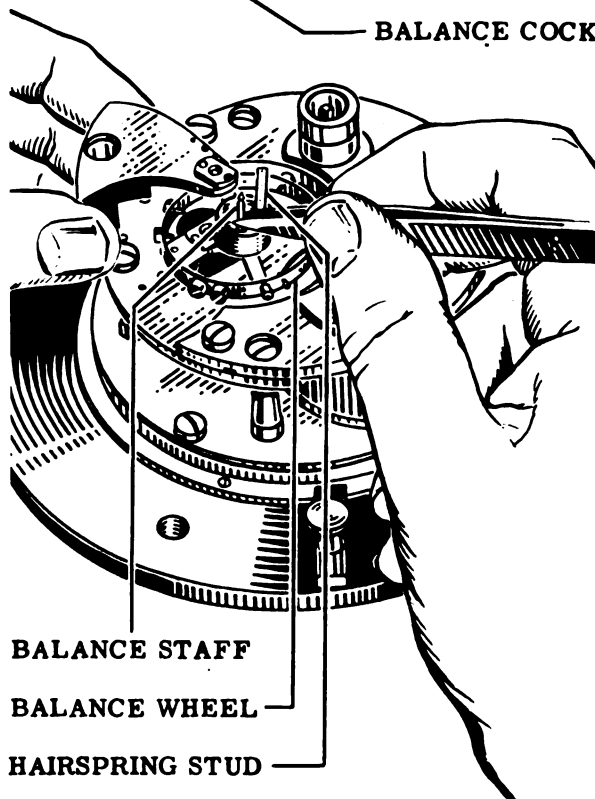
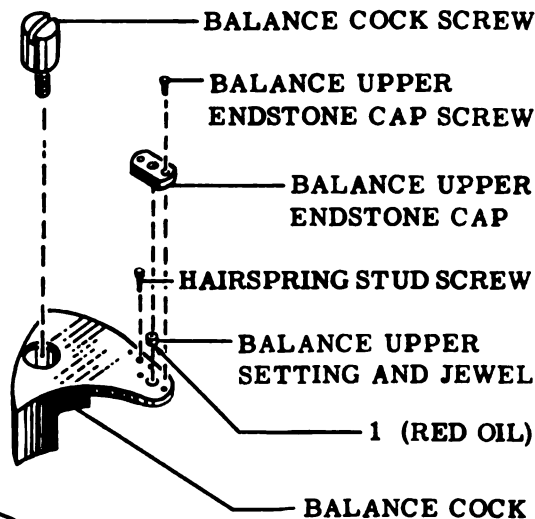


Figure 85

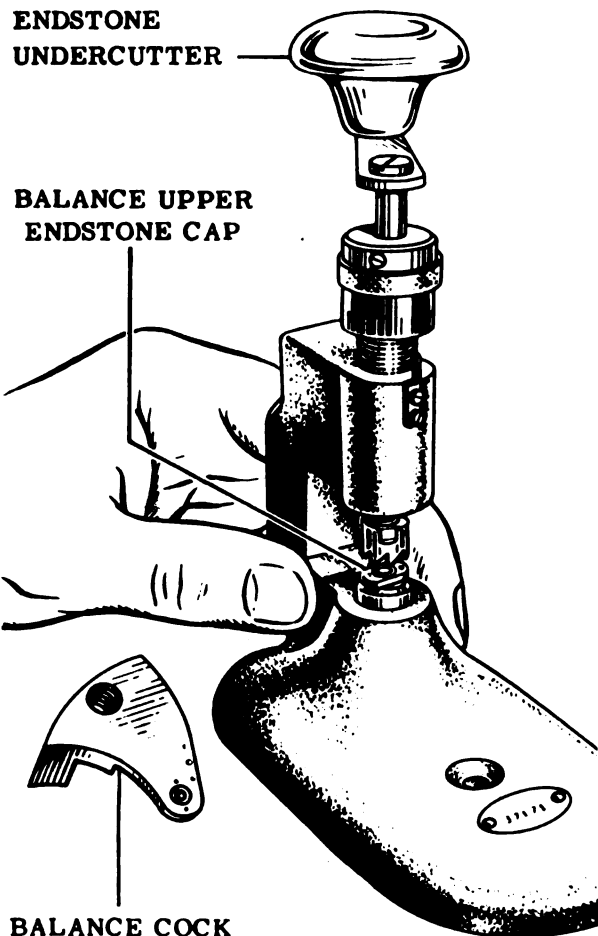


Figure 86

75. Start the hairspring stud screw in its screw hole. Carefully turn the balance unit clockwise until the hairspring stud is in the correct location. Lift the stud into place. Fasten the hairspring stud screw as shown in Fig. 87. Disengage the train blocking screw by backing it up as far as possible and tightening it in this position.
76. Move the balance wheel back and forth to see that each tooth of the escape wheel drops into the large portion of the crescent of the impulse roller where it is arrested by the impulse jewel. If a tooth misses the impulse jewel completely, it is an indication that the jewel is too short or the escape wheel is worn. In such case, replace escape wheel or use larger roller. Check the impulse jewel for uprightness and to see that it is not chipped on the edge. See Section VII, Fig. 38.

NOTE

In all late model chronometers, detents are made of beryllium

copper and trip springs of Hamilton Elinvar. If instruments are presented for overhaul that are otherwise equipped, they should be changed to this combination of metals.

In making adjustments and parts replacements in the detent assembly, incidental to the following operations, consult the disassembly and reassembly operations for the detent in Section VII "Inspection and Repair."

CAUTION

Do not use lubricant on any part of the detent assembly.

77. Turn the movement over. Check detent assembly and screws for rust. Check tightness of all screws. Check for upright jewel and for straightness of detent and trip springs.
78. Check the spring action of detent spring in spring testing fixture, Tool No. 8. Hang weight of 0.770 grams on the locking jewel. See Fig. 88. This weight should separate the detent spring from the stop button

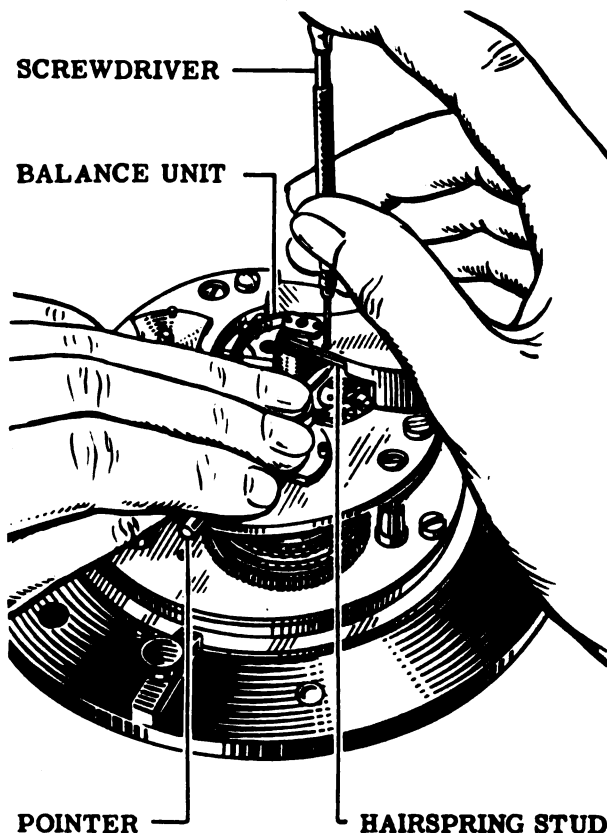


Figure 87

by approximately 0.005 in. If necessary, carefully bend the detent springs until proper adjustment is obtained, making sure that the locking jewel has not been thrown out of upright in the process.

79. Observe the action of the trip spring. Very slight force should be required for flexing. This spring should be set so that it does not fly back violently. It should lie flat and rest lightly on the abutment arm.
80. Insert the detent, wedging it in place with Tool No. 11. See Fig. 89.
81. Turn movement over. Install detent support block screw and remove wedge.

BALANCE LOWER BRIDGE

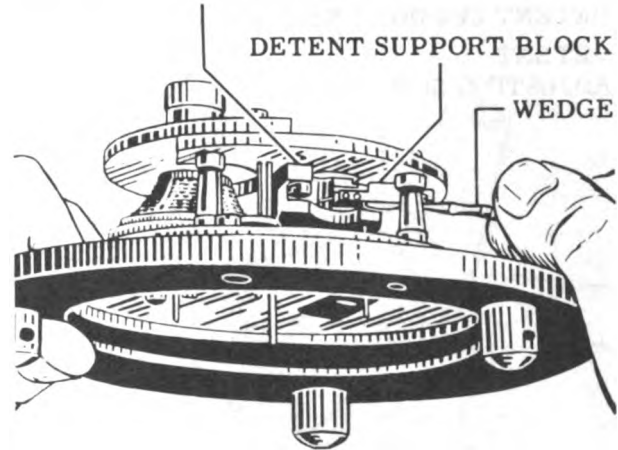


Figure 89

NOTE

Whenever necessary to install the detent assembly, this is done in the manner indicated in Operations 79 and 80.

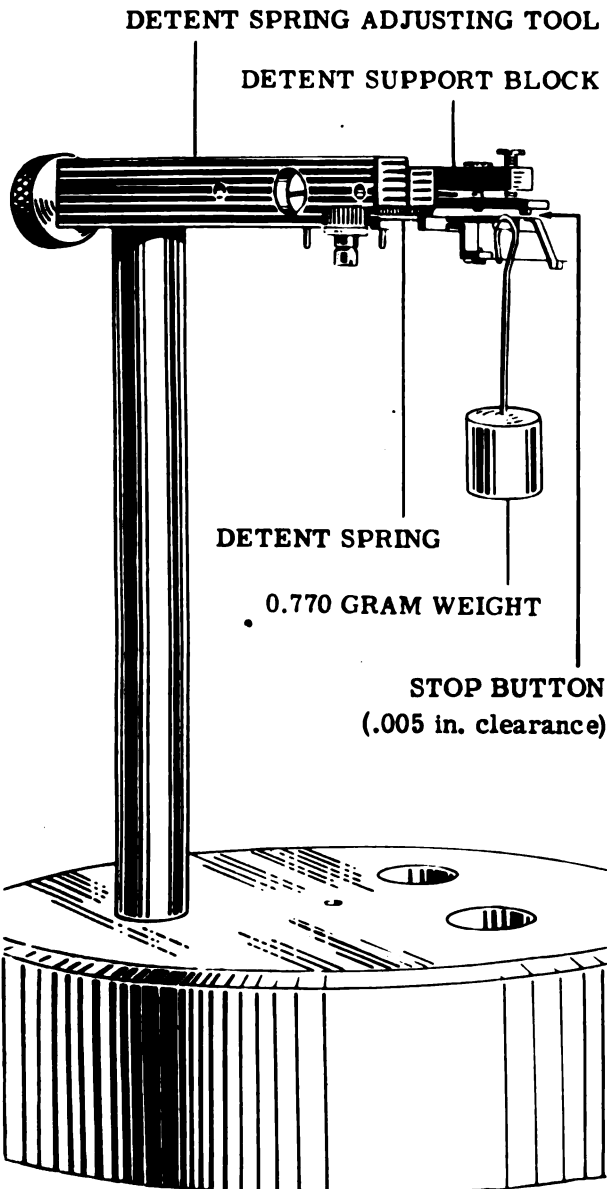


Figure 88

82. Check height of the escape wheel on the impulse jewel. The escape wheel should be high enough to clear the detent, but should be centered on the impulse jewel, with an equal portion of the jewel above and below the escape wheel. See Fig. 14 in Section IV "Principles of Operation."

NOTE

By using the stump and punch of a watchmaker's staking set, the escape wheel on its hub may be driven a small amount up or down on its pinion. This permits centering the escape wheel on the impulse jewel.

83. Check to be sure that the escape teeth are striking the face of the impulse jewel. These teeth should not contact the edge of the jewel or at any time enter the small portion of the crescent. Use spanner wrench to decrease the angular distance between impulse and unlocking jewels if this defect appears. Use Tool No. 42.
84. Check roller shake with special illuminator, Tool No. 104. See Fig. 90. Roller shake is the clearance between the impulse roller and the ends of the two escape teeth nearest to the impulse roller, when the escape wheel is in locked position. This clearance should be about 0.002 in. and approximately equal, or (preferably) be slightly more on the outgoing tooth. If the

HAMILTON SHIP CHRONOMETER

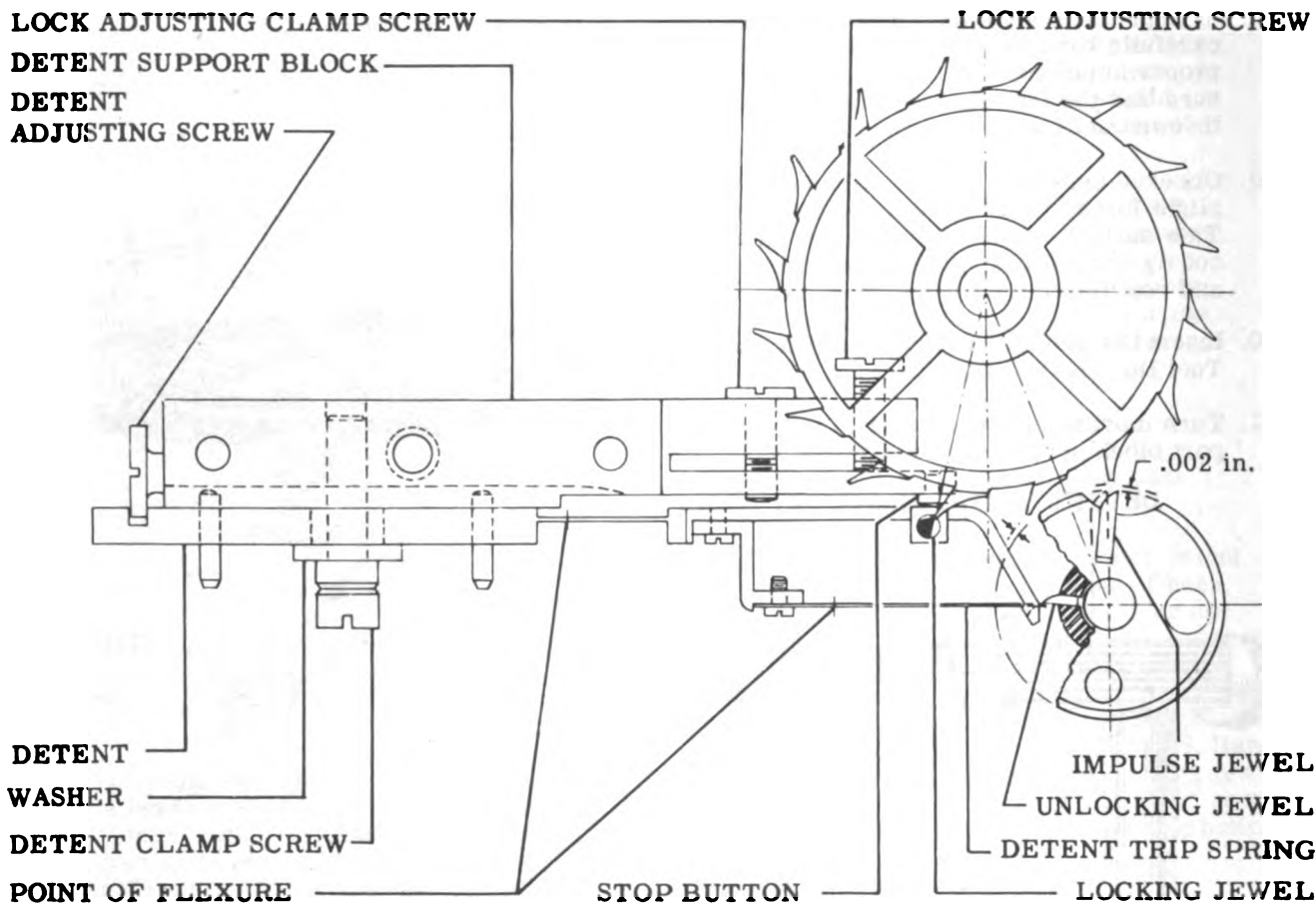


Figure 90

combined shake is appreciably more or less than 0.004 in., select and install a new roller. If about 0.004 in. total clearance, adjust for approximately equal shake by loosening the detent clamp screw and turning the detent adjusting screw. Be sure to tighten the detent clamp screw when this operation is completed.

85. Check the lock on every tooth with special illuminator, Tool No. 104. Place graduated sector (Tool No. 12) on upper train bridge under the edge of balance wheel in the position shown in Fig. 91. Place slotted pin, with mark on the side for gaging (Tool No. 13), near the middle of the rim of the balance wheel, when it is at rest. See Fig. 91. To determine lock, turn balance clockwise with pointer (Tool No. 87) until the detent trip spring returns to its normal position. Then turn the balance counterclockwise until light can be seen between the detent and the stop button. Observe the gage at this point (refer to Fig. 91) and continue to turn the balance until the escape tooth, which is resting on the locking jewel, drops off. Again, observe the gage. The lock should measure

approximately six degrees. Adjust, if necessary, by loosening the detent lock adjusting screw and turning the detent lock adjusting screw. Refer to Fig. 90 for the location of these screws.

NOTE

Where it appears that the trip spring is too short to unlock certain teeth, this failure might be also due to a bent lower pivot on the balance staff.

86. Check the let off (arc traversed from the point of unlock to the point of detent release). To determine let off, turn the balance clockwise until the detent trip spring returns to its normal position. Then turn the balance counterclockwise until the escape tooth, which is resting on the locking jewel, drops off. Observe the gage reading from this point until the detent falls back into its normal position. The let off should be a minimum of six degrees. If less than six degrees, this indicates that the trip spring is too short, and will be corrected in Operation 87. In special

cases, the fault may lie in misalignment of the detent. If trip spring is too long, this will be adjusted as a result of performing Operation 87.

87. Check detent trip spring for correct length by observing overall travel. Overall is the balance arc equivalent to the total distance covered by the trip spring. To determine overall, turn balance clockwise until the trip spring falls off the unlocking jewel and returns to its normal position. Stop the balance with Tool No. 87 at this point and observe gage. Turn balance counterclockwise until the detent falls back into its normal position. Observe the gage. The difference in readings should be from 26 to 30 degrees. If necessary to adjust trip spring length, this will be done in Operations 90 and 91.

88. Check the unlocking jewel for uprightness. Check clearance between detent abutment arm and unlocking jewel. The abutment arm should clear by about 0.010 in.

89. Remove detent by inserting wedge (Tool No. 11) and removing detent support block screw. Then turn movement over and straddle detent with tweezers. Remove wedge and detent.

NOTE

Whenever necessary to remove the detent assembly, this is done in the manner indicated in Operation 89.

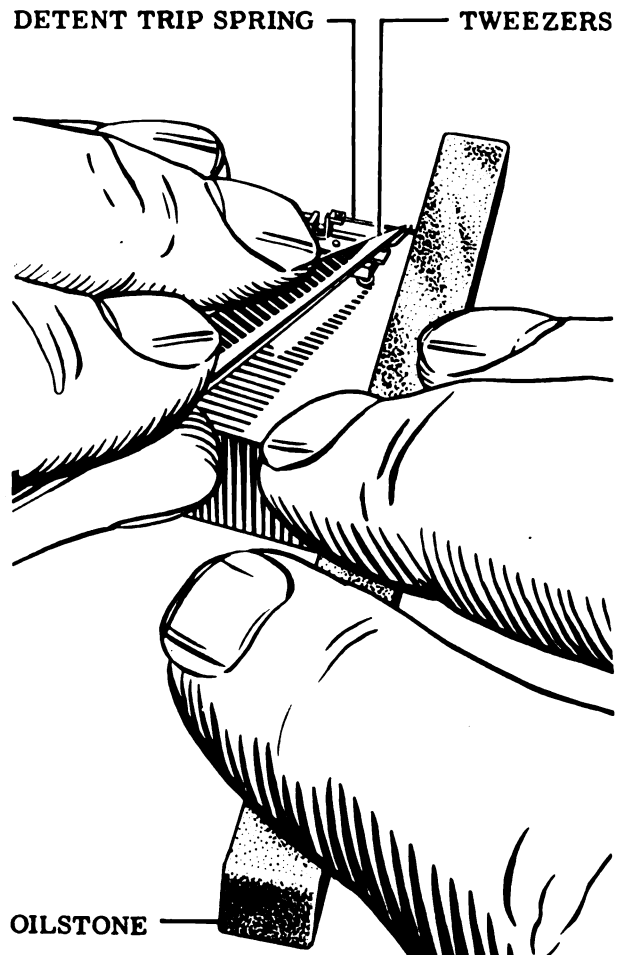


Figure 92

90. If, in Operation 87, the detent trip spring was found to be too short, change to a new one. See the instructions for detent disassembly and reassembly in Section VII "Inspection and Repair."

91. If, in Operation 87, the detent trip spring was found too long, it must be stoned off to proper length. This is done as shown in Fig. 92, by clamping down the detent and grasping the trip spring in the tweezers to steady it, at the same time using the stone, Tool No. 10. Stone off straight and remove all burrs.

92. Replace detent and check overall, lock and let off. The detent may have to be removed several times until the trip spring is adjusted to the proper length.

93. Recheck the spring action of both detent spring and trip spring after overall is satisfactory. See Operations 78 and 79. Make sure that the detent clamp screw is tight and that the detent adjusting screw is screwed in against the detent.

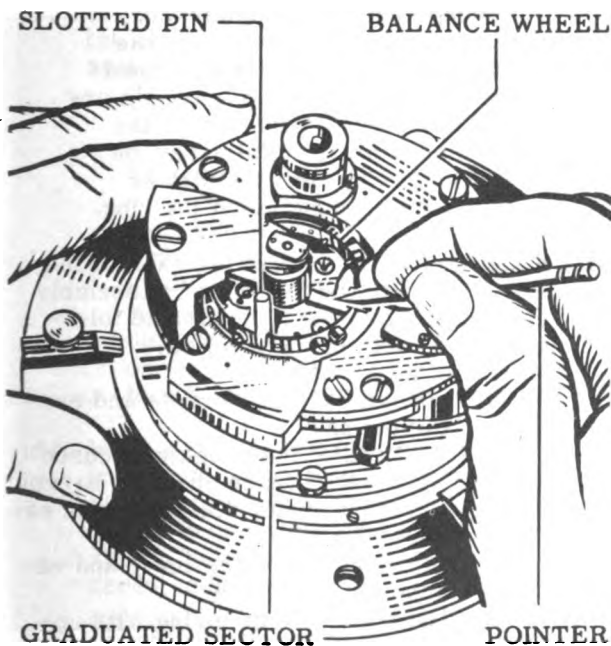


Figure 91

94. Clean detent by dipping in naphtha and drying in warm air.
95. Replace detent.
96. Check overall, lock and let off.
97. Check and adjust drop, by gaging with graduated sector and slotted pin. Refer to Fig. 91. To check drop, turn the balance counterclockwise until the escape tooth falls on the impulse jewel. Stop balance at this point and observe gage. Turn balance clockwise until the escape tooth falls off the impulse jewel. Observe gage. Allowances must be made for variations in the length of escape wheel teeth when setting the drop, to make sure that the shorter teeth do not entirely miss the impulse jewel. For the average escape wheel approximately two degrees on the graduated sector will be about right. To adjust the drop, use spanner, Tool No. 42, to change the angular distance between impulse and unlocking jewels. Increase this angular distance if the drop is too great and decrease it when the escape tooth fails to contact the impulse jewel.

NOTE

If the one-piece roller is being used, a slight variation in drop can be adjusted by reshaping the trip spring, but the amount of reshaping should be limited so that the appearance of the spring is not appreciably altered.

98. Move the balance until an escape tooth rests on the locking jewel.
99. Turn the movement over. Holding the balance wheel lightly with the finger so it does not oscillate, and using small punch, Tool No. 38, install the second hand so it is positioned exactly in line with one of the second markers. See Fig. 93. If new second hand is used, it may be necessary to broach, using Tool No. 82. If necessary, use second hand closer, Tool No. 105. Do not force the hand tightly into place until after Operation 100.
100. Oscillate the balance and note the position of the second hand relative to the second markers for one complete revolution of the hand. If the hand does not fall evenly on the seconds, adjustment must be made to the hand with the small punch (Tool No. 38) at the same time holding the balance. Repeat Operations 98, 99 and 100 until the desired result is obtained. Then, use the small punch to

SECOND HAND PUNCH

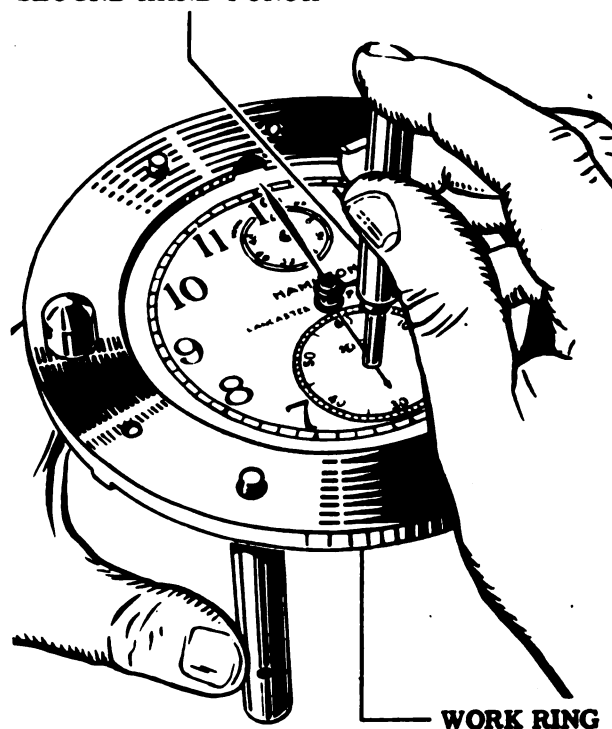


Figure 93

secure the second hand in correct position, being sure to allow clearance between the hand and the dial.

NOTE

Under certain circumstances, tolerances permit the dial or the movement to shift on the mounting ring, causing the second's orbit to be out of center with the second's pivot. This may be corrected by shifting dial or movement on the mounting ring.

101. Examine the instrument for each of the items noted in the following reassembly check list. Use the methods and tolerances set forth in this section.
 - A. Balance endshake
 - B. Endshake of train wheels and sustaining pawl
 - C. Tightness of fusee chain on fusee
 - D. Alignment of fusee chain on barrel
 - E. Alignment of impulse jewel and escape wheel
 - F. Alignment of locking jewel and escape wheel
 - G. Alignment of trip spring with unlocking jewel and abutment arm
 - H. Detent for general appearance and tightness of screws

- I. Setting of escapement
 - a. Drop, lock, overall, let off
 - b. Tightness of balance screws
 - c. Tightness of timing and vernier weights
 - d. Proper action of detent spring and trip spring
- J. Cleanness and proper amount of oil in jewels and bushings
- K. Setup of mainspring
- L. Dial side
 - a. Location of wind indicator hand
 - b. Alignment of hour and minute hands
 - c. Alignment of second hand
- M. General appearance
 - a. Fingermarks, dust and lint
 - b. Other marks and marred screws. The latter should be of particular concern if on the balance wheel.
 - c. Tightness of screws other than mentioned in the foregoing. Use Tool No. 34 to check tightness of balance lower bridge screws.

NOTE

The instrument is now ready for regulation and isochronal testing. See Section IX "Test and Adjustment."

- 102. Take the movement from the work ring.
- 103. Insert the movement in the case. Start the chronometer, and set hands in preparation for pre-test, using Tool No. 110.

NOTE

The instrument is now ready for pre-testing and performance testing. See Section IX "Test and Adjustment." After that work is performed, the chronometer may be fitted into its mounting box as indicated in Operations 104 and 105, or prepared for storage. See Section X "Storage and Shipment."

- 104. Lightly grease the four pivots of the gimbal mount. Loosen gimbal ring rear pivot screw. Use Tool No. 102. Place the chronometer on pivots in the gimbal ring. Tighten pivot screw and pivot screw locknut against the gimbal ring.
- 105. Check freedom of chronometer in gimbals. If required, correct endshake by either underturning the head of the rear

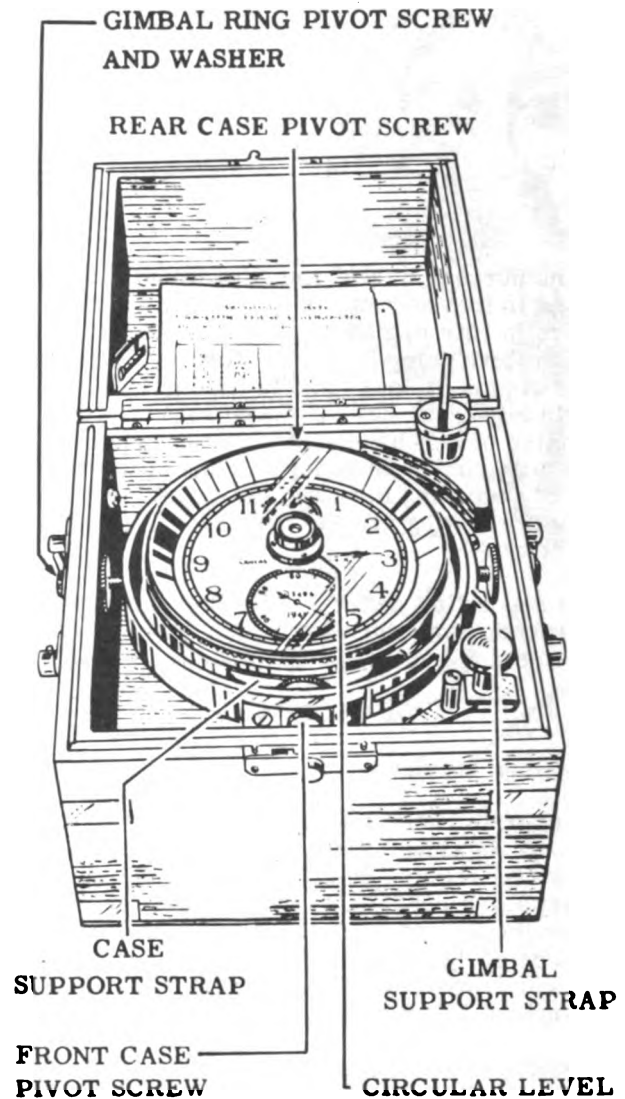


Figure 94

pivot screw or cutting back the pivot shoulder of the same screw. If, though free, the chronometer is out of level, adjust case and gimbal support straps to level the instrument. Use Tool No. 103. The case support strap at "6" position is used for centering and leveling the case. The gimbal ring support strap at "3" position is used for leveling the gimbal ring. Exact horizontal carriage may be attained by the use of a circular level with bubble indicator (Tool No. 101), placed in the center of the bezel. If the gimbal ring is not free, or is too free, in the mounting box, cup or reduce thickness of one of the gimbal ring pivot screw washers, as required. See Fig. 94.

After successfully passing its tests and following reassembly into the mounting box, the chronometer is ready for use.

SECTION IX

TEST AND ADJUSTMENT

The purpose of test and adjustment, as discussed in this section, is to find and correct errors in rate by means of regulation and temperature compensation. Rate really means "rate of gain or loss" and is time gained or lost in a 24-hour day. A "good rate" is the objective of all chronometer work, which makes important an understanding of the method of recording the rate, the details of performance testing and the meaning of certification of rate.

In connection with the following paragraphs, reference should be made to the "Navigational Timepiece Record," NavShips 3587.

DISCUSSION

Recording the Rate

Accurate time indications from a chronometer are dependent upon careful observation of the dial reading and the application of an accumulated error factor to the dial reading. Under these conditions the maintenance of records of the chronometer rate is essential. The following general instructions will indicate the method by which rate is determined.

Daily, or at any other convenient intervals, check the dial reading of the chronometer against a master time standard, such as an authorized radio signal. It is, of course, convenient to employ a comparing watch for intercomparison with the time signal and the chronometer, although it should be remembered that an additional observational error is introduced by use of this method. By either means, it is very simple to arrive at the dial error factor, the difference between exact time and that indicated by the chronometer. Since the chronometer hands advance in one-half second increments, the dial error can best be recorded to the nearest one-half second. In overhaul stations, where greater accuracy and speed is desirable, the dial error is determined by an electronic device, the Hamilton comparator, to which reference will be made in subsequent paragraphs.

Determine the rate (that is the gaining or losing rate of the chronometer) by calculating the difference of the dial error between successive readings. If these dial error readings

are recorded daily, the rate then, of course, is an indication of the time gained or lost per day. When observations are made several days apart, the rate should be reduced to a per diem (24-hour) basis by dividing the difference in dial errors by the number of days involved. Even when the rates are recorded daily, it is generally necessary to average the rates for an extended number of days to arrive at the mean rate of the chronometer, because of the fractional errors occurring through recognition of the dial reading to only the nearest one-half second. It is this mean daily rate established over an interval of days which is employed for the computation of exact time when the chronometer has not been checked against a master time source for an extended period.

This method, as described in the foregoing, can be easily understood from the accompanying Table I. Under "Remarks," as shown in the table, it is well to record any unusual circumstances, such as when the chronometer has been started or stopped, set or moved in location. These disturbances of the instrument may each exert an influence upon the rate and, hence, should be noted for the sake of identification of possible causes of fluctuations in rate.

TABLE I
COMPUTATION OF RATE

Date	Dial Error + = Fast - = Slow		Daily Rate + = Gain - = Loss	Mean Deviation in Daily Rate	Remarks
	Min	Sec			
Oct 1948					
3	+0	2			Started+Set
4	+0	2½	+½		
5	+0	2½	0	¼	
6	+0	3	+½	¼	
7	+0	3	0	¼	
8	+0	3½	+½	¼	
9	-	-	-	-	Not wound
10	+0	4	+¼	-	2 day avg.

(Mean daily rate = +1/4 second)

In Table I, there will be noted a column headed "Mean Deviation in Daily Rate." The



Figure 95

HAMILTON SHIP CHRONOMETER

mean deviation in daily rate is one-half the difference between rates computed for two consecutive days. This indicates the observational error plus error resulting from changing rate. The temperature compensation error of the Hamilton make of Ship Chronometer, when properly adjusted, is negligible; thus the maintenance of a temperature record for this instrument while in service is not necessary unless otherwise required.

Performance Testing

The performance test, made prior to issuance of the chronometer, is a test of thirty (30) consecutive days duration. This test is made in a dust-free test room, favorably isolated from shocks and vibration, with room temperature controlled at 72-1/2 degrees F and relative humidity at 40 percent. During the time of the test a daily check of the rate of the instrument on an electronic comparator is made. The electronic comparator, a Hamilton development illustrated in Fig. 95, is used at overhaul stations because it determines the dial error accurately and speedily. The accuracy of dial reading with this instrument is well under ± 0.02 seconds.

The thirty-day period of the test is divided into six periods of five days. Each of these

is a period at which the chronometer is rated at a specified temperature. One convenient type of electrical temperature cabinet, located within the test room, is illustrated in Fig. 96. The temperature of the periods, two periods each, are 90, 72-1/2 and 55 degrees F. The temperature periods are arranged so that the two 90 degree periods are at the beginning and at the end of the test; the two 55 degree periods are in the middle of the test; and the two 72-1/2 degree periods are placed between the 90 and the 55 degree periods.

The purpose of arranging the temperature periods in the above described manner is to provide a method of checking the chronometer for recovery, that is, to determine whether or not the chronometer will repeat the rating result for periods of like temperature after having been subjected in the interim to one or more different temperatures. While the arrangement of the temperature periods provides means of checking the recovery, the different temperature periods' primary purpose is to discover the temperature error.

On figuring the rate for the various periods of the test, the five daily rates are added and then divided by the number of days (five) in the period, which gives the mean daily rate for that particular period. From these six

MARINE CHRONOMETER PERFORMANCE TEST

Chronometer Number: 3390				Timing Specifications: 18C7d				Inspector: 24.8				Test Completed			
Tested For: Bu. Ships #85310				DEC. 28, 1940				Test Started: (0 Day) FEB 19 44				Chronometer Delivered:			

Period I 90° F.				Period III 55° F.				Period V 72 1/2° F.				Isochronism				Combine with any Day in Periods II or V 72 1/2° F.	Passed ☒ Failed ☐ Approved C22K
Day	Error	Rate	Dev.	Day	Error	Rate	Dev.	Day	Error	Rate	Dev.	Day	Error	Rate			
0	58.58	---	---	10	58.84	---	---	20	58.86	---	---	7	58.60	---			
1	58.55	-0.03	.01	11	58.84	+0.00	.01	21	58.93	+0.07	.04	12 Hr.	58.66	+0.06			
2	58.50	-0.05	.01	12	58.78	-0.06	.07	22	58.96	+0.03	.00	8	58.72	+0.12			
3	58.39	-0.12	.08	13	58.80	+0.02	.01	23	59.00	+0.04	.01	Dif. 12 Hr. & 1/4 24 Hr. Rate	0.00				
4	58.40	-0.02	.06	14	58.84	+0.04	.03	24	59.02	+0.02	.01						
5	58.40	+0.00	.04	15	58.90	+0.06	.05	25	59.00	-0.02	.05						
Sum	-0.18	.20		Sum	+0.06	.17		Sum	+0.14	.11							
Mean	-0.04	.04		Mean	+0.01	.03		Mean	+0.03	.02							

Period II 72 1/2° F.				Period IV 55° F.				Period VI 90° F.				Performance		Actual	Tolerance
Day	Error	Rate	Dev.	Day	Error	Rate	Dev.	Day	Error	Rate	Dev.				
5	58.40	---	---	15	58.90	---	---	25	59.02	---	---	Regulation:			
6	58.50	+0.10	.01	16	58.90	+0.00	.01	26	59.04	+0.02	.02	Largest Mean Daily Rate Any Period	0.09	1.55	
7	58.60	+0.10	.01	17	58.95	+0.05	.06	27	59.06	+0.02	.02	Rating:			
8	58.72	+0.12	.03	18	58.95	+0.00	.01	28	59.04	-0.02	.02	Mean Deviation of Rates	0.03	0.50	
9	58.75	+0.03	.06	19	58.95	+0.00	.01	29	58.98	-0.06	.06	Largest Dif. Any Two Daily Rates in same Period	0.14	0.75	
10	58.84	+0.09	.00	20	58.86	-0.09	.08	30	59.00	+0.02	.02	Temperature Compensation:			
Sum	+0.44	.11		Sum	-0.04	.17		Sum	+0.02	.14		Dif. Between Mean Daily Rates at 72 1/2° F. (Periods I & VI) and 72 1/2° F. (Periods II & V)	0.08	0.75	
Mean	+0.09	.02		Mean	-0.01	.03		Mean	-0.00	.03		Dif. Between Mean Daily Rates at 55° F. (Periods III & IV)	0.06	0.75	
												Dif. Between Mean Daily Rates at 90° F. (Periods I & VI) and 55° F. (Periods III & IV)	0.02	1.20	
												Recovery:			
												Largest Dif. Between Mean Rates For Different Periods at Same Temp	0.06	0.70	
												Isochronism:			
												Dif. 12 Hr. and 1/4 24 Hr. Rates	0.00	0.50	

Note: All Observations in Seconds. All Rates in Seconds Per Day Plus (+) Rate = Gaining Minus (-) Rate = Losing
 Testing Laboratories HAMILTON WATCH COMPANY Lancaster, Pennsylvania, U. S. A.

Factory Performance Test Card for Hamilton Chronometer No. 3390

mean daily rates is determined the value of the chronometer's most important criteria.

These factors, which are of vital consequence in determining the performance of the chronometer, are as follows: Regulation — the largest mean daily rate for any period; Temperature Compensation — the difference between the mean daily rate at one test temperature (two periods) and the mean daily rate at another test temperature (two periods); and Recovery — the largest difference between the the mean rate of any two periods at the same temperature. The allowable tolerances as specified by the U. S. Bureau of Ships for these classifications are: Regulation ± 1.55 seconds per day; temperature compensation 1.20 seconds per day for a difference of 35 degrees F and 0.75 seconds per day for a difference of 17-1/2 degrees F; and recovery 0.70 seconds per day.

In addition to the above criteria, the chronometer performance card lists the mean deviation of rates for all periods and the largest difference between any two daily rates in the same period. For these, the tolerances are 0.50 seconds per day for deviation and 0.75 seconds per day for largest difference. These two factors, just mentioned, are important in determining the excellence of a chronometer. A steady rating chronometer, which is the desired objective, will show a low figure for mean deviation of rate and for largest difference between two daily rates in the same period. The isochronism check, shown on card, indicates the difference between the 12 hour rate and one-half the 24 hour rate at 72-1/2 degrees F.

Certification of Rate

The purpose of rate certification is to provide information which will accompany the instrument in order to show how the chronometer may be expected to rate under average conditions. See the "Navigational Timepiece Record," NavShips 3587.

When preparing the certification of rate, the method of reading the chronometer's error is the same as that described in the foregoing paragraphs except that the rating is done entirely at room temperature of 72-1/2 degrees F, and subsequent to the completion of the performance test. Fifteen consecutive daily rates are taken at this temperature during the period immediately preceding the issuance of the instrument for service. Here again, the daily rates, in this case 15, are averaged, and the mean rate at room temperature obtained. This rate is an indication of how much the chronometer will gain or lose each day if maintained at this temperature.

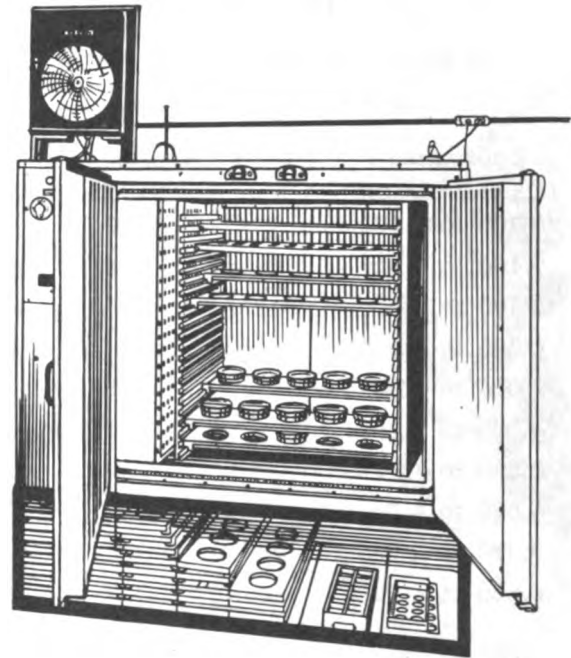


Figure 96

The temperature coefficient, which also is indicated in the chronometer record, is obtained from the performance test and is indicated in seconds per day per degree F. This figure is the amount that the mean rate at room temperature will change for every degree change in temperature from a room temperature of 72-1/2 degrees F.

TIMING OPERATIONS

Timing operations are concerned with the removal and replacement of the balance unit, as covered in Sections V and VIII, and also involve assembly and disassembly of this unit itself as described in Section VII. The instructions given in those sections are applicable to the corresponding operations incidental to timing and temperature compensation.

Rough timing is essential only after the balance unit has been disassembled for some reason, such as to change the hair-spring. If it has been unchanged in overhaul and has been cleaned as a unit, timing can begin with Operation 6 in the following schedule. Where rough timing is required, begin with Operation 1. Step-by-step instructions for this work are given in the text accompanying Tables II and III on the next page.

The instrument is uncased and, for convenience, it is mounted in its work ring (Tool No. 98), in the same condition as at Operation 101, Section VIII "Reassembly of Chronometer."

TABLE II

SCREW CHANGE EQUIVALENTS

(For Pairs of Screws)

Approximate Head Height, In.*	Approximate Minutes per Day
0.100 to 0.080	16
0.100 to 0.060	34
0.100 to 0.050	43
0.100 to 0.040	50
0.080 to 0.060	16
0.080 to 0.050	25
0.080 to 0.040	34
0.060 to 0.050	6
0.060 to 0.040	13
0.050 to 0.040	7

One full turn of timing weight equals about 40 seconds. One full turn of vernier timing weight equals about 2.8 seconds.

Table III, used with Table II, makes possible combinations to permit close regulation, easily coming within the tolerances specified in Operation 5. These tables contemplate replacement of screws and washers at the same holes from which screws were removed.

After Changing Hairspring or Balance

1. Make certain timing weights and vernier timing weights are in middle position. Start the chronometer.
2. Count the balance oscillations for one or more minutes, using stop watch. The count should be 240 oscillations or 120 impulses per minute. If fast or slow, remove the balance unit.
3. Convert the error into minutes per day. Regulate by disassembling the balance unit, and making necessary screw or washer changes according to Table II and Table III.
4. Bring the balance wheel into perfect poise. Use Tool No. 24. Reassemble and reinstall the balance unit.
5. Repeat Operations 1, 2, 3 and 4 until the chronometer times within two minutes per day. Then, continue with Operations 6 through 12.

NOTE

The work of Operations 1 through 5 may be accomplished much faster if a Hamilton oscillometer is available, into which the balance unit may be inserted for rough timing.

TABLE III

WASHER CHANGE EQUIVALENTS

(For Pairs of Washers)

Washer Thickness Inches*	Daily Time Effect		Difference		Combinations		
	Min	Sec	Min	Sec	Inches	Min	Sec
0.002	1	10			0.002 + 0.003	3	4
				44			
0.003	1	54			0.002 + 0.004	4	28
			1	24			
0.004	3	18			0.003 + 0.004	5	12
			1	6			
0.006	4	24			0.002 + 0.006	5	34
			2	6			
0.008	6	30			0.003 + 0.006	6	18
			2				
0.010	8	30			0.002 + 0.008	7	40

*See Section XI "Parts Catalog" for corresponding part number designation.

*See Section XI "Parts Catalog" for corresponding part number designation.

TAPE RECORDER PICK-UP

TAPE RECORDER

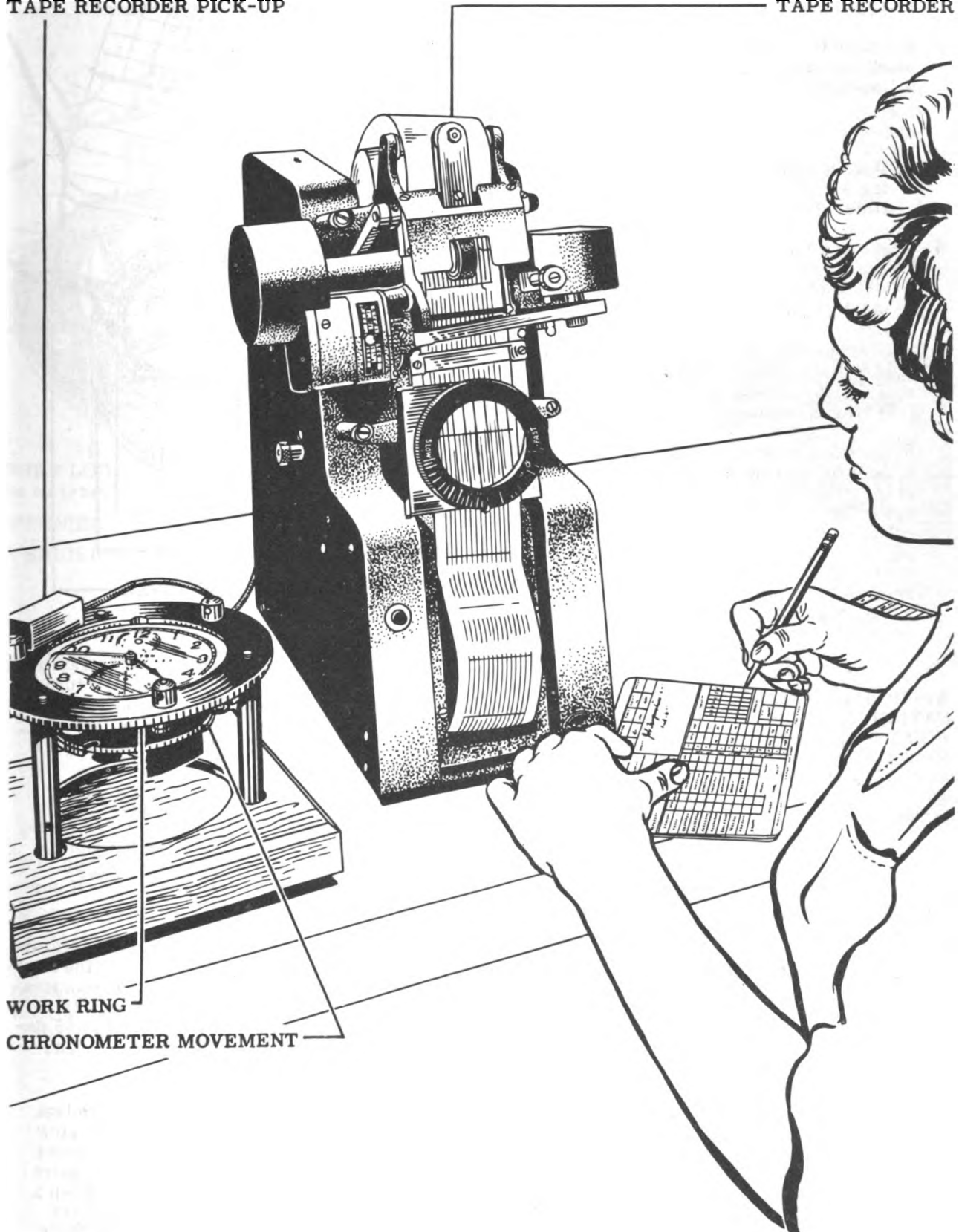


Figure 97

HAMILTON SHIP CHRONOMETER

After Overhaul

6. Regulate to within ± 0.5 seconds per day, using continuous tape recorder reading to 0.1 seconds. See Fig. 97.

NOTE

For Operation 6, use only the timing weights to bring balance to time. The screwdriver is Tool No. 61.

7. Test and adjust for isochronism at the following balance motions: $1-1/8$, $1-1/4$, $1-3/8$, $1-1/2$ turns. These motions may be obtained by varying the power supply.

Record the rate at each of these motions on the tape recorder. The allowed tolerance is 1.5 seconds between $1-1/8$ and $1-1/2$ turns, preferably faster at $1-1/8$ turns.

In any case where the chronometer is found to operate outside the tolerance mentioned in the preceding paragraph, corrections are to be made by turning the hairspring collet on the balance staff, thereby changing the position of the impulse jewel when the balance is at rest. This is sometimes referred to as the "zero point." The following examples "A" and "B" show procedure for isochronal adjustment.

A

APPROXIMATE LOCATION OF ZERO POINT WHEN CORRECTION IS MADE FOR A FAST RATE AT $1-1/8$ TURNS OF MOTION.

B

APPROXIMATE LOCATION OF ZERO POINT WHEN CORRECTION IS MADE FOR A SLOW RATE AT $1-1/8$ TURNS OF MOTION.

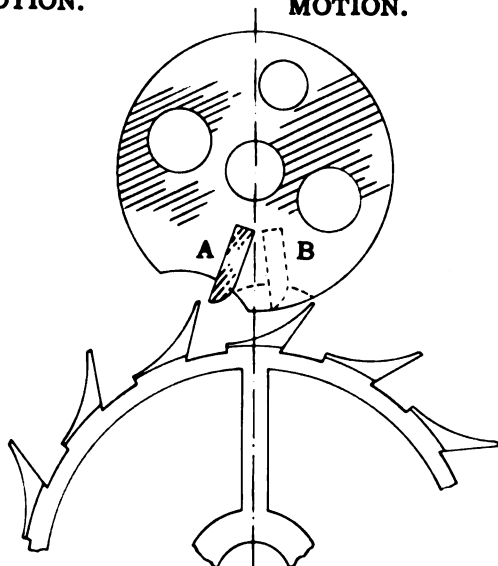


Figure 98



Figure 99

Example "A": If the chronometer rate is faster at $1-1/8$ turns than at $1-1/2$ turns, correct as follows: Turn the hairspring collet slightly counterclockwise. See Fig. 98. Check again on the rate recorder. If correction is insufficient, repeat.

Example "B": If the chronometer is slower at $1-1/8$ turns than at $1-1/2$ turns, correct as follows: Turn the hairspring collet slightly clockwise. See Fig. 98. Check again on the rate recorder. If correction is insufficient, repeat.

8. Place the chronometer on pre-temperature test. In the pre-temperature test, the chronometer is timed in the following temperatures for the periods indicated and its daily rate carefully noted: Three days in 55 degrees F; two days in 90 degrees F; and two days in 55 degrees F.

Tolerances for the pre-temperature test are 0.5 seconds fast or 0.8 seconds slow per day. If outside these limits, correct the balance by moving one or more pairs of screws according to Table IV. Keep a record of the movement of each pair of screws, so they can be returned to their former location if desired. See Fig. 99, showing numbered block, Tool No. 54. Use

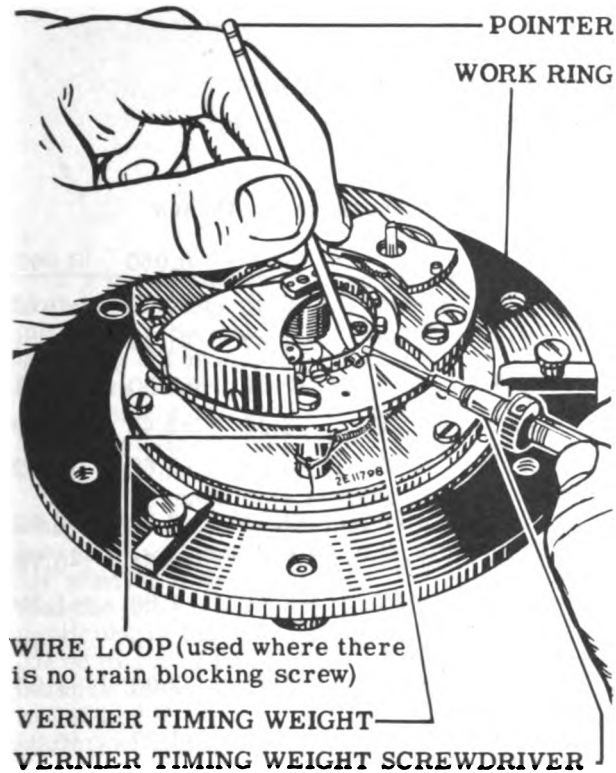


Figure 100

screwdriver known as Tool No. 63. After each screw change, poise the balance.

CAUTION

If corrections for temperature are made in the pre-temperature test (necessitating removal of the hair-spring in order to poise the balance), the test for isochronism must be repeated as set forth in Operation 7. This is true whenever the collet is removed from the staff.

9. Repeat the pre-temperature test as outlined in Operation 8 until the tolerances are met as indicated in that operation.
10. In 90 degrees F temperature, regulate the chronometer to within ± 0.5 seconds. See Table II.

NOTE

If regulation has been carried out accurately at room temperature, previous to the pre-temperature test, then regulating in Operation 10 can be accomplished by moving only the vernier timing weights. By the use of the vernier screwdrivers (Tool Nos. 60 or 62), fine adjustments of the vernier timing

weights can be made to 0.1 second per day. See Fig. 100.

11. Place the chronometer on performance test as described earlier in this section under "Performance Testing." If the instrument fails to pass the performance test, check for cause of failure. Make corrections as necessary. Repeat testing procedures as required.
12. Rate the chronometer for 15 days in 72-1/2 degrees F. See heading entitled "Certification of Rate," appearing earlier in this section.

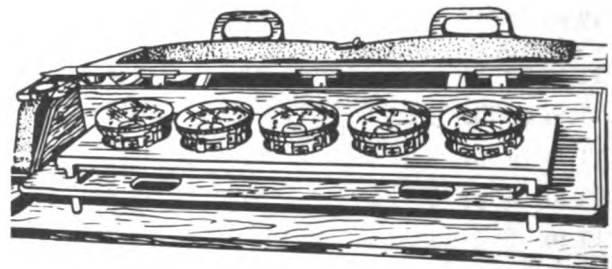


Figure 101

NOTE

Where many chronometers are undergoing test, a winding rack is a great convenience. Such a fixture is illustrated in two positions in Figs. 101 and 102. Fig. 102 also shows a timesaving winding key (winder).

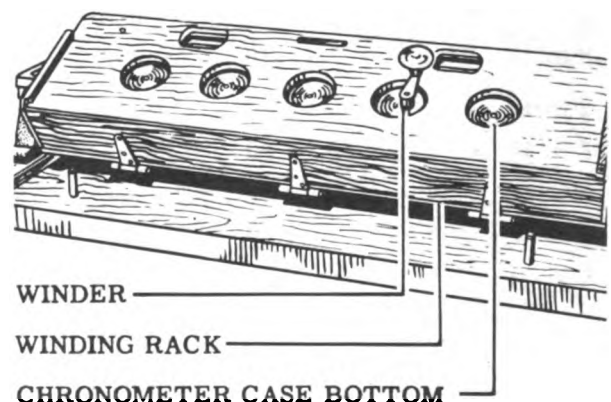


Figure 102

Following completion of the work of regulation and temperature compensation as outlined in the preceding twelve operations, the instrument is ready to be prepared for storage or shipment, or for gimbaling and leveling. See Operations 104 and 105 in Section VIII "Reassembly of Chronometer."

TABLE IV

TABLE OF TEMPERATURE COMPENSATION CORRECTIONS*
CORRESPONDING TO CHANGES IN BALANCE SCREW POSITIONS

One Pair of Screws Moved:		Head Height of Balance Screws — Measured in Inches							
Hole:	Hole:	0.120	0.100	0.090**	0.080	0.070**	0.060	0.050	0.040
12 to 11		-1.69	-1.41	-1.10	-0.79	-0.66	-0.53	-0.32	-0.12
12 to 10		-3.55	-2.96	-2.55	-2.14	-1.90	-1.67	-1.27	-0.91
12 to 9		-4.99	-4.16	-3.67	-3.17	-2.85	-2.54	-2.00	-1.51
12 to 8		-5.89	-4.91	-4.37	-3.82	-3.46	-3.09	-2.46	-1.89
12 to 7		-6.19	-5.16	-4.60	-4.04	-3.66	-3.28	-2.62	-2.02
11 to 12		+1.69	+1.41	+1.10	+0.79	+0.66	+0.53	+0.32	+0.12
11 to 10 or 3 to 4		-1.86	-1.55	-1.45	-1.35	-1.24	-1.14	-0.95	-0.79
11 to 9 or 3 to 5		-3.30	-2.75	-2.57	-2.38	-2.20	-2.01	-1.68	-1.39
11 to 8 or 3 to 6		-4.20	-3.50	-3.26	-3.03	-2.80	-2.56	-2.14	-1.77
11 to 7		-4.50	-3.75	-3.50	-3.25	-3.00	-2.75	-2.30	-1.90
10 to 12		+3.55	+2.96	+2.55	+2.14	+1.90	+1.67	+1.27	+0.91
10 to 11 or 4 to 3		+1.86	+1.55	+1.45	+1.35	+1.24	+1.14	+0.95	+0.79
10 to 9 or 4 to 5		-1.44	-1.20	-1.11	-1.03	-0.95	-0.87	-0.73	-0.60
10 to 8 or 4 to 6		-2.34	-1.95	-1.82	-1.68	-1.55	-1.42	-1.19	-0.98
10 to 7		-2.64	-2.20	-2.05	-1.90	-1.76	-1.61	-1.35	-1.11
9 to 12		+4.99	+4.16	+3.67	+3.17	+2.86	+2.54	+2.00	+1.51
9 to 11 or 5 to 3		+3.30	+2.75	+2.57	+2.38	+2.20	+2.01	+1.68	+1.39
9 to 10 or 5 to 4		+1.44	+1.20	+1.11	+1.03	+0.95	+0.87	+0.73	+0.60
9 to 8 or 5 to 6		-0.90	-0.75	-0.70	-0.65	-0.60	-0.55	-0.46	-0.38
9 to 7		-1.26	-1.01	-0.96	-0.87	-0.80	-0.74	-0.62	-0.51
8 to 12		+5.89	+4.91	+4.37	+3.82	+3.46	+3.09	+2.46	+1.89
8 to 11 or 6 to 3		+4.20	+3.50	+3.26	+3.03	+2.80	+2.56	+2.14	+1.77
8 to 10 or 6 to 4		+2.34	+1.95	+1.82	+1.68	+1.55	+1.42	+1.19	+0.98
8 to 9 or 6 to 5		+0.90	+0.75	+0.70	+0.65	+0.60	+0.55	+0.46	+0.38
8 to 7		-0.31	-0.26	-0.24	-0.22	-0.21	-0.19	-0.16	-0.13
7 to 12		+6.19	+5.16	+4.60	+4.04	+3.66	+3.28	+2.62	+2.02
7 to 11		+4.50	+3.75	+3.50	+3.25	+3.00	+2.75	+2.30	+1.90
7 to 10		+2.64	+2.20	+2.95	+1.90	+1.76	+1.61	+1.35	+1.11
7 to 9		+1.26	+1.01	+0.96	+0.87	+0.80	+0.74	+0.62	+0.51
7 to 8		+0.31	+0.26	+0.24	+0.22	+0.21	+0.19	+0.16	+0.13

*Corrections are shown in seconds per day. Plus corrections compensate for minus rates, and vice versa.

**Equivalent screw heights for 0.090 in. and 0.070 in. are obtained by screw and washer combinations. See pages 82 and 88.

SECTION X

STORAGE AND SHIPMENT

If a chronometer is to be transported, even for short distances by hand, lock the gimbal suspension by means of the latch lever. Free-swinging chronometers are vulnerable to damage from violent oscillations of the gimbal suspension system. Care is necessary to avoid jarring the instrument at any time.

If the chronometer is to be kept running while in transit for a period of 24 hours or more, it is essential that provision be made for winding it daily. If there is any chance that the daily winding may be omitted, and particularly that the instrument may be allowed to run down, it is necessary that the balance assembly of the chronometer be immobilized (corked or locked) before transportation is begun.

PREPARING FOR SHIPMENT

The condition of a chronometer prepared for shipment should be the same as that in which it was received. See "Operation When Received" in Section III. In preparing for shipment, the procedure in succeeding paragraphs should be carefully followed with the aid of an experienced repairman, if possible, particularly for the immobilization of the balance assembly.

First, allow the chronometer to run 48 hours down. Then, disassemble the cased movement from the gimbal ring by backing off the case rear pivot screw in the gimbal ring. Remove the movement from its case, carefully following instructions given for this operation in Section III under "Taking Movement from Case."

Stop the balance wheel by applying a very light pressure against the top edge of the wheel with a clean, non-metallic pointer or with any soft, dry material which will not injure the balance. Let the wheel come to rest in its dead center position so that the hairspring no longer tends to pull the wheel in either direction.

CAUTION

Do not exert any pressure against the balance wheel or damage to the pivots may result.

Gently lock the balance wheel locking arm, illustrated in Section III, Fig. 9, by loosening its screw, turning the arm into the locked position, and tightening the screw. To place the locking arm over the timing weight in position of lock, it is necessary to lead the balance wheel slightly counterclockwise.

NOTE

Former practice, before locking arms were installed, made use of cork wedges or wedges made from folded plastic or tissue. These were placed beneath the rim of the balance wheel at the extremities of the wheel arm, at the same time holding the balance wheel lightly in zero position. Tails of the wedge (not the fold end) were thrust in position with tweezers, taking care that they would neither fall out easily, nor exert an extreme force against the wheel. See Section III, Fig. 10. Chronometers not equipped with locking arms should have balance wheels immobilized in this manner.

After the balance is locked, replace the movement in its case and screw the bezel on the case.

CAUTION

In the event that it is necessary to unlock (or uncork) the balance for any reason, the instrument should be fully wound before the locking device is released. On a chronometer equipped with balance locking arm, it is important to make certain that the screw holding this arm is tight when the arm is in either the locked or unlocked position.

Shipment by Carrier

Pack the chronometer in the special shipping container designed for the purpose, enclosing only the cased movement. This container, of aluminum alloy, is identified by Bureau of Ships Navigational Equipment Section Sketch Number 624-1 Alt. 1, and is illus-

HAMILTON SHIP CHRONOMETER

trated in Fig. 103. Sponge rubber inserts, supplied with the container, hold the chronometer securely in place and isolate it from shock. The aluminum shipping container is to be protected on all faces by the use of two sheets of 200 pound test corrugated fibreboard scored to fit the container across the top and down the sides, and another piece of the same size to fit across the bottom and up the sides. The container is then to be over-packed in an RSC 275 pound fibreboard carton, conforming to the requirements of Federal Specification LLL-B-631, of size required to accommodate the above package. The over-pack carton is to be marked "Delicate Instrument" in conspicuous red letters. Instruments so packed may be shipped by Air Express, Railway Express or Parcel Post.

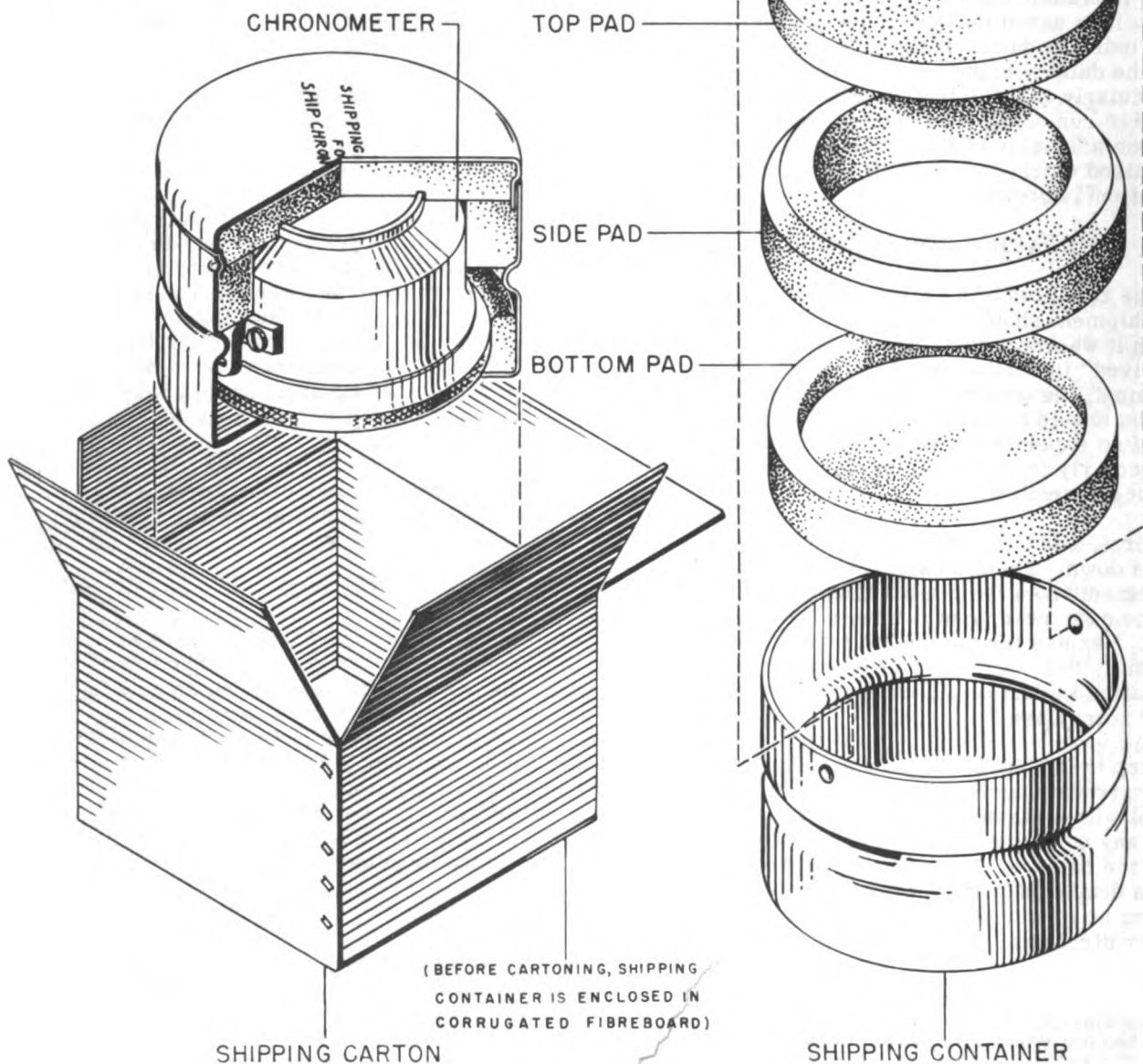


Figure 103

PREPARING FOR STORAGE

Chronometers are always stored in an ungimbaled condition. When preparing chronometers for storage after overhaul, there are two circumstances which determine the procedure to be followed: (1) If intended for use in the immediate future, the chronometers should be wound daily and kept running in a rack at normal room temperature, with a relative humidity not in excess of 40 percent. (2) If to be out of use for a considerable period, allow the chronometer to run 48 hours down, lock the balance wheel, place the instrument in the shipping container, seal and store at the same temperature and humidity as above.

Conditions in the storage room should be the same as those in the room where performance testing is conducted. See Section IX under "Performance Testing." One convenient design of rack for open storage is illustrated in Fig. 104.

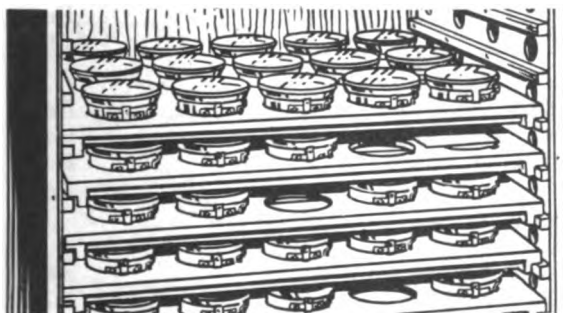


Figure 104

Storage at chronometer pools is preceded by an initial ten day run and followed by subsequent periodic trials which are made in accordance with the procedure outlined under "Storage Procedure."

Storage Procedure

When chronometers are received at chronometer pools from the repair activity, they are unpacked, wound and unlocked (or uncorked). They are then started, set to Greenwich Civil Time and given a ten day run, with daily winding and rating prior to storage. The rate is entered in the "Navigational Timepiece Record," NavShips 3587.

After the instruments have completed their ten day run, they are allowed to run 48 hours down, then stopped, corked or locked, placed in the shipping container, sealed with scotch or masking tape, and stored. While in storage, they should be inverted every 30 days in order to redistribute the oil on the lubricated surfaces.

Periodic trials are made every six months, or at intervals not to exceed eight months, while the chronometers are in storage. To conduct the trial, the chronometer is removed from its container, wound, uncorked or unlocked, and run for five days with daily winding. After it has operated for a period of two days to settle down, it is rated for three days, and the rate entered in the "Navigational Timepiece Record," NavShips 3587. At the conclusion of the five day trial period, the instrument is allowed to run 48 hours down, stopped, corked or locked, placed in the shipping container, sealed and returned to storage.

For additional information regarding maintenance of navigational timepieces, refer to current Bureau of Ships directives, "Navigational Timepiece, Maintenance Program."

CHRONOMETER PARTS

In preparing chronometer parts for shipment or storage, the important considerations are protection from (1) mechanical injury and (2) corrosion (chiefly from moisture).

As convenience dictates, parts are packaged in tin boxes of various sizes, cardboard boxes, paper envelopes and cellophane envelopes. Interior wrapping or padding of tissue is used where indicated. Detents are packed in glass vials with metal caps.

Labels supply complete identification. Where parts are factory-coated with rust preventative, a label is attached, "Before using, clean with petroleum or chlorinated solvents."

Vapor Phase Inhibitor paper (VPI 260, a product of Shell Development Company) is included with packaged parts where indicated. Where steel and brass are in contact in packages containing VPI, tissue wrapping is used around the parts themselves.

Wheels with pinions are packaged in cardboard boxes with false bottoms which are perforated to admit and hold each assembly. Balance units, when packaged in quantity, are clamped in position in a special holder. For details see Fig. 105.

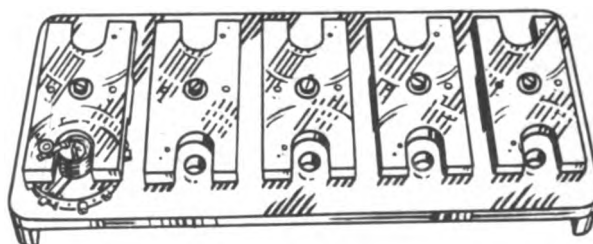


Figure 105

SECTION XI

MAINTENANCE PARTS CATALOG

Introduction

The primary purpose of this Maintenance Parts Catalog is to make easy the ordering of parts which may be required for the maintenance of the Hamilton Ship Chronometer. Such maintenance parts are under the cognizance of the Ships Parts Control Center, Mechanicsburg, Pa.

This catalog comprises a Group Assembly Parts List detailing all main assemblies, sub-assemblies, component parts and attaching parts; a series of exploded views illustrating the Group Assembly Parts List; and a Numerical Parts List. These are described in subsequent paragraphs.

(continued on page 80)

GROUP ASSEMBLY PARTS LIST

Figure and Index Number	SPCC-MFR Number	Hamilton Part Number	Part Name	Units Per Assy.
	F.S.S.C. 18-C-815	42301	Ship Chronometer Assembly	1
	F.S.S.C. 18-C-815	+42213	Ship Chronometer Assembly	Ref.
	F.S.S.C. 18-C-815	+42212	Ship Chronometer Assembly	Ref.
106-1	H18-HM-10002	+42202	Case - Carrying, complete	Ref.
106-2	H18-HM-10003	+42130	Case - Carrying, complete	Ref.
	H18-HM-10004	42297	Mounting Box and Gimbal Assembly	1
106-3	H18-HM-10005	42201	Box - Mounting, complete	1
106-4	H18-HM-10006	+42129	Box - Mounting, complete	Ref.
106-5	H18-HM-10007	42044	Key - Winding, complete	1
	H18-HM-10008	42100A	Gimbal Assembly	1
106-6	H18-HM-10009	42106	Ring - Gimbal, complete with bushing	1
106-7	H18-HM-10010	42214	Bushing - Pivot screw	1
106-8	H18-HM-10011	42107	Strap - Gimbal support, complete with bushing	1
106-9	H18-HM-10012	42116	Screw - Gimbal support strap	2
106-7	H18-HM-10010	42214	Bushing - Pivot screw	1
106-10	H18-HM-10013	42108	Strap - Case support	1
106-9	H18-HM-10012	42116	Screw - Case support strap	2
106-11	H18-HM-10014	42118	Screw - Case rear pivot	1
106-12	H18-HM-10015	42121	Nut - Pivot screw lock	1
106-13	H18-HM-10016	42119	Screw - Case front pivot	1
106-12	H18-HM-10015	42121	Nut - Pivot screw lock	1
106-14	H18-HM-10017	42120	Screw - Gimbal ring pivot	2
106-15	H18-HM-10018	42122	Washer - Gimbal ring pivot screw	2
106-12	H18-HM-10015	42121	Nut - Pivot screw lock	2
106-16	H18-HM-10019	42109	Bracket - Latch support	1
106-17	H18-HM-10020	42110	Bracket - Latch clamping	1
106-18	H18-HM-10021	42123	Washer - Latch support bracket screw	2
106-19	H18-HM-10022	42115	Screw - Latch support bracket	2
106-20	H18-HM-10023	42111	Lever - Latch	1
106-21	H18-HM-10024	42277	Spring - Latch lever take-up	1
106-22	H18-HM-10025	42276	Screw - Latch lever take-up spring	2
106-23	H18-HM-10026	42114	Screw - Latch clamping	1
106-24	H18-HM-10027	42112	Handle - Latch lever	1
106-25	H18-HM-10028	42298	Case and Movement Assembly	1

+See note on page 88.

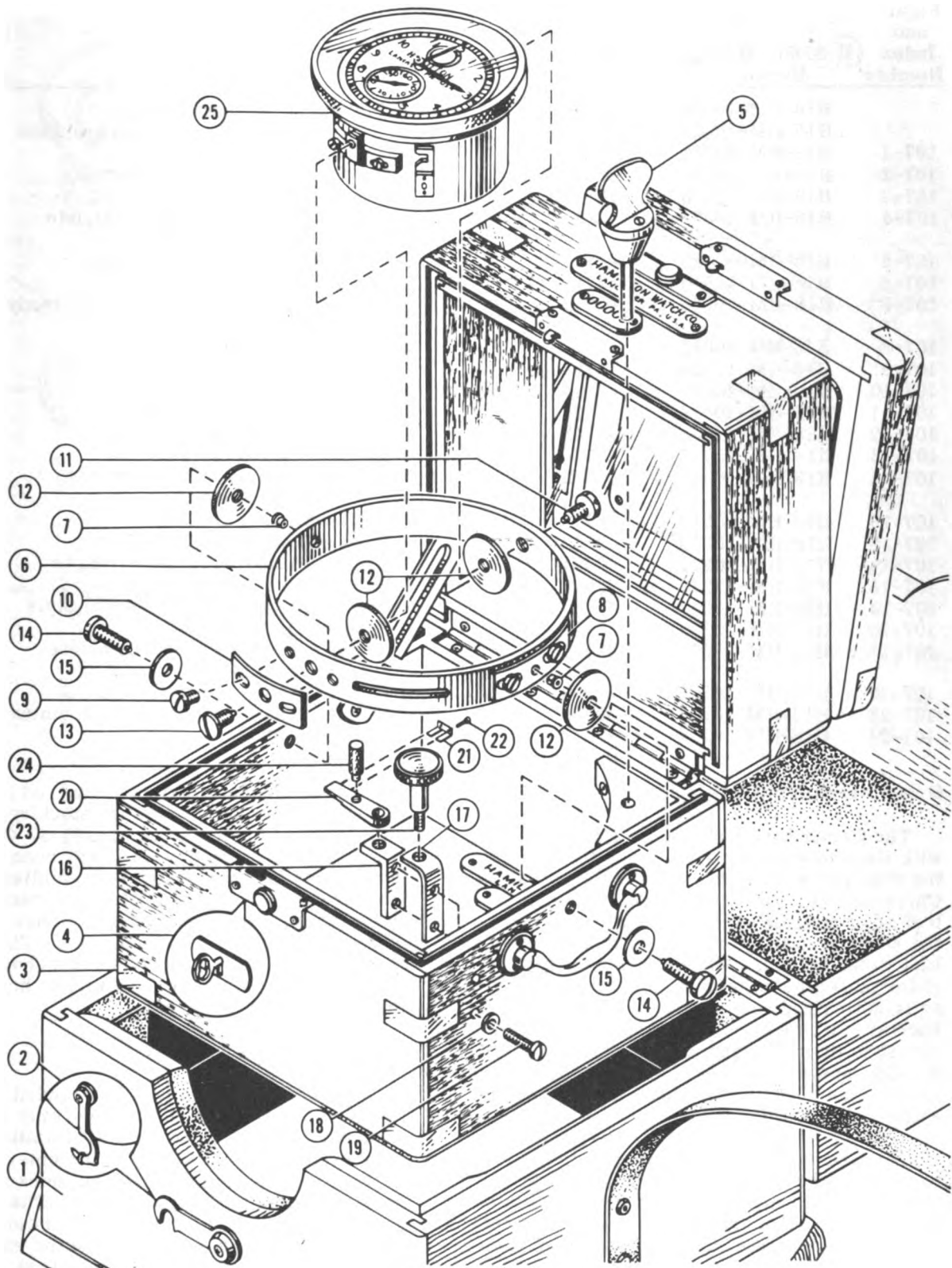


Figure 106

GROUP ASSEMBLY PARTS LIST

Figure and Index Number	SPCC-MFR Number	Hamilton Part Number	Part Name						Units Per Assy.
	H18-HM-10028	42298	Case and Movement Assembly						Ref.
	H18-HM-10029	42296	Case - Complete with bezel and brackets						1
107-1	H18-HM-10030	42101	Case - Chronometer						1
107-2	H18-HM-10039	*42102	Bezel - Complete with crystal						1
107-3	H18-HM-10040	42103	Crystal						1
107-4	H18-HM-10037	42105	Bracket - Case support, complete with bushing						2
107-5	H18-HM-10038	42117	Screw - Case support bracket						4
107-6	H18-HM-10010	42214	Bushings - Pivot screw						2
107-7	H18-HM-10035	42113	Keeper - Latch, complete with steady pins						1
107-8	H18-HM-10012	42116	Screw - Latch keeper						1
107-9	H18-HM-10036	42128	Washer - Separating						1
107-10	H18-HM-10032	42104	Plate - Shield						1
107-11	H18-HM-10034	42124	Screw - Shield plate shoulder						1
107-12	H18-HM-10033	42125	Screw - Shield plate stop						1
107-13	H18-HM-10031	42126	Spring - Shield plate return						1
107-14	H18-HM-10041	*42000	Movement Assembly - Chronometer, 85-size, model 21						1
107-15	H18-HM-10216	42033	Hand - Minute						1
107-16	H18-HM-10215	42032	Hand - Hour						1
107-17	H18-HM-10217	42034	Hand - Second, complete with pin						1
107-18	H18-HM-10218	42035	Hand - Wind indicator						1
107-19	H18-HM-10219	42030	Dial - Modern arabic numeral style						1
107-20	H18-HM-10220	35756	Screw - Dial						4
107-21	H18-HM-10221	42081	Wheel - Wind indicator, complete with hub						1
107-22	H18-HM-10222	42080	Wheel - Hour, complete with hub						1
107-23	H18-HM-10223	42078	Wheel - Minute, complete with pinion						1
107-24	H18-HM-10224	42077	Pinion - Cannon						1

Group Assembly Parts List

The Group Assembly Parts List, together with its exploded views, lists and illustrates the complete disassembly of the Hamilton Ship Chronometer. This breakdown covers, in logical sequence, all operations beginning with the removal of the chronometer from its gimbals to the removal of the bushings from the pillar plate, including some operations ordinarily not performed in chronometer overhaul. Such arrangement not only allows for a complete and logical listing of maintenance parts, but also serves as a reliable supplement to the sections on disassembly, Section V, and reassembly, Section VIII.

The Group Assembly Parts List is indented to indicate the correct relationship of each part to its next higher assembly. The attaching parts for each assembly are listed directly following that assembly, before the assembly is "broken down."

*Not recommended for stocking.

Some component parts are starred (*) with the note "not recommended for stocking." This note is added for one of two reasons. One reason may be that the manufacturer recommends some parts be purchased as assemblies rather than as component parts because assembling the parts may require special equipment not ordinarily found at repair stations. The other reason may be that experience has shown that some starred parts are not likely to be needed for replacement purposes.

Numerical Parts List

A Numerical Parts List is included as Section XII to aid identification when only the Hamilton part number is known. All Hamilton numbered parts are included, in numerical sequence, together with figure and index numbers referring to that component. The figure and index numbers indicate immediately the location of the component in both the exploded views and Group Assembly Parts List.

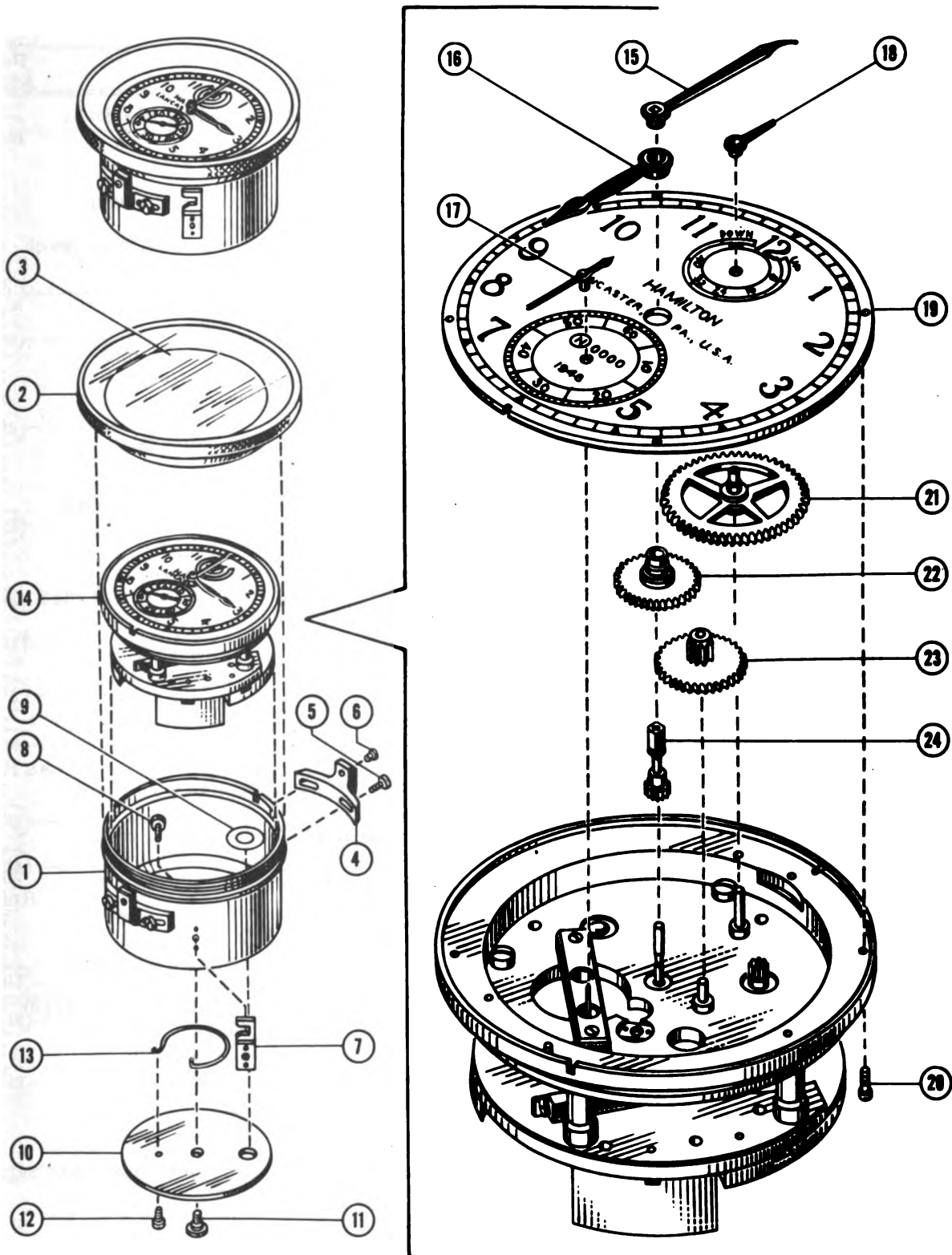


Figure 107

GROUP ASSEMBLY PARTS LIST

Figure and Index Number	SPCC-MFR Number	Hamilton Part Number	Part Name	Units Per Assy.
	H18-HM-10041	42000	Movement Assembly - Chronometer, 85-size, model 21	Ref.
108-1	H18-HM-10225	42066	Cock - Balance, complete with pins	1
108-2	H18-HM-10226	42192	Screw - Balance cock	1
108-3	H18-HM-10227	42160	Cap - Balance upper endstone, with jewel	1
108-4	H18-HM-10228	20762	Screw - Balance upper endstone cap	2
108-5	H18-HM-10229	42155	Setting - Balance upper endstone, with jewel	1
108-6	H18-HM-10230	42162	Setting - Balance, complete with jewel	1
108-7	H18-HM-10231	42193	Balance and Hairspring Assembly	1
108-8	H18-HM-10232	27760	Screw - Hairspring stud	1
108-9	H18-HM-10233	42188	Hairspring	1
108-10	H18-HM-10237	42189	Stud - Hairspring, complete with pins	1
108-11	H18-HM-10236	42147	Pin - Hairspring stud clamp wedge	1
108-12	H18-HM-10235	42191	Clamp - Hairspring stud	1
108-13	H18-HM-10234	42190	Collet - Hairspring	1
108-11	H18-HM-10236	42147	Pin - Hairspring collet clamp wedge	1
108-12	H18-HM-10235	42191	Clamp - Hairspring collet	1
	H18-HM-10238	*42195	Balance Wheel Assembly, with screws	1
108-14	H18-HM-10239	42178	Wheel - Balance, complete with spoke	1
108-15	H18-HM-10240	42177	Screw - Timing weight	2
108-16	H18-HM-10241	42176	Weight - Timing, 93 ± 1 mg.	2
108-17	H18-HM-10242	42197	Screw - Vernier timing weight	2
108-18	H18-HM-10118	37115	Weight - Vernier timing, 10.5 ± 0.3 mgs.	2
108-19	H18-HM-10250	**42171	Screw - Balance, 125 to 130 mgs., 0.049 in. head height	4-6
108-20	H18-HM-10252	**42173	Screw - Balance, 200 to 205 mgs., 0.080 in. head height	2
108-21	H18-HM-10253	**42174	Screw - Balance, 250 to 255 mgs., 0.101 in. head height	2
108-22	H18-HM-10243	**42181	Washer - Timing, 4.0 to 5.0 mgs., 0.002 in. thick	As Req'd.
108-23	H18-HM-10255	42252	Roller - Unlocking, complete with jewel	1
108-24	H18-HM-10256	287	Jewel - Unlocking	1
108-25	H18-HM-10257	**42263	Roller - Impulse, complete with jewel, (Identity No. 3; O.D. .249)	1
108-26	H18-HM-10258	286	Jewel - Impulse	1
108-27	H18-HM-10561	***42180	Roller - One piece	Ref. 1
108-28	H18-HM-10264	++42248	Cap - Balance wheel	1
108-29	H18-HM-10263	++42249	Screw - Balance wheel hold-down	2
108-30	H18-HM-10248	42186	Hub - Balance wheel, complete with staff	1
108-31	H18-HM-10196	42299	Arm - Balance wheel locking	1
108-32	H18-HM-10278	37204	Screw - Flat head fillister (NAVOBSY No. 4618)	1
108-33	H18-HM-10277	42251	Washer - (NAVOBSY No. 55-3)	1
108-34	H18-HM-10214	42300	Pin - Balance wheel locking arm stop	1
108-35	H18-HM-10280	+++42029	Plate - Setup cover	1

**See tables on page 88 for additional variations.

***Part No. 42180 should be replaced with Parts Nos. 42252 and 42263 whenever in need of repair or replacement.

++Not included in early model chronometers.

+++In disassembly, the escape upper bridge, detent assembly, and escape wheel are removed ahead of the setup cover plate. Therefore, to follow disassembly exactly, refer to parts 110-1 through 110-18, in Fig. 110.

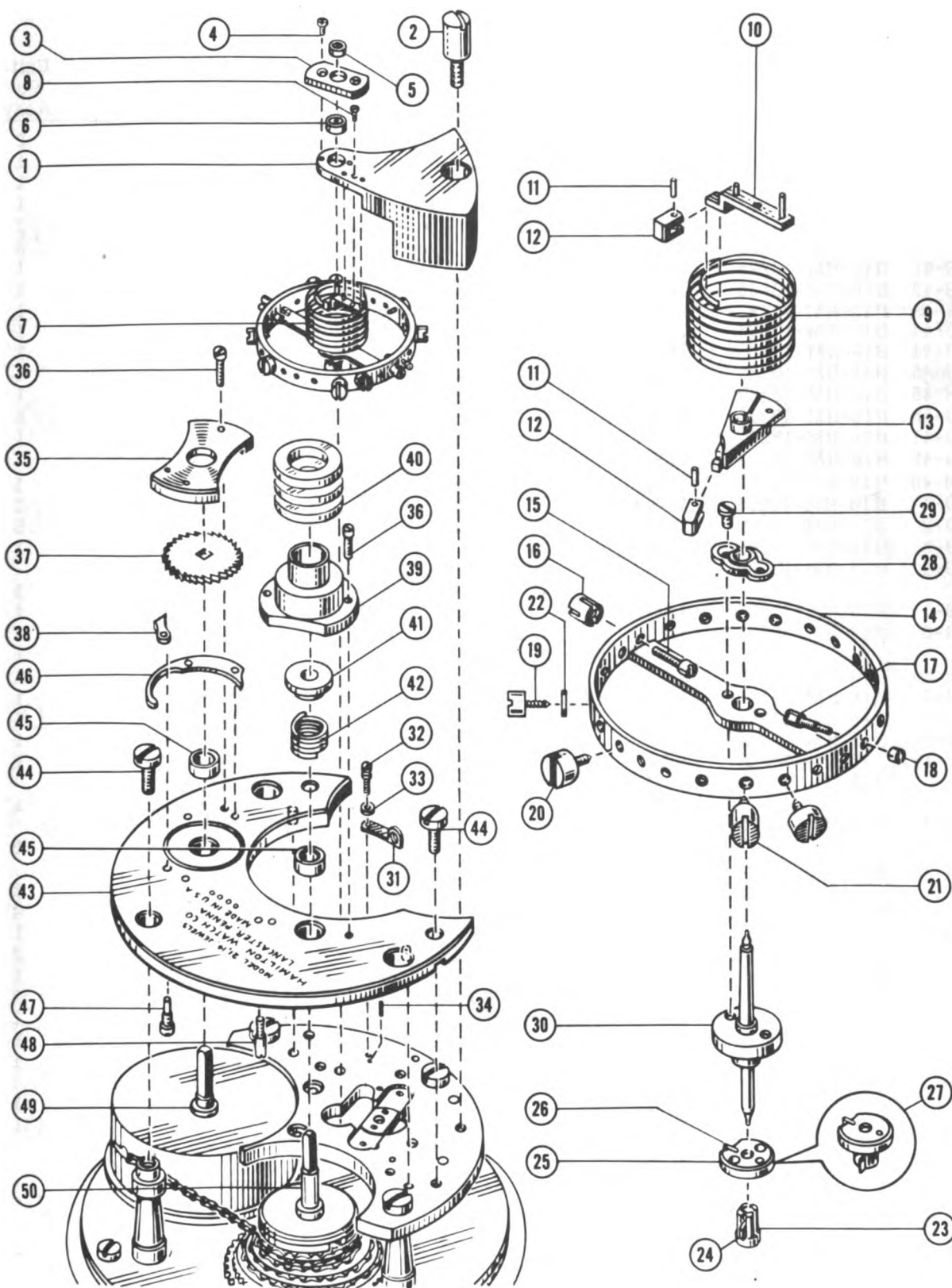


Figure 108

GROUP ASSEMBLY PARTS LIST

Figure and Index Number	SPCC-MFR Number	Hamilton Part Number	Part Name	Units Per Assy.
			1 2 3 4 5 6	
108-36	H18-HM-10109	42056	Screw - Setup cover plate	2
108-37	H18-HM-10281	42026	Wheel - Setup ratchet	1
108-38	H18-HM-10282	42027	Pawl - Setup	1
108-39	H18-HM-10283	42051	Seal - Dust	1
108-36	H18-HM-10109	42056	Screw - Dust seal	2
108-40	H18-HM-10284	42054	Ring - Packing	3
108-41	H18-HM-10285	42052	Ring - Seal	1
108-42	H18-HM-10286	42053	Spring - Helical seal	1
108-43	H18-HM-10287	42061	Bridge - Barrel, complete with pins	1
108-44	H18-HM-10288	42055	Screw - Barrel bridge pillar	1
108-44	H18-HM-10288	42055	Screw - Barrel bridge	2
108-45	H18-HM-10289	42164	Bushing - Barrel upper	1
108-45	H18-HM-10289	42164	Bushing - Fusee upper	1
108-46	H18-HM-10290	42028	Spring - Setup pawl, complete with pins	1
108-47	H18-HM-10291	42036	Screw - Setup pawl pivot	1
108-48	H18-HM-10292	42099	Stop - Fusee winding stop-bar	1
108-49	H18-HM-10293	42242	Barrel Assembly	1
109-1	H18-HM-10294	*42169	Cap - Barrel	1
109-2	H18-HM-10102	37023	Screw - Barrel cap	5
109-3	H18-HM-10295	42170	Arbor - Barrel	1
109-4	H18-HM-10296	42038	Mainspring - Complete with anchor pin, 0.0165 $\pm$ $\frac{5}{10}$ in. thick	1
109-5	H18-HM-10297	42037	Brace - Mainspring	1
109-6	H18-HM-10298	*42168	Barrel - Mainspring, complete with anchor pin	1
108-50	H18-HM-10299	42006	Fusee Assembly	1
109-7	H18-HM-10300	42020	Pin - Taper, fusee assembly	1
109-8	H18-HM-10301	42019	Plate - Fusee end	1
109-9	H18-HM-10302	42015	Wheel - Fusee	1
109-10	H18-HM-10303	42016	Spring - Sustaining, complete with pins	1
	H18-HM-10304	42009A	Wheel Assembly - Sustaining ratchet, com- plete with pawls, springs and screws	1
109-11	H18-HM-10305	42009	Wheel - Sustaining ratchet, complete with pawls	1
109-12	H18-HM-10306	42007	Spring - Winding pawl, complete with pins	2
109-13	H18-HM-10307	42012	Screw - Winding pawl spring	4
109-14	H18-HM-10308	42013	Wheel - Winding ratchet	1
109-15	H18-HM-10309	42014	Screw - Winding ratchet wheel	2
109-16	H18-HM-10310	42008	Plate - Fusee top	1
109-17	H18-HM-10232	27760	Screw - Fusee top plate	2
109-18	H18-HM-10311	42025	Spring - Winding stop-bar	1
109-19	H18-HM-10312	42024	Stop-bar - Winding	1
109-20	H18-HM-10313	42021	Fusee - Complete with arbor	1
109-21	H18-HM-10314	*42022	Arbor - Fusee, complete with wind indi- cator pinion	1
109-22	H18-HM-10315	42001	Fusee Chain Assembly	1

*Not recommended for stocking.

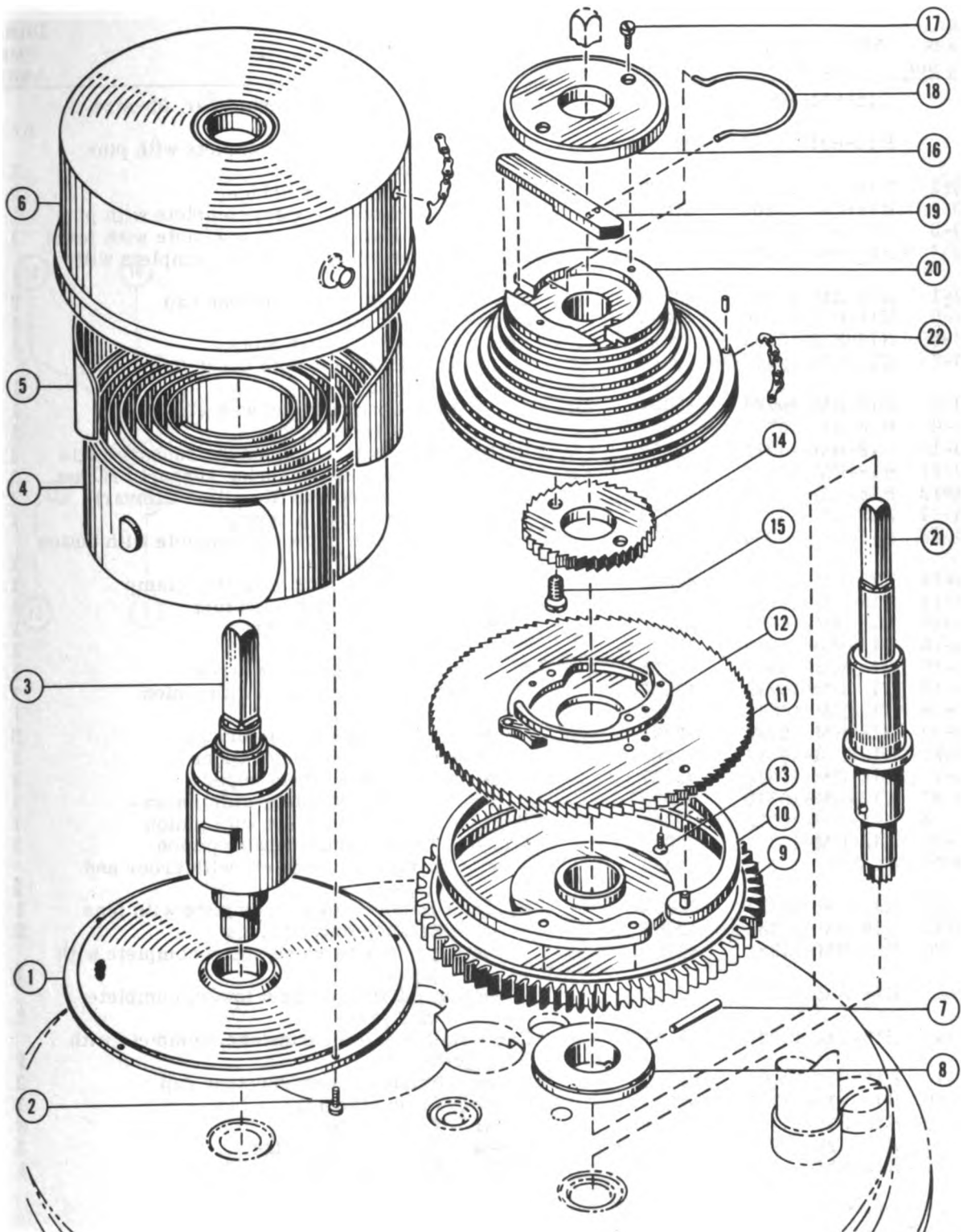


Figure 109

GROUP ASSEMBLY PARTS LIST

Figure and Index Number	SPCC-MFR Number	Hamilton Part Number	Part Name	Units Per Assy.
	H18-HM-10041	42000	Movement Assembly - Chronometer, 85-size, model 21	Ref.
	H18-HM-10265	*42064A	Bridge - Escape upper, complete with pins and setting	1
110-1	H18-HM-10228	20762	Screw - Escape upper bridge	2
110-2	H18-HM-10266	42064	Bridge - Escape upper, complete with pins	1
110-3	H18-HM-10230	42162	Setting - Escape upper, complete with jewel	1
110-4	H18-HM-10267	42159	Cap - Escape upper endstone, complete with jewel	1
110-1	H18-HM-10228	20762	Screw - Escape upper endstone cap	2
110-5	H18-HM-10268	42093	Detent Assembly	1
110-6	H18-HM-10109	42056	Screw - Detent support block	1
110-7	H18-HM-10269	***42087	Detent - Complete with jewel (beryllium copper)	1
110-8	H18-HM-10270	42089	Pin - Locking jewel wedge	1
110-9	H18-HM-10271	285	Jewel - Locking	1
110-10	H18-HM-10272	42092	Bracket - Detent trip spring support angle	1
110-11	H18-HM-10113	1770	Screw - Detent trip spring support bracket	1
110-12	H18-HM-10273	***42088	Spring - Detent trip (Hamilton Elinvar)	1
110-11	H18-HM-10113	1770	Screw - Detent trip spring	1
110-13	H18-HM-10274	42086	Block - Detent support, complete with button and pins	1
110-14	H18-HM-10275	42091	Screw - Detent lock adjusting clamp	1
110-14	H18-HM-10275	42091	Screw - Detent lock adjusting	1
110-15	H18-HM-10276	20756	Screw - Detent adjusting	1
110-16	H18-HM-10278	37024	Screw - Detent clamp	1
110-17	H18-HM-10277	42251	Washer - Detent clamp screw	1
110-18	H18-HM-10279	42076	Wheel - Escape, complete with pinion	1
110-19	H18-HM-10316	42062	Bridge - Upper train	1
110-20	H18-HM-10288	42055	Screw - Pillar upper train bridge	3
110-21	H18-HM-10317	42166	Bushing - Center wheel upper	1
110-22	H18-HM-10318	42167	Bushing - Third wheel upper	1
110-23	H18-HM-10319	42073	Wheel - Fourth, complete with pinion	1
110-24	H18-HM-10320	42068	Wheel - Center, complete with pinion	1
110-25	H18-HM-10321	42071	Wheel - Third, complete with pinion	1
110-26	H18-HM-10322	42096	Pawl - Sustaining, complete with arbor and springs	1
110-27	H18-HM-10323	42065	Bridge - Balance lower, complete with pins	1
110-20	H18-HM-10288	42055	Screw - Balance lower bridge	2
110-28	H18-HM-10324	42161	Setting - Fourth wheel upper, complete with jewel	1
110-3	H18-HM-10230	42162	Setting - Balance wheel lower, complete with jewel	1
110-4	H18-HM-10267	42159	Cap - Balance lower endstone, complete with jewel	1
110-1	H18-HM-10228	20762	Screw - Balance lower endstone cap	2
110-29	H18-HM-10325	**42247	Screw - Train blocking	1
110-30	H18-HM-10326	42059	Pillar - Train bridge	3
110-20	H18-HM-10288	42055	Screw - Pillar train bridge	3
110-31	H18-HM-10327	42058	Pillar - Barrel bridge	1

*Not recommended for stocking.

**Part No. 42247, Train Blocking Screw, is provided in Hamilton chronometer movements beginning with Serial No. 4003.

***Beryllium copper detents shall use only Elinvar detent trip springs.

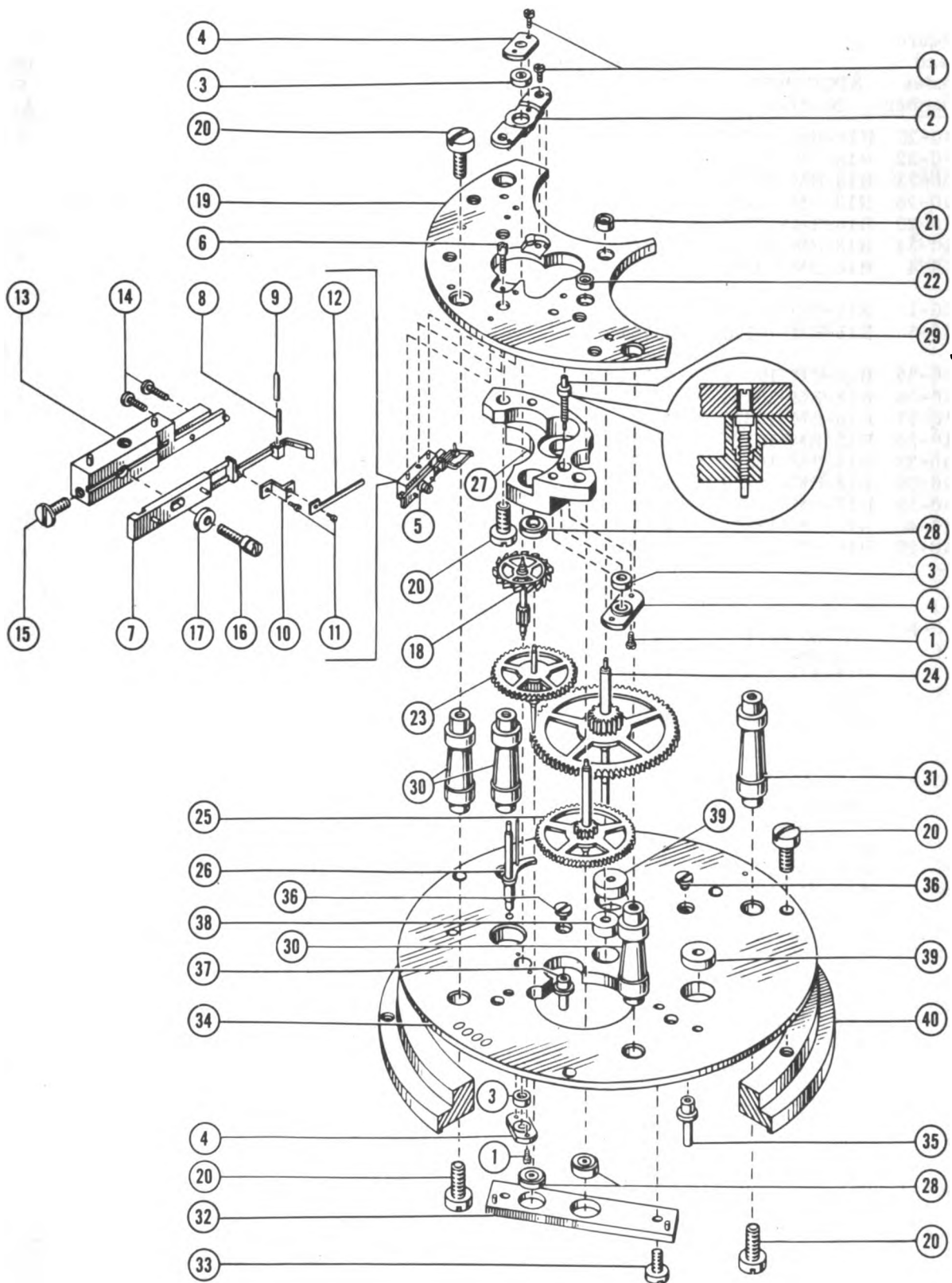


Figure 110

GROUP ASSEMBLY PARTS LIST

Figure and Index Number	SPCC-MFR Number	Hamilton Part Number	Part Name	Units Per Assy.
110-20	H18-HM-10288	42055	Screw - Pillar barrel bridge	1
110-32	H18-HM-10328	42063	Bridge - Lower train, complete with pins	1
110-33	H18-HM-10329	42163	Screw - Lower train bridge	2
110-28	H18-HM-10324	42161	Setting - Third lower, complete with jewel	1
110-28	H18-HM-10324	42161	Setting - Fourth lower, complete with jewel	1
110-34	H18-HM-10330	*42060	Plate - Pillar, complete with pins	1
110-4	H18-HM-10267	42159	Cap - Escape lower endstone, complete with jewel	1
110-1	H18-HM-10228	20762	Screw - Escape lower endstone cap	2
110-3	H18-HM-10230	42162	Setting - Escape wheel lower, complete with jewel	1
110-35	H18-HM-10331	42084	Post - Wind indicator wheel	1
110-36	H18-HM-10164	35779	Screw - Wind indicator wheel post	1
110-37	H18-HM-10332	42085	Post - Minute wheel	1
110-36	H18-HM-10164	35779	Screw - Minute wheel post	1
110-38	H18-HM-10333	42165	Bushing - Center wheel lower	1
110-39	H18-HM-10289	42164	Bushing - Barrel lower	1
110-39	H18-HM-10289	42164	Bushing - Fusee lower	1
110-40	H18-HM-10334	42057	Ring - Mounting	1
110-20	H18-HM-10288	42055	Screw - Mounting ring	3

Table of Additional Variations of Balance Screws

SPCC-MFR Number	Part Number	Weight	Size
H18-HM-10249	42271	100 to 105 mgs.	0.039 in. head height
H18-HM-10251	42172	150 to 155 mgs.	0.060 in. head height
H18-HM-10254	42175	300 to 305 mgs.	0.121 in. head height

Table of Additional Variations of Timing Washers

SPCC-MFR Number	Part Number	Weight	Size
H18-HM-10244	42182	6.0 to 7.0 mgs.	0.003 in. thick
H18-HM-10245	42183	8.0 to 9.0 mgs.	0.004 in. thick
H18-HM-10246	42184	12.0 to 13.0 mgs.	0.006 in. thick
H18-HM-10247	42185	16.0 to 17.0 mgs.	0.008 in. thick
H18-HM-10248	42256	20.0 to 21.0 mgs.	0.010 in. thick

Table of Additional Variations of Impulse Rollers

SPCC-MFR Number	Part Number	Outside Diameter	Identity No. On Part
H18-HM-10259	42264	.250	4
H18-HM-10260	42265	.251	5
H18-HM-10261	42266	.252	6
H18-HM-10262	42267	.253	7

*Not recommended for stocking.

+Not currently supplied. Ship Chronometer Assembly Nos. 42213 and 42212, which included Carrying Case No. 42202 or 42130, are no longer furnished. For further information see Section I and Section II. Ship Chronometer Assembly No. 42212 differed from No. 42213 in the fastening devices used on the Carrying Case and Mounting Box. Parts Nos. 42212, 42130 and 42129 were assigned Serial Nos. 1 to 337. Parts Nos. 42213, 42202 and 42201 were assigned Serial Nos. 338 to 1941.

SECTION XII

NUMERICAL PARTS LIST

Hamilton Part Number	Figure and Index No.	Quantity	Hamilton Part Number	Figure and Index No.	Quantity	Hamilton Part Number	Figure and Index No.	Quantity
285	110-9	1	42058	110-31	1	42126	107-13	1
286	108-26	1	42059	110-30	3	42128	107-9	1
287	108-24	1	42060	110-34	1	42129	106-4	Ref.
1770	110-11	2	42061	108-43	1	42130	106-2	Ref.
20756	110-15	1	42062	110-19	1	42147	108-11	2
20762	108-4	2	42063	110-32	1	42155	108-5	1
	110-1	8	42064	110-2	1	42159	110-4	3
27760	108-8	1	42065	110-27	1	42160	108-3	1
	109-17	2	42066	108-1	1	42161	110-28	3
35756	107-20	4	42068	110-24	1	42162	108-6	1
35779	110-36	2	42071	110-25	1		110-3	3
37115	108-18	2	42073	110-23	1	42163	110-33	2
37023	109-2	5	42076	110-18	1	42164	108-45	2
37024	110-16	1	42077	107-24	1		110-39	2
37204	108-32	1	42078	107-23	1	42165	110-38	1
42000	107-14	1	42080	107-22	1	42166	110-21	1
42001	109-22	1	42081	107-21	1	42167	110-22	1
42006	108-50	1	42084	110-35	1	42168	109-6	1
42007	109-12	2	42085	110-37	1	42169	109-1	1
42008	109-16	1	42086	110-13	1	42170	109-3	1
42009	109-11	1	42087	110-7	1	42171	108-19	4-6
42012	109-13	4	42088	110-12	1	42173	108-20	2
42013	109-14	1	42089	110-8	1	42174	108-21	2
42014	109-15	2	42091	110-14	2	42176	108-16	2
42015	109-9	1	42092	110-10	1	42177	108-15	2
42016	109-10	1	42093	110-5	1	42178	108-14	1
42019	109-8	1	42096	110-26	1	42180	108-27	1
42020	109-7	1	42099	108-48	1	42181	108-22	As Req.
42021	109-20	1	42101	107-1	1	42186	108-30	1
42022	109-21	1	42102	107-2	1	42188	108-9	1
42024	109-19	1	42103	107-3	1	42189	108-10	1
42025	109-18	1	42104	107-10	1	42190	108-13	1
42026	108-37	1	42105	107-4	2	42191	108-12	2
42027	108-38	1	42106	106-6	1	42192	108-2	1
42028	108-46	1	42107	106-8	1	42193	108-7	1
42029	108-35	1	42108	106-10	1	42197	108-17	2
42030	107-19	1	42109	106-16	1	42201	106-3	1
42032	107-16	1	42110	106-17	1	42202	106-1	1
42033	107-15	1	42111	106-20	1	42214	106-7	2
42034	107-17	1	42112	106-24	1		107-6	2
42035	107-18	1	42113	107-7	1	42242	108-49	1
42036	108-47	1	42114	106-23	1	42247	110-29	1
42037	109-5	1	42115	106-19	2	42248	108-28	1
42038	109-4	1	42116	106-9	4	42249	108-29	2
42044	106-5	1		107-8	1	42251	108-33	1
42051	108-39	1	42117	107-5	4		110-17	1
42052	108-41	1	42118	106-11	1	42252	108-23	1
42053	108-42	1	42119	106-13	1	42263	108-25	1
42054	108-40	3	42120	106-14	2	42276	106-22	2
42055	108-44	3	42121	106-12	4	42277	106-21	1
	110-20	12	42122	106-15	2	42298	106-25	1
42056	108-36	4	42123	106-18	2	42299	108-31	1
	110-6	1	42124	107-11	1	42300	108-34	1
42057	110-40	1	42125	107-12	1			

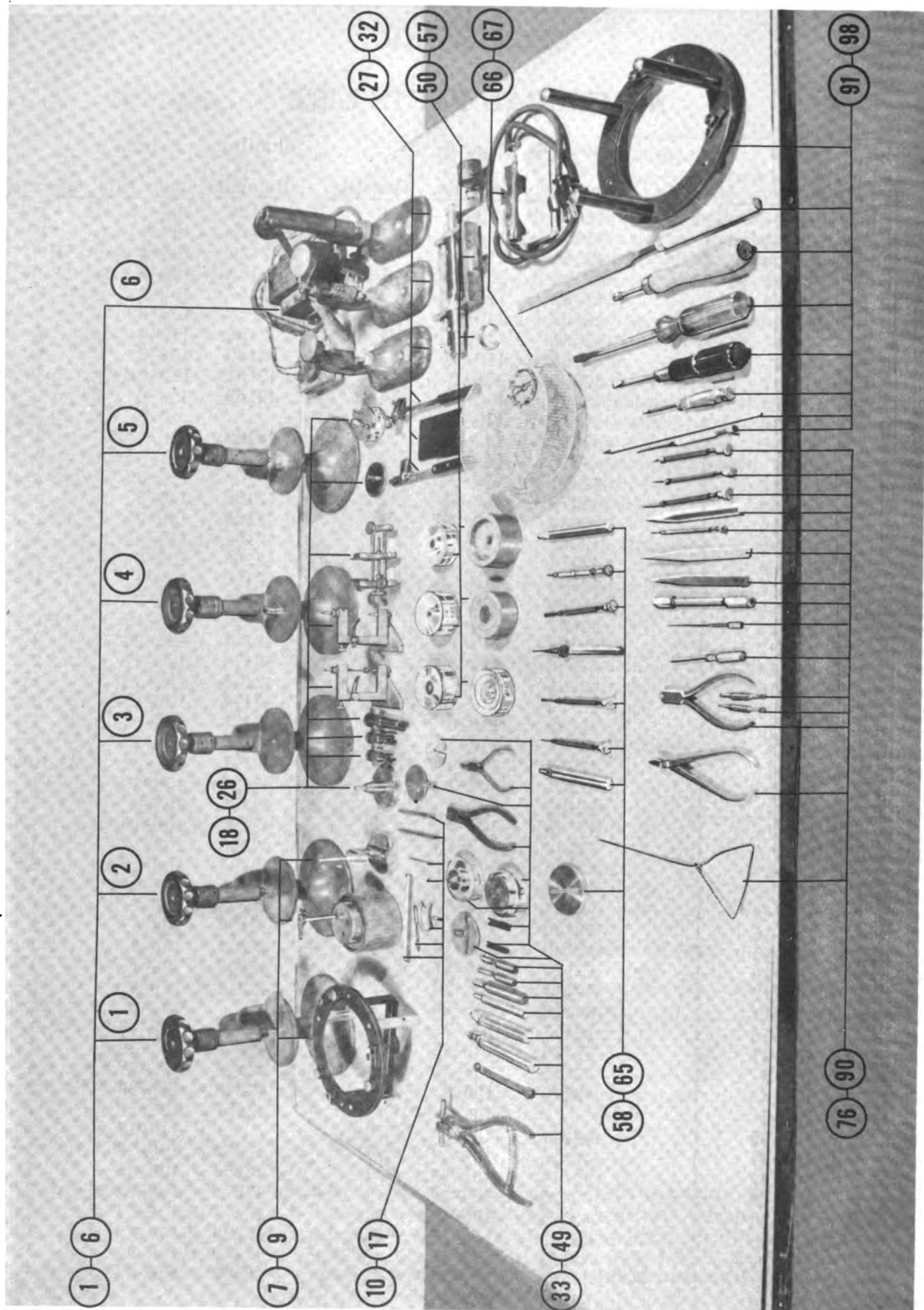


Figure 111

SECTION XIII

SERVICE TOOLS

In the course of developing and mass producing a ship chronometer for the U. S. Navy, the Hamilton Watch Company found it not only expedient but necessary to evolve special service tools, gages and fixtures. See Figs. 111 and 112. These devices, used in conjunction with commercial tools, were the means by which quality was rigidly controlled at no sacrifice in production. The tools and techniques, as identified and set forth in this book, are the results of modern methods engineering and work simplification. They are used by the Bureau of Ships as a basis for establishing control of chronometer overhaul and repair methods and procedures.

Contained herein is a listing of all special tools, gages and fixtures, together with other service tools. The words "Hamilton Make" denote that, if it is desired to procure the item, the special tool (or special inserts for a standard tool) can be obtained only from the Hamilton Watch Company; otherwise, all items listed are representative of commercial equipment which can be acquired through various sources. The name of each tool has been made as functional as possible, so that the reader will have an idea of its application. All tools have been appropriately referenced and are used throughout the overhaul and repair procedures described in this handbook.

SHIP CHRONOMETER

Service Tool List

Hamilton Index and Tool No.	Tool Name and Application
1	Stand - Endshaking (to adjust endshake of center upper and third upper bushings), Hamilton make.
2	Stand - Endshaking (to adjust endshaking of bar-hole jewels), Hamilton make.
3	Press - Roller (to press on balance rollers), Hamilton make.
4	Press - Roller (impulse roller removing stand), Hamilton make.
5	Stand - Endshaking (to adjust endshake of barrel and fusee bushings), Hamilton make.
6	Heater - Impulse roller (to heat impulse roller before applying shellac), Hamilton make.
7	Ring - Dialing (to hold movement during dialing), Hamilton make.
8	Fixture - Detent spring testing, with weight, (to hold detent while testing trip spring with .770 gram weight), Hamilton make.
9	Tool - Wedge pin staking, including small punch (used in colleting and studding with Tool No. 17; used in jewel detent with punch as supplied), Hamilton make.
10	Stone - Arkansas oil (to stone down detent trip spring or collet and stud taper pin).
11	Wedge - Detent removing (to hold detent while removing detent support block screw), Hamilton make.
12	Sector - Graduated (to measure balance wheel angle in detent adjustment), Hamilton make.
13	Pin - Gage (to serve as an indicator of balance wheel position, used with Tool No. 12), Hamilton make.
14	Washer - 0.010 in. thick (to support detent in disassembly).
15	Punch - Taper pin staking (to stake taper pin in colleting and studding).
16	Punch - Balance staff removing, Hamilton make.

SHIP CHRONOMETER

Service Tool List

Hamilton Index and Tool No.	Tool Name and Application
17	Punch - Taper pin removing (to remove taper pin in colleting and studding -- used with Tool No. 9).
18	Key - Let-down (to let down mainspring), Hamilton make.
19	Remover - Minute hand, double jaw-screw pressure type, Hamilton make.
20	Remover - Hour hand, double jaw-screw pressure type, Hamilton make.
21	Remover - Cannon pinion, double jaw-screw pressure type, Hamilton make.
22	Caliper - Hairspring truing, Hamilton make.
23	Caliper - Balance truing, Hamilton make.
24	Tool - Balance poising, Hamilton make.
25	Gage - Impulse roller jewel, Hamilton make.
26	Gage - Unlocking roller jewel, Hamilton make.
27	Tool - Setting, unlocking roller jewel (to set unlocking roller jewel in position in jewelng operation), supplied with clamp nut wrench, Hamilton make.
28	Mat - Rubber, roller jewel replacement, Hamilton make.
29	Tool - Setting, impulse roller jewel (to set impulse roller jewel in position in jewelng operation), Hamilton make.
30	Undercutter - Balance screw, Hamilton make.
31	Undercutter - Endstone, Hamilton make.
32	Press - Endstone, Hamilton make.
33	Closer - Cannon pinion (used in dialing operation), Hamilton make.
34	Wrench - Balance lower bridge screw, Hamilton make.
35	Punch - Hour hand, lucite (to be used in dialing operation), Hamilton make.
36	Punch - Minute hand, lucite (to be used in dialing operation), Hamilton make.
37	Punch - Wind indicator hand, lucite (to be used in dialing operation), Hamilton make.
38	Pusher - Second hand (to be used in dialing operation), Hamilton make.
39	Broach - Hour wheel, Hamilton make.
40	Broach - Hour hand, Hamilton make.
41	Tool - Locking jewel locating (to locate locking jewel in jewelng the detent), Hamilton make.
42	Tool - Drop adjusting (to adjust angle of unlocking roller), Hamilton make.
43	Tool - Roller alignment (to adjust angle of impulse roller), Hamilton make.
44	Block - Colleting, lucite, metal insert, Hamilton make.
45	Block - Studding, lucite, metal insert, Hamilton make.
46	Pliers - Cutting (used in colleting and studding), Hamilton make.
47	Key - Winding, Hamilton make.
48	Pliers - Unlocking roller removing, Hamilton make.
49	Shield - Dial (to protect dial when removing wind indicator and second hand), Hamilton make.
50	Block - Balance staff fitting, lucite, metal insert, Hamilton make.
51	Block - Sustaining ratchet wheel assembly, lucite, Hamilton make.

SHIP CHRONOMETER

Service Tool List

Hamilton Index and Tool No.	Tool Name and Application
52	Block - Balance work, lucite, Hamilton make.
53	Block - Fusee assembly, Hamilton make.
54	Block - Balance, numbered, lucite, metal insert (used in balance wheel screw adjustments), Hamilton make.
55	Block - Barrel assembly, Hamilton make.
56	Ring - Mainspring (to hold wound mainspring when out of barrel), Hamilton make.
57	Winder - Mainspring, Hamilton make.
58	Block - Collet taper pin filing, Hamilton make.
59	Punch - Balance staff staking, Hamilton make.
60	Screwdriver - Vernier timing weight adjustment, Hamilton make.
61	Screwdriver - Timing weight adjustment, Hamilton make.
62	Screwdriver - Bubble gage type, vernier timing weight, Hamilton make.
63	Screwdriver - Balance wheel screw adjustment, Hamilton make.
64	Screwdriver - Hairspring wedge (to remove and replace hairspring assembly), Hamilton make.
65	Holder - Impulse roller reamer, lucite, (to be used with Tool No. 81), Hamilton make.
66	Holder - Balance cleaning, Hamilton make.
67	Heater - Roller and detent (to heat roller and detent in jewelng operation), Hamilton make.
76	Wire - Dip (to hand clean chronometer parts), Hamilton make.
77	Pliers - End cutting (to cut unlocking jewel pin).
78	Pliers - Side cutting (used in colleting and studding).
79	Remover - Wind indicator and second hand, claw at one end.
80	Remover - Wind indicator and second hand, claw at one end.
81	Reamer - Impulse roller (used with Tool No. 65).
82	Broach - Second hand.
83	Vise - Pin (used in colleting and studding).
84	Tweezers - Steel, No. 3 size (used to hold smallest parts).
85	Tweezers - Brass.
86	Screwdriver - Small (to repair detent).
87	Pointer - Lucite (for use on balance wheel when adjusting detent).
88	Screwdriver - Balance cap screws.
89	Screwdriver - Hairspring stud and train blocking screw.
90	Screwdriver - Endstone cap screws.
91	Screwdriver - Dial screws.
92	Pointer - Nickel wire, pointed (to remove shellac in jewelng operation).
93	Screwdriver - Fusee ratchet wheel screws.
94	Screwdriver - Lower train bridge screws.
95	Screwdriver - Balance cock screw.

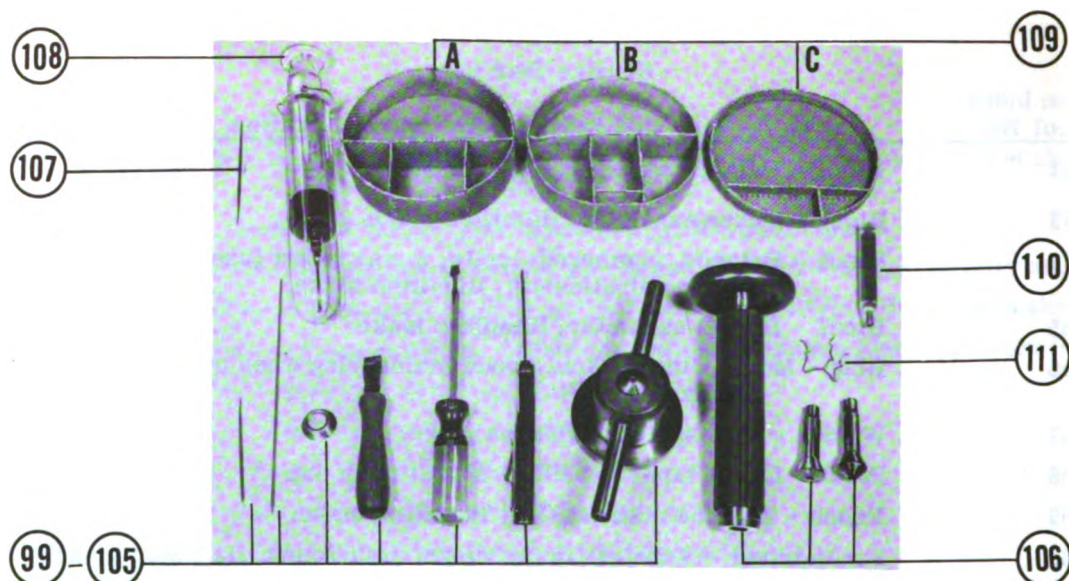


Figure 112

SHIP CHRONOMETER

Service Tool List

Hamilton Index and Tool No.	Tool Name and Application
96	Screwdriver - Shell (bridge and pillar screw), Hamilton make.
97	File - Fine cut (to file taper pins in colleting and studding).
98	Ring - Work, with legs (to hold movement in disassembly and reassembly), Hamilton make.
99	Tool - Unlocking roller jewel insertion (filed down at one end to fit in the unlocking roller jewel slot), Hamilton make.
100	Tool - Roller holding (turned down at one end to hold roller on heater in jeweling operation), Hamilton make.
101	Level - Circular (to level chronometer in gimbals).
102	Screwdriver - Wide blade, gimbal pivot screws.
103	Screwdriver - Gimbal support strap screw, ground round behind end of blade, Hamilton make.
104	Illuminator - Special, small bulb at end of thin tube, (to illuminate detent and rollers during adjustment), Hamilton make.
105	Closer - Second hand (to close hole in second hand to tighten fit), Hamilton make.
106	Closer - Minute and hour hand, two closers and handle (to close hole in the minute and hour hand to tighten fit), Hamilton make.
107	File - Rat tail (to file fusee end plate channel to adjust fusee endshake).
108	Syringe - Mainspring grease.
109a.	Basket - Cleaning, four compartments, Hamilton make.
109b.	Basket - Cleaning, five compartments, Hamilton make.
109c.	Basket - Cleaning, three compartments, Hamilton make.
110	Tool - Special setting (to set hour and minute hands), Hamilton make.
111	Tripod - Balance (to hold balance for special cleaning), Hamilton make.

SECTION XIV

CHRONOMETER SERVICE BULLETINS

Introduction

This section is provided for the handbook as a place to incorporate the various ship chronometer service bulletins as they are issued by the Bureau of Ships. It is contemplated that such bulletins will be the medium by which chronometer repair activities are informed of changes and developments in overhaul and repair techniques. Hence, it is important that all such information on chronometer repair be placed immediately in this section of the handbook. Users should be advised of the importance of keeping up-to-date on all changes. Good work can come only from well-informed personnel who are experienced in their duties.

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